AI33-MPOPT · 2026

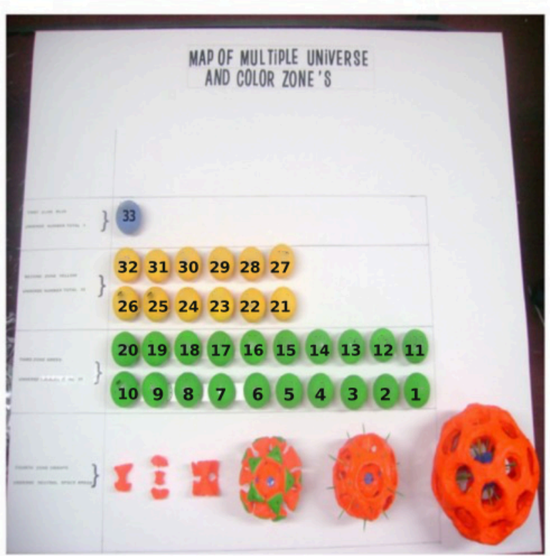
The 33-Multiverse Map

One geometry, mapped from every zone — the icosahedral architecture where **Universe #33**, the central hub, is surrounded by **12 inner**, **20 outer**, and **20 curvilinear** domains.

Built from the physical model, mapped figure by figure, with its sizes compared to published cosmology.

989T ly	32 throats	Milky Way ± 7%
full-capacity diameter	converge on the hub	compared with public JWST, Planck and DESI data

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



- Zone 1 — Core, Universe #33
- Zone 2 — 12 inner, Universes #21–32
- Zone 3 — 20 outer, Universes #1–20
- Zone 4 — 20 Curvilinears, the neutral space between

WHY THE NUMBERS, ZONES AND COLORS

Every universe has a number and a color so the map is easy to see, explain and measure. **Blue** is the hub, #33. **Yellow** is the 12 inner universes, #21–32. **Green** is the 20 outer universes, #1–20. **Orange** is the 20 curvilinears between them. If everything were one color, no one could tell which part you mean or point to a distance. A fixed number and zone give each universe an address, so any distance can be measured, checked and repeated, and future researchers can add their own measurements and build on this map without losing their place.

Rolando Rivero · Independent Researcher · AI33-MPOPT

GEOMETRICAL SCALE · FULL CAPACITY

COMPUTED

How Big Is the 33-Multiverse?

989 trillion ly

total diameter, all zones

2.14×10⁴¹ ly³

total volume, cone-derived

510 million×

the observable universe's volume

1 in 15.6M

filled today, by volume

DIAMETER — LINEAR SUM

Zone 1 · Core (#33)	23 trillion ly
Zone 2 · Inner 12	276 trillion ly
Zone 3 · Outer 20	460 trillion ly
Zone 4 · Curvilinears (×20)	230 trillion ly
Total diameter	**989 trillion ly**

VOLUME — CONE-DERIVED

Zone 1 · Core (#33)	6.37×10³⁹ ly³
Zone 2 · Inner 12	7.64×10⁴⁰ ly³
Zone 3 · Outer 20	1.27×10⁴¹ ly³
Zone 4 · Curvilinears (bicone)	3.98×10³⁹ ly³
Total volume	**2.14×10⁴¹ ly³**

FIGURE 1

The Complete 33-Multiverse Geometrical System

ALL 33 UNIVERSES

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

1–20

green, Zone 3, outer

21–32

yellow, Zone 2, inner

33

blue, Zone 1, the hub

DESCRIPTION

Every universe has a number and a color so the map is easy to see, explain and measure.

- Zone 1 — Core, Universe #33
- Zone 2 — 12 inner, Universes #21–32
- Zone 3 — 20 outer, Universes #1–20

What you cannot see: the blue center hub #33 sits inside. The photo shows the north side facing you; the other numbers are on the back, or rotary, side, the south side, or hidden behind a neighbor. Bring any other universe of the same group to the front and the geometry looks the same, but each universe keeps its own number and its own place.

THE NUMBERS

Count how many universes each of the 120 symmetries of this shape leaves in place and average them: the result is exactly 3, the number of groups: the hub, the twelve, and the twenty. Within each group the geometry gives every universe the same role: nothing in the structure alone can tell one apex universe from another, or one outer universe from another. What each universe does still depends on where a process starts (Rivero-Zeta, page 30). That is what ‘one structure’ means: the same geometry in every member of a group, not the same behavior. · Rivero-Zeta / Cosmic Web Atlas



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub
Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe’s own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0) · **Yellow, the 12:** all 23.00T ly out, at latitudes 90°, 26.57°, −26.57°, −90°
Green, the 20: all 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62°

FIGURE 2

Where Two Rings Interlock

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

12 + 20

inner and outer, nested

Ω_Λ≈0.70

dark-energy fraction

Ω_m≈0.30

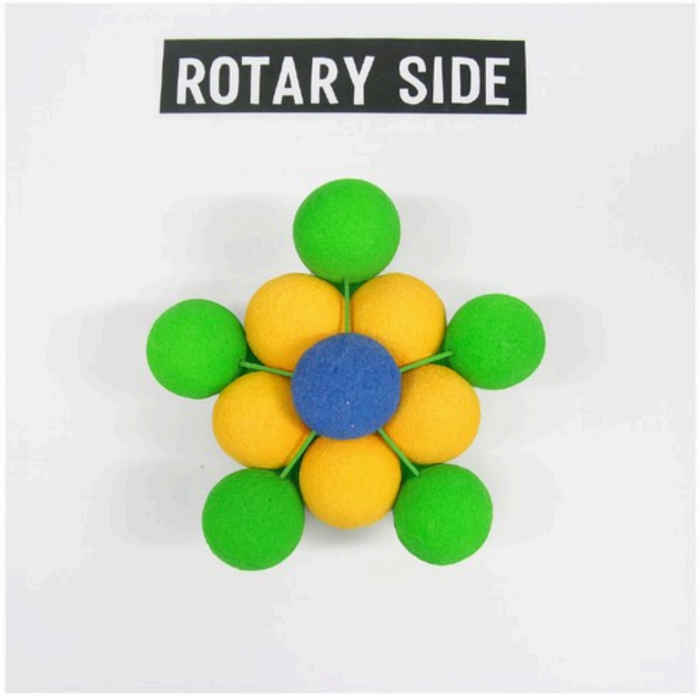
dark-matter fraction

DESCRIPTION

Yellow Zone 2 spheres nest within the connecting green Zone 3 triangles. The geometric tension between the two zones is what breaks the initial symmetry and gives birth to the four forces. At each green apex, dark energy pushes inward through the coherence field between the zones, feeding the blue hub at inception; at each yellow apex, dark matter pulls toward that same hub. The two forces do not meet head-on — they arrive in a curving, spinning rotation around the hub. The sign in the picture reads ROTARY SIDE: the rotary side, the south side (green #1–10 and yellow #21–26), is pulled out completely, so you see the inside of the north side, the inside back side, where the curvilinears work (green #11–20 and yellow #27–32). The sticks are the energy lines: the transparent curvilinear throats.

THE NUMBERS

One rotation step per domain: 360° ÷ 33 = 10.9091°. Per throat: 360° ÷ 32 = 11.25°. Five-fold at an apex is 72° and three-fold at a curvilinear is 120°; all close exactly. · Rivero-Zeta



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.

Blue #33, the hub: origin (0, 0, 0)

Shown, the inside of the north side: yellow #27–32, 23.00T ly out: #27 at latitude 90°, #28–32 at 26.57° · green #11–20, 36.55T ly out: #11–15 at 52.62°, #16–20 at 10.81°

Pulled out, the rotary side, the south side: yellow #21–26: #21 at latitude −90°, #22–26 at −26.57° · green #1–10: #1–5 at −10.81°, #6–10 at −52.62°

FIGURE 3

An Empty Shell of the Curvilinears

ZONE 1 HUB | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

12 missing

yellow, dark matter — left out on purpose

20

curvilinears, the shell itself

Clear, not orange

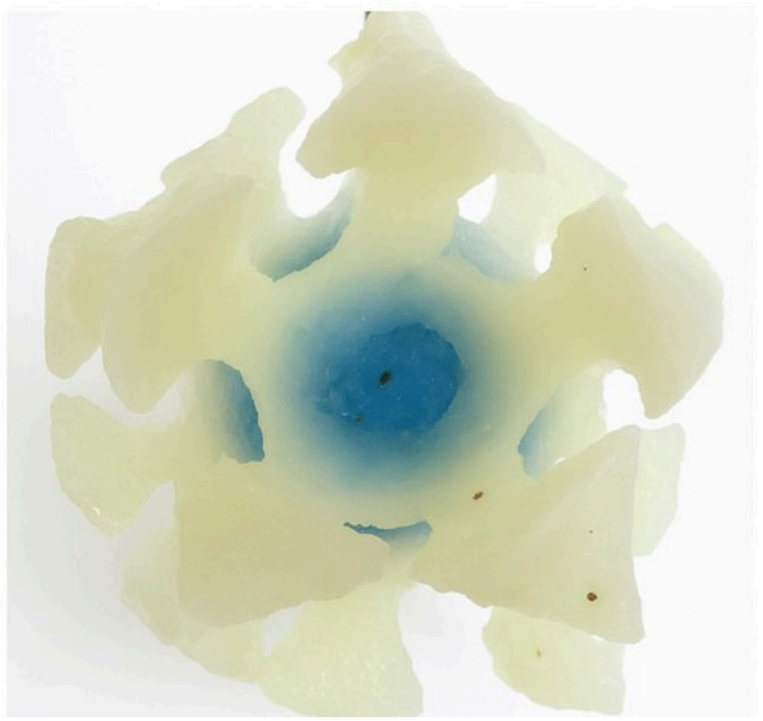
rendered to signal dark energy

DESCRIPTION

This is the shell of the 20 Curvilinears alone — no spheres are shown. Rendered translucent, not solid orange, it marks dark energy pushing inward through the throats toward the blue center, Universe #33, where all 32 throats meet: 20 pushing, 12 pulling, all converging inward, none outward. The gaps are deliberate: that is where the 12 yellow, dark-matter domains would sit. In the full system the 20 Curvilinears are all interconnected: each one touches its three neighbors along the three sides of its base, 20 × 3 ÷ 2 = 30 contacts, so together they close into one continuous shell around the hub.

THE NUMBERS

152 links − 33 domains + 1 = 120 independent loops. Net displacement zero, net turn 360° — the shell completes a full circulation without the spheres it is missing. · Rivero-Zeta



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Orange, the 20 curvilinears C1–C20: on the same 20 directions as green #1–20, each centered 18.28T ly from the hub and 23.00T ly long
Yellow, the 12 (left out): they would sit 23.00T ly out, at latitudes 90°, 26.57°, −26.57°, −90°

FIGURE 4

The Outer Twenty Drive the Expansion

ZONE 1 HUB | ZONE 3 OUTER 20 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

20
universes, #1–20

460T ly
zone diameter, summed

ρ_Λ ~10⁻²⁷
dark-energy magnitude, kg/m³

DESCRIPTION

Universes 1–20 arrange into their stable icosahedral shell — Zone 3. Each green face is one outer, dark-energy domain, numbered here 1 through 5; the orange body is the Curvilinear structure joining them, and the small blue center visible through the gap is the hub, Universe #33. As on the empty shell shown earlier, the push here runs inward, through the throats toward that center — not outward.

THE NUMBERS

Each of the 20 faces is flat inside: 0.0000° of curvature. Curvilinear throat: half-angle 21.8475°, mouth 0.4513 sr, deficit 226.0303°. Dark-energy scale carried inward: ρ_Λ ~ 10^{-27} kg/m³. · Rivero-Zeta / GENESYS



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Green #1–5: 36.55T ly out, latitude −10.81°: #1 18.0° · #2 90.0° · #3 162.0° · #4 234.0° · #5 306.0° longitude
Green, the 20: all 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62° (5 at each)
Orange, C1–C20: each centered 18.28T ly from the hub and 23.00T ly long

FIGURE 5

The Inner Twelve Hold the Line

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

12
universes, #21–32

276T ly
zone diameter, summed

#27–32
shown in this cluster

DESCRIPTION

Universes 21–32 form Zone 2, shown here close-up (27–32). This inner layer provides the stabilizing, cohesive pull of Dark Matter, preventing runaway expansion driven by the outer shell. This is the same structure as the empty Curvilinear shell on the previous page, now shown the other way round: the 12 yellow domains are fully present, and the small green wedges visible between them mark where each outer, dark-energy apex would sit — present, but only at the edge of view.

THE NUMBERS

Each of the 12 carries a 60° defect — with $\delta = 8\pi G\mu/c^2$ that is about 5.6×10^{25} kg/m. Apex throat: half-angle 17.6289°, mouth 0.2951 sr. The green slivers shown here belong to the other class entirely: curvilinear throat, half-angle 21.8475°, mouth 0.4513 sr — two different geometries meeting at the same seam. · Rivero-Zeta, c^2 restored

LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub
Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Yellow #27–32: 23.00T ly out · #27 latitude 90° · #28–32 latitude 26.57°, longitudes 162.0°, 234.0°, 306.0°, 18.0°, 90.0°
Blue #33, the hub: origin (0, 0, 0)



FIGURE 6

Sixteen Throats, One Epicenter

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#33

the Creator Core

8 + 8 visible

of 32 — the rest sit behind this angle

Before Singularity 10.33

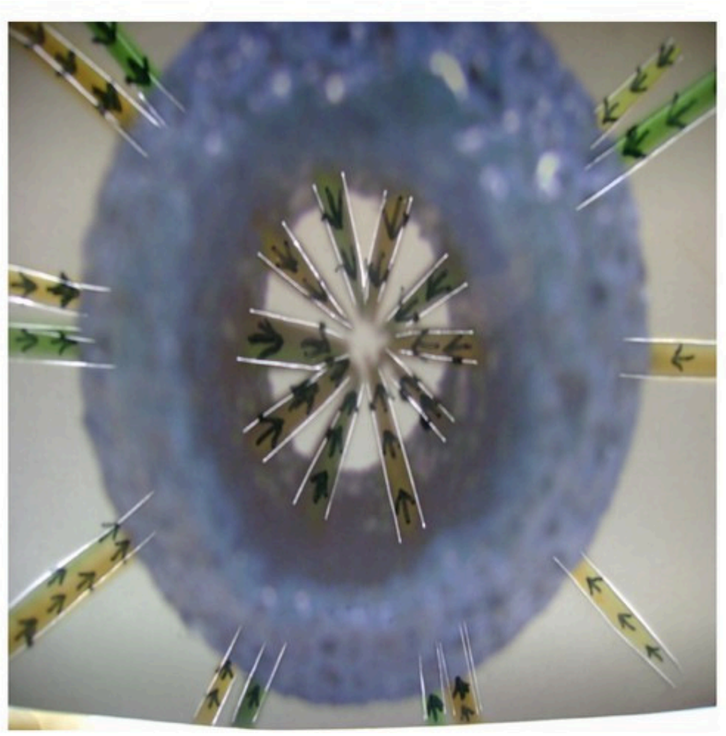
converging, not yet burst

DESCRIPTION

The 32 multiverses are intentionally not shown — only their central energy lines, each marking the exact point where a universe would sit. Lines are missing here for two reasons: some fall behind others at this angle; others were left off the front and back faces entirely, so the view stays open rather than closed off. 8 green and 8 yellow are countable, converging toward the 'Singularity 10.33' point — the instant before it fires, not the burst itself.

THE NUMBERS

Inside a throat the wall is flat (K = 0). All the curvature sits at the tip as an angle deficit: 250.9735° at an apex throat, 226.0303° at a curvilinear throat. Two mouths, two values. · Rivero-Zeta



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Yellow throat lines, 12: all 23.00T ly out, at latitudes 90°, 26.57°, −26.57°, −90°
Green throat lines, 20: all 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62°

FIGURE 7

Ten and Ten, Aimed at One Point

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#33

the Creator Core

Singularity 10.33

the epicenter, now lit

10 + 10 visible

of 32 — the rest sit behind this angle

DESCRIPTION

The central blue sphere, #33, is the Creator Core, cut away on purpose so the burst is visible, not hidden behind a solid surface. Lines are missing for two reasons: some sit behind the yellow at this angle; others were left off the front and back faces, so the cutaway stays open. 10 green and 10 yellow are countable. This is the 'Singularity 10.33' point itself, arriving from the previous page: inception, not its setup.

THE NUMBERS

The hub joins all 32 others: 32 throats + 120 sibling links = 152 links · Rivero-Zeta. Energy sufficiency: 20 throats reach E_g/U = 0.80 (short), 32 reach 1.28, 33 reach 1.38 · GENESYS

LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Yellow throat lines, 12: all 23.00T ly out, at latitudes 90°, 26.57°, −26.57°, −90°
Green throat lines, 20: all 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62°

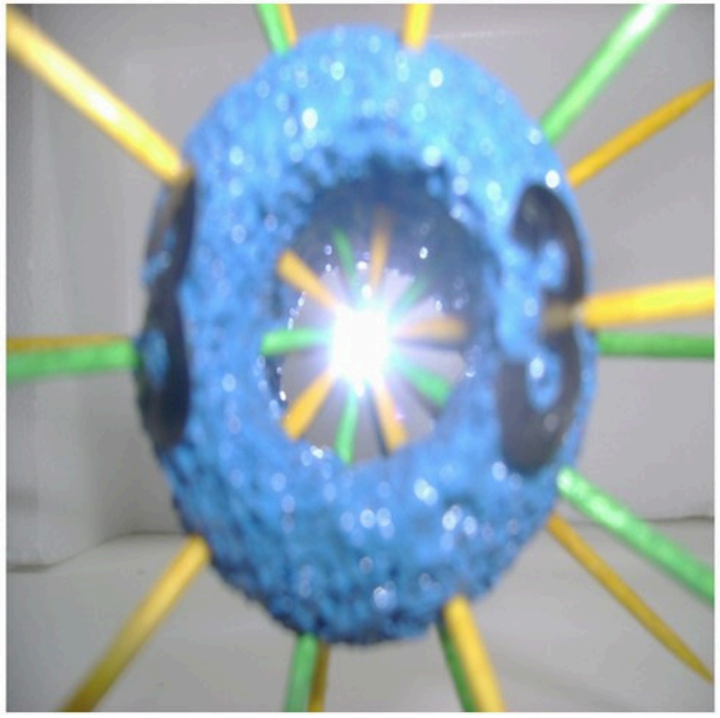


FIGURE 8

How a Curvilinear Joins Three Zones

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

20
curvilinears, Zone 4

230T ly
zone diameter, summed

3-fold
apex domains joined per piece

DESCRIPTION

One orange Curvilinear piece, sitting exactly where three zones meet: the blue hub (#33), a yellow inner domain, and a green outer domain, all touching at once. The concave notch cut into the piece is not decoration — it is shaped to cradle one specific apex, curving around the space that sphere occupies. A second yellow domain sits just beyond it, left outside this particular notch, showing that each Curvilinear fits its own apexes and no others.

THE NUMBERS

Each curvilinear has three-fold local symmetry: a 120° turn returns the same view, so it joins exactly 3 apex domains. 60 apex-curvilinear edges run across the shell, identical couplings but different surroundings — why one piece cradles one apex and leaves its neighbor untouched. · GENESYS. Zone 4 volume 3.98×10^{39} ly³, waist assumed at ¼ of the length.



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub
Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Example C1: on green #1 (−10.81°, 18.0°, 36.55T ly), centered 18.28T ly out, 23.00T ly long
Touches the hub and yellow #22 (−26.57°, 54.0°), #26 (−26.57°, 342.0°), #31 (26.57°, 18.0°), all 23.00T ly out

FIGURE 9

One Curvilinear, Built in Full

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

20
curvilinears, Zone 4

1 of 20
shown sculpted in full 3D

#27 · #33
the yellow apex and the blue hub

DESCRIPTION

Only three pieces here are built in complete 3D: the orange Curvilinear vein, yellow universe #27, and the blue hub, #33 — each raised, rounded, and fully sculpted. Everything else in view is a shell: numbers painted flat onto a textured surface, marking where each universe belongs before it has been given its own physical form. This is the model mid-construction, concept and finished piece side by side.

THE NUMBERS

The flat shell obeys the same math as the sculpted vein: three-fold symmetry at every curvilinear, coupling to its own apex at 2.2×10^{-17} cross-talk to any other — zero, to machine precision — and a least-action check of $\delta S = 0$, no ghost modes. Painted or sculpted, the geometry doesn't change. · Rivero-Zeta / GENESYS

LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Yellow #27: latitude 90.0°, 23.00T ly from the hub
Orange, the sculpted curvilinear: one of C1–C20, centered 18.28T ly from the hub and 23.00T ly long



FIGURE 10

Three Curves, One Waveform

ZONE 2 INNER 12 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

3	1.3×10⁻⁴ ρ_Λ	~0.2 Gyr
curves united	beat modulation amplitude	coupling window

DESCRIPTION

Three curves join into one continuous shape, and together they read as a single string-like line of energy, like one note on a string. The exact part is in the throats: each is a cone with an angle deficit, the same geometry a cosmic string (a line defect) leaves in space.

THE NUMBERS

Deficit δ = 8πGμ/c²: 4.3803 rad at an apex throat, 3.9450 rad at a curvilinear throat · Rivero-Zeta. An analogy only: real cosmic strings are limited by the sky to Gμ/c² below about 10^{-7}; here it is δ/8π ≈ 0.16 to 0.17.

LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.

Three yellow universes touching: each pair 24.18T ly apart, center to center, all 23.00T ly from the hub; the curved gap between them holds one curvilinear

Example, the face of green #1: yellow #22 (−26.57°, 54.0°) · #26 (−26.57°, 342.0°) · #31 (26.57°, 18.0°); gap on the axis of #1 (−10.81°, 18.0°)

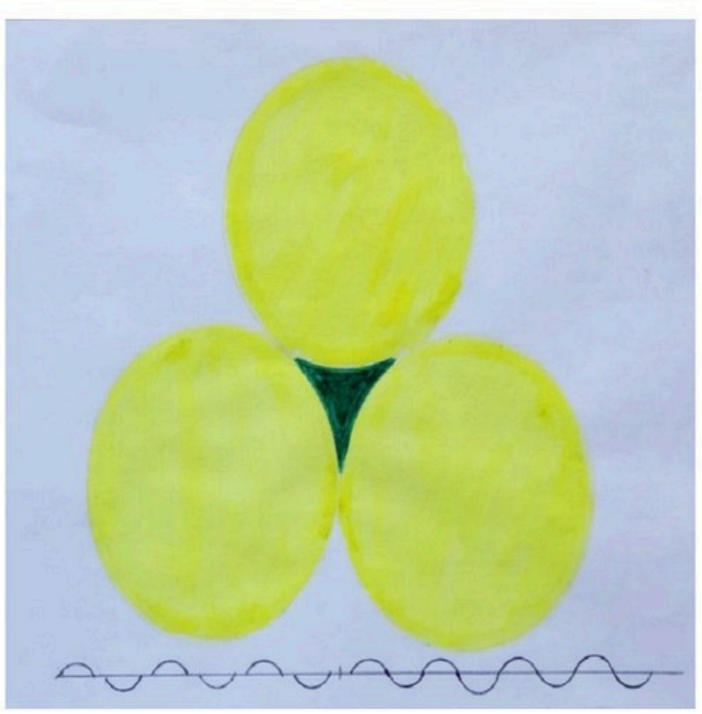


FIGURE 11

One Formation, Just Before the Spark

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

20

universes, #1–20

#33

the hub, barely visible

Invisible

the neutral space, still void

DESCRIPTION

Green and yellow spheres pack tightly around the small blue center, #33, barely visible at the middle. The lines running between them mark where the Curvilinears — the neutral, connecting domains of Zone 4 — will eventually sit. Right now that space is empty: the curvilinears are not yet there, invisible, the same neutral void carried elsewhere in this deck. This is one perfect geometric formation, captured the instant before the spark, not the burst itself.

THE NUMBERS

Two particles here begin at zero entanglement, climbing toward $\log_2(33) = 5.044$ bits once they interact — a joint state no classical arrangement can produce, quantum and invisible until disturbed. · Rivero-Zeta



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33, the hub: origin (0, 0, 0)
Yellow, the 12: all 23.00T ly out, at latitudes 90°, 26.57°, −26.57°, −90°
Green, the 20: all 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62°

FIGURE 12

First Light, Stage Three

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20 | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Stage 3

Rebound & Illumination

p = 0.781

pause factor, ±0.013

4π sr

32 mouths tile the sky

DESCRIPTION

Look closely at the light: rays fan out from one bright point in every direction. In the model this is the Stage 3 rebound, the Core's first burst of light and the birth of electromagnetism. The fan mirrors the geometry: the 32 throat cones spread from the hub and their mouths tile the whole sky, 12 × 0.2951 + 20 × 0.4513 = 12.5664 sr = 4π (Rivero-Zeta). Gravity then pauses as the forces separate. GENESYS fits that pause at p = 0.781 ± 0.013, early expansion about 21.9% slower, giving structure about 28% more time at z ~ 10–20. This is the start of the expansion with gravity paused: the light fans outward while the siblings hold together, and that pause is what slows the early expansion (shown in full on pages 15 and 16).

THE FOUR FORCES, AS THIS MODEL ASSIGNS THEM

Electromagnetism
Born in this flash. Carried by the orange Curvilinear feed channels.

Gravity
The geometry of the whole system. Paused here: H = p·H_ΛCDM, p = 0.781.

Strong
The outer 20, Zone 3, green. Compressive force on the Core.

Weak
The inner 12, Zone 2, yellow. Cohesive pull that holds the balance.

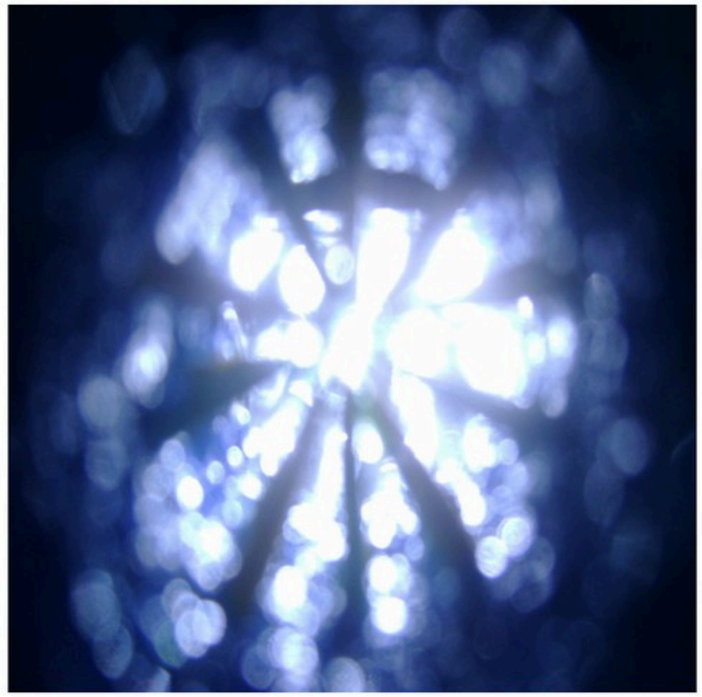


FIGURE 13

The Burst, Seen Closer

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#33	Stage 3	0.999892
the Creator Core	Rebound & Illumination	one flux, both readings

DESCRIPTION

The same rebound event, seen closer in: dual-photon resonances inside the Core, the light burst that marks the transition from implosion to expansion. Here you can see the full effect, before, during and after: before the burst, the 32 lines converge on the central spark and the Core implodes (page 17); during it, this burst of light; after it, the expansion, with gravity paused (pages 14 and 16).

THE NUMBERS

Information and curvature both start at zero and leave zero together; one flux carries both (0.999892), and Λ and F are locked at −1.0000 · Rivero-Zeta. Zone partition scans support p ≈ 0.781 · GENESYS



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.

Blue #33, the hub: origin (0, 0, 0)

Yellow, the 12, touching the hub: 23.00T ly center to center, at latitudes 90°, 26.57°, −26.57°, −90°

Green, the 20, reaching it through their curvilinears: 36.55T ly out, at latitudes 52.62°, 10.81°, −10.81°, −52.62°

FIGURE 14

Where Every Universe Would Sit

ZONE 1 HUB | ZONE 4 CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

32

directions, 12 yellow + 20 green

63.4349° · 41.8103°

between neighbouring yellow · green

Stage 3

Rebound & Illumination

DESCRIPTION

The blue Core sits at the center. The orange is the Curvilinear frame, the neutral edges and ends, with open windows where the spheres are left out on purpose. The green and yellow lines carry the Core's potential energy outward, each aimed at the exact place where an outer (green) or inner (yellow) universe sits. This is an example of the moment after inception, with the energy expanding outward; the spark itself is not seen here. It also shows the direction in which paused gravity begins: the yellow and green universes are removed so the outward direction shows, and the empty shell edges mark where they would be if they were still there, siblings held tight while the expansion slows.

THE NUMBERS

The 12 yellow lines aim at the icosahedron's vertices, the 20 green at its face centers. In C60, a measured molecule, 12 pentagons and 20 hexagons face those same directions. 12 × 60° = 720° = 4π · Rivero-Zeta. After the rebound, H = p·H_ΛCDM, p = 0.781 · GENESYS

LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub

Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.

Blue #33, the hub: origin (0, 0, 0)

12 yellow lines aim at the vertices: 23.00T ly, 63.4349° between neighbors, at latitudes 90°, 26.57°, −26.57°, −90°

20 green lines aim at the face centers: 36.55T ly, 41.8103° between neighbors, at latitudes 52.62°, 10.81°, −10.81°, −52.62°



FIGURE 15

Before the Throats Open

ZONE 1 HUB | ZONE 2 INNER 12 | ZONE 3 OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

4π sr

shell curvature, already present

F = 0, R = 0

information and curvature, both zero

Singular

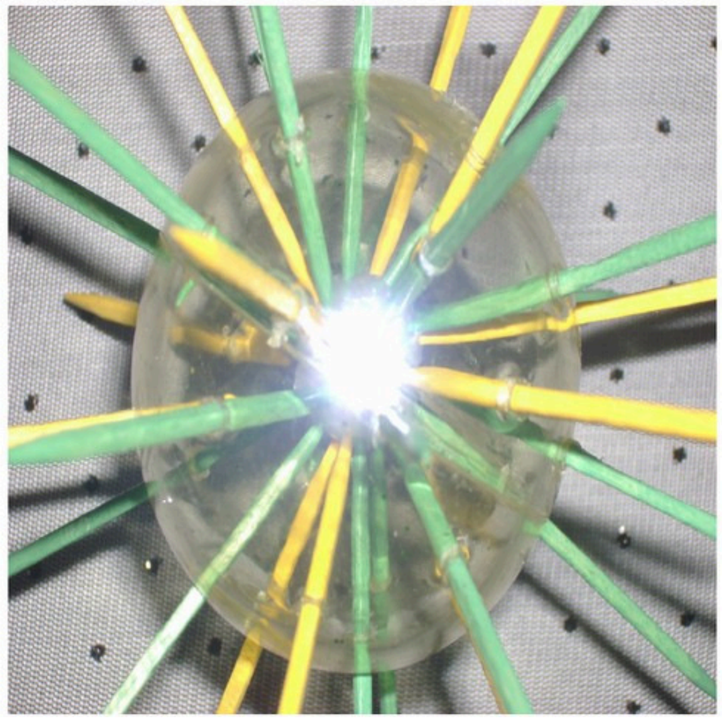
operator has no inverse, no propagator

DESCRIPTION

This is the Core rendered clear and transparent, its energy lines reaching toward where every other multiverse will sit and meeting at the central spark. The shell already carries its full curvature, 12 × 60° = 720° = 4π, computed by two separate routes that agree to every digit. What it does not yet have is a single relation: no distance, no duration, and no throat yet delivering the spark to anywhere. The clear sphere plays two roles at once, the real side and the invisible, quantum side; the Rivero-Zeta deck follows the invisible side (page 32). In this model, green, the outer 20, is dark energy pushing inward through the orange curvilinears toward the blue hub; yellow, the inner 12, is dark matter pulling inward.

THE NUMBERS

QUANTUM GRAVITY, COMPUTED: 1 spark, 2 possible states, 1.000000 probability of return. The hub behaves as a two-state quantum system — only 0 or 33 — so a single spark of energy placed there comes back with perfect, exact probability every 2π/33 = 0.1904 radians. Not approximate: computed. One spark, on the invisible side of this picture, answering on an exact schedule. · Rivero-Zeta



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub
Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
Blue #33: origin (0, 0, 0) · its 32 lines: 12 yellow 23.00T ly long, 20 green 36.55T ly long
Latitudes: yellow 90°, 26.57°, −26.57°, −90° · green 52.62°, 10.81°, −10.81°, −52.62°

FIGURE 16

The Invisible Half, in Six Dimensions

ALL 33 UNIVERSES

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

0.707107

the shared radius, seen and unseen

63.43° · 116.57°

the inner twelve's two neighbour angles

6D → 3D + 3D

one half visible, one half not

DESCRIPTION

The 33 transparent, glass-like spheres you can see are only half the object. In six dimensions, a cube has twelve nearest neighbours. Projected down to the three dimensions we live in, those twelve land exactly on this icosahedron, the frame of all 33. The other three dimensions are not empty: the same 33 spheres form a second, complete 33-multiverse there — identical to this one, invisible, and unreachable from inside 3D. In this model, green, the outer 20, is dark energy pushing inward through the orange curvilinears toward the blue hub; yellow, the inner 12, is dark matter pulling inward.

THE NUMBERS

QUANTUM GRAVITY, ENTANGLED: 32 points, 1 instant, 0 delay. Send that spark from the hub and all 32 points on the invisible side light up identically at once, equal to machine precision. This is entanglement: a change at one point is a change at all 32, with nothing sent between them after. · Rivero-Zeta

The shared radius is the uncertainty minimum: 0.707107 × 0.707107 = ½ (ħ = 1). Seen half as position, unseen half as momentum is the framework's reading (page 73).



LOCATION DIAGRAM · COORDINATES latitude, longitude, distance from the hub
Whole structure: 33 universes × 3 coordinates = 99 numbers; each universe's own are on pages 24 and 25.
All 33 universes, here transparent: blue #33 at the origin · yellow 12 at 23.00T ly · green 20 at 36.55T ly
Latitudes: yellow 90°, 26.57°, −26.57°, −90° · green 52.62°, 10.81°, −10.81°, −52.62°
The invisible half: the same 33 in the other three dimensions, shared radius 0.707107 (not in this frame)

THE 32 THROATS, COMPARED

Compared with Public Data

COMPARED · PUBLIC DATA

6.4×10^{10}

Milky Way: model 6.4×10^{10} M☉ · ~6×10^{10} measured

14 / 14

JWST black holes inside the model's ceiling, z ≈ 4–10.6

0.613 · 0.639

Clustering: observed galaxies · the architecture · random sky 0.495

WHAT WAS TESTED

A continuous feed through the 32 throats, switched on at the first galaxies (z ≈ 20), was compared with public Planck, DESI and Pantheon+ data, with three JWST galaxies under 300 Myr old, with 14 JWST black holes tested at z ≈ 4–10.6, and with two JWST protoclusters — then carried forward, unchanged, to the Local Group we live in.

THE NUMBERS

Andromeda: within 1.4×. Local Group together: within 1.2×. Three JWST galaxies under 300 Myr old, compared with 10^{39} W of continuous delivery. 2 of 2 protoclusters land inside the prediction, which was recorded before the comparison. One fitted feed rate, η = 0.053%, is held fixed for star formation from z ≈ 14.5 to 9. · Continuous Throat Feed

WHERE THIS PAGE STANDS

AI33-MPOPT proposes that observable physical effects may emerge from a deeper symmetry-driven geometric transport architecture. This page shows the architecture's numbers beside published measurements, so each can be checked.

WHAT BACKS IT

Mathematics (proved): 33 nodes (1 hub, 12 apex, 20 curvilinear), 32 throats, 152 edges, |Aut| = 120, enumerated exhaustively. Compared with data: the occupation ratio is 2.5490 against Planck's 2.5893 ± 0.0562; clustering is 0.639 for the architecture, 0.613 observed, 0.495 for a random sky. Recorded before the comparison: 2 of 2 JWST protoclusters. Fitted: η = 0.053%, p = 0.781 ± 0.013 and γ = 0.62, the only tuned numbers.

COORDINATES · ALL ZONES

The Coordinate Frame

HUB AT THE ORIGIN

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#33
blue, Zone 1, the hub at the origin

#21–32
yellow, Zone 2, 12 inner vertices

#1–20
green, Zone 3, 20 outer face centers

C1–C20
orange, Zone 4, 20 curvilinears

How to read: the blue hub is the center (0, 0, 0). Every other universe sits around it in 3D. Distances are in T ly = trillions of light-years. Straight line, center to center.

WHAT THIS SHOWS

Every universe in this deck, placed by exact icosahedral geometry. The hub, Universe #33, sits at the origin. The 12 inner domains take the icosahedron's vertices and the 20 outer domains its face centers, the same 12 + 20 directions as Figure 14. Each of the 20 curvilinears sits on the axis of one face, between the hub and its outer domain: C1 on #1, C2 on #2, and so on.

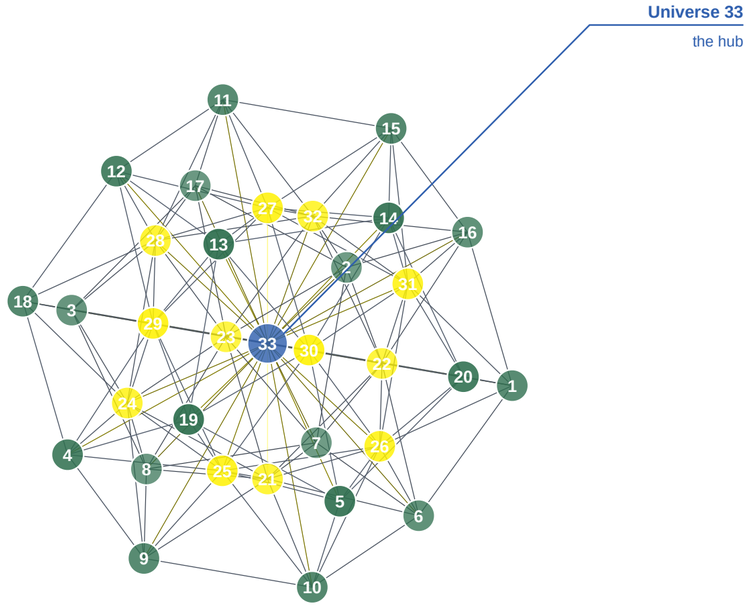
The +z axis points to #27, with #28–32 ringing it (Figure 5) and #11–15 the five faces around it (Figure 1) . Colatitude is measured from +z, azimuth from +x; lengths are in T ly (10^{12} ly).

THE NUMBERS

All 33 spheres are 23T ly across (276 ÷ 12 = 460 ÷ 20 = 23). Inner centers sit at 23.0T and touch the hub; outer centers at 36.55T, the closest radius where they touch the inner spheres without overlap; curvilinear centers at 18.28T on the face plane. Packed edge to edge, the structure spans 96.1T ly; the 989T figure is the sum of all diameters. Neighbor angles: 63.4349° vertex to vertex, 37.3774° vertex to face, 41.8103° face to face.

How to read an address. Like Earth: latitude, longitude and distance. This is the same kind of coordinate used to map the real sky (right ascension and declination, or galactic longitude l and latitude b); the frame here is this atlas's own. In it, north is the pole toward #27 (latitude +90°), south is the pole toward #21 (−90°), and east is the direction of increasing longitude, measured from +x. Distance is from the hub in trillions of light-years. Example: #13 is 52.62° north, 270.0° longitude, 36.55T ly out.

THE 33 CENTERS AND THEIR 152 LINKS



How the real sky is mapped. Each object is placed by two angles on the sky, like latitude and longitude, plus a distance.

COORDINATES · ZONES 1 + 2

The Hub and the Inner Twelve, Placed

Lengths in T ly from the hub at the origin · +z toward #27 · colatitude and azimuth in degrees · hub gap = surface to surface

ZONE 1 · CORE

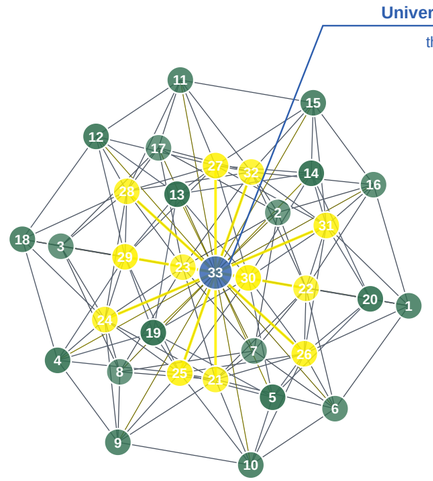
ZONE 2 · INNER 12

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#	Zone	Kind	x	y	z	Radius	Colat°	Az°	Dia	Volume ly³	Throat	Mouth sr	Hub gap
#33	Zone 1	Hub	0.0	0.0	0.0	0.0	–	–	23.0	6.371×10^{39}	32 throats	0	0.0
#21	Zone 2	Inner (yellow)	0.0	0.0	−23.0	23.0	180.0	0.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#22	Zone 2	Inner (yellow)	12.0918	16.643	−10.2859	23.0	116.5651	54.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#23	Zone 2	Inner (yellow)	−12.0918	16.643	−10.2859	23.0	116.5651	126.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#24	Zone 2	Inner (yellow)	−19.565	−6.357	−10.2859	23.0	116.5651	198.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#25	Zone 2	Inner (yellow)	0.0	−20.5718	−10.2859	23.0	116.5651	270.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#26	Zone 2	Inner (yellow)	19.565	−6.357	−10.2859	23.0	116.5651	342.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#27	Zone 2	Inner (yellow)	0.0	0.0	23.0	23.0	0.0	0.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#28	Zone 2	Inner (yellow)	−19.565	6.357	10.2859	23.0	63.4349	162.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#29	Zone 2	Inner (yellow)	−12.0918	−16.643	10.2859	23.0	63.4349	234.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#30	Zone 2	Inner (yellow)	12.0918	−16.643	10.2859	23.0	63.4349	306.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#31	Zone 2	Inner (yellow)	19.565	6.357	10.2859	23.0	63.4349	18.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0
#32	Zone 2	Inner (yellow)	0.0	20.5718	10.2859	23.0	63.4349	90.0	23.0	6.371×10^{39}	apex throat	0.2951	0.0

ROUTES ON THIS PAGE

12 straight lines, hub to each inner center: 23.0T ly each



- 12 hub → inner routes
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: each row is one universe's address from the hub. x, y, z are the 3D directions; Radius is the straight-line distance from the hub. Straight line, center to center.

COORDINATES · ZONE 3

The Outer Twenty, Placed

Lengths in T ly from the hub at the origin · +z toward #27 · colatitude and azimuth in degrees · hub gap = surface to surface

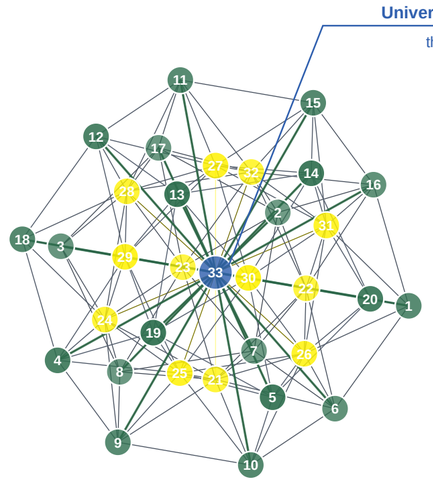
ZONE 3 · OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#	Zone	Kind	x	y	z	Radius	Colat°	Az°	Dia	Volume ly³	Throat	Mouth sr	Hub gap
#1	Zone 3	Outer (green)	34.1478	11.0953	−6.8573	36.5541	100.8123	18.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#2	Zone 3	Outer (green)	0.0	35.9052	−6.8573	36.5541	100.8123	90.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#3	Zone 3	Outer (green)	−34.1478	11.0953	−6.8573	36.5541	100.8123	162.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#4	Zone 3	Outer (green)	−21.1045	−29.0479	−6.8573	36.5541	100.8123	234.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#5	Zone 3	Outer (green)	21.1045	−29.0479	−6.8573	36.5541	100.8123	306.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#6	Zone 3	Outer (green)	21.1045	6.8573	−29.0479	36.5541	142.6226	18.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#7	Zone 3	Outer (green)	0.0	22.1906	−29.0479	36.5541	142.6226	90.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#8	Zone 3	Outer (green)	−21.1045	6.8573	−29.0479	36.5541	142.6226	162.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#9	Zone 3	Outer (green)	−13.0433	−17.9526	−29.0479	36.5541	142.6226	234.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#10	Zone 3	Outer (green)	13.0433	−17.9526	−29.0479	36.5541	142.6226	306.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#11	Zone 3	Outer (green)	−13.0433	17.9526	29.0479	36.5541	37.3774	126.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#12	Zone 3	Outer (green)	−21.1045	−6.8573	29.0479	36.5541	37.3774	198.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#13	Zone 3	Outer (green)	0.0	−22.1906	29.0479	36.5541	37.3774	270.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#14	Zone 3	Outer (green)	21.1045	−6.8573	29.0479	36.5541	37.3774	342.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#15	Zone 3	Outer (green)	13.0433	17.9526	29.0479	36.5541	37.3774	54.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#16	Zone 3	Outer (green)	21.1045	29.0479	6.8573	36.5541	79.1877	54.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#17	Zone 3	Outer (green)	−21.1045	29.0479	6.8573	36.5541	79.1877	126.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#18	Zone 3	Outer (green)	−34.1478	−11.0953	6.8573	36.5541	79.1877	198.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#19	Zone 3	Outer (green)	0.0	−35.9052	6.8573	36.5541	79.1877	270.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541
#20	Zone 3	Outer (green)	34.1478	−11.0953	6.8573	36.5541	79.1877	342.0	23.0	6.371×10³⁹	curvilinear throat	0.4513	13.5541

ROUTES ON THIS PAGE

20 straight lines, hub to each outer center: 36.55T ly each



- 20 hub → outer routes
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: each row is one green universe's address from the hub. All 20 are the same distance away (36.55T ly), each in a different direction. Straight line, center to center.

The Twenty Curvilinears, Placed

Centers in T ly on the face axes of #1–#20 (C1 on #1) · 23T long, 5.75T waist · tips reach 11.5T either side of the center

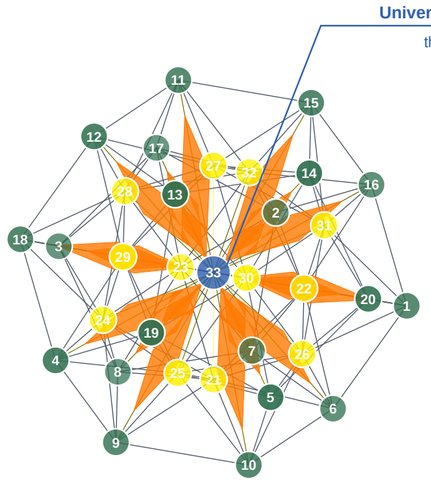
ZONE 4 · CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#	Zone	Kind	x	y	z	Radius	Colat°	Az°	Dia	Volume ly³	Throat	Mouth sr	Hub gap
C1	Zone 4	Curvilinear (bicone)	17.0739	5.5477	−3.4286	18.2771	100.8123	18.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C2	Zone 4	Curvilinear (bicone)	0.0	17.9526	−3.4286	18.2771	100.8123	90.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C3	Zone 4	Curvilinear (bicone)	−17.0739	5.5477	−3.4286	18.2771	100.8123	162.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C4	Zone 4	Curvilinear (bicone)	−10.5523	−14.5239	−3.4286	18.2771	100.8123	234.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C5	Zone 4	Curvilinear (bicone)	10.5523	−14.5239	−3.4286	18.2771	100.8123	306.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C6	Zone 4	Curvilinear (bicone)	10.5523	3.4286	−14.5239	18.2771	142.6226	18.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C7	Zone 4	Curvilinear (bicone)	0.0	11.0953	−14.5239	18.2771	142.6226	90.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C8	Zone 4	Curvilinear (bicone)	−10.5523	3.4286	−14.5239	18.2771	142.6226	162.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C9	Zone 4	Curvilinear (bicone)	−6.5217	−8.9763	−14.5239	18.2771	142.6226	234.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C10	Zone 4	Curvilinear (bicone)	6.5217	−8.9763	−14.5239	18.2771	142.6226	306.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C11	Zone 4	Curvilinear (bicone)	−6.5217	8.9763	14.5239	18.2771	37.3774	126.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C12	Zone 4	Curvilinear (bicone)	−10.5523	−3.4286	14.5239	18.2771	37.3774	198.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C13	Zone 4	Curvilinear (bicone)	0.0	−11.0953	14.5239	18.2771	37.3774	270.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C14	Zone 4	Curvilinear (bicone)	10.5523	−3.4286	14.5239	18.2771	37.3774	342.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C15	Zone 4	Curvilinear (bicone)	6.5217	8.9763	14.5239	18.2771	37.3774	54.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C16	Zone 4	Curvilinear (bicone)	10.5523	14.5239	3.4286	18.2771	79.1877	54.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C17	Zone 4	Curvilinear (bicone)	−10.5523	14.5239	3.4286	18.2771	79.1877	126.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C18	Zone 4	Curvilinear (bicone)	−17.0739	−5.5477	3.4286	18.2771	79.1877	198.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C19	Zone 4	Curvilinear (bicone)	0.0	−17.9526	3.4286	18.2771	79.1877	270.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–
C20	Zone 4	Curvilinear (bicone)	17.0739	−5.5477	3.4286	18.2771	79.1877	342.0	23 (length)	1.99×10³⁸	curvilinear throat	0.4513	–

ROUTES ON THIS PAGE

20 curvilinears, each a bicone along the axis from the hub to its outer universe



- 20 curvilinears C1–C20
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: each orange curvilinear sits on the line from the hub to its green universe (C1 on #1), 18.28T ly from the hub. Straight line, center to center.

Curvilinear section: pages 31–37

TRAVEL ATLAS · COORDINATES IN LIGHT-YEARS

The Hub and the Inner Twelve, in Light-Years

ZONE 1 · CORE ZONE 2 · INNER 12

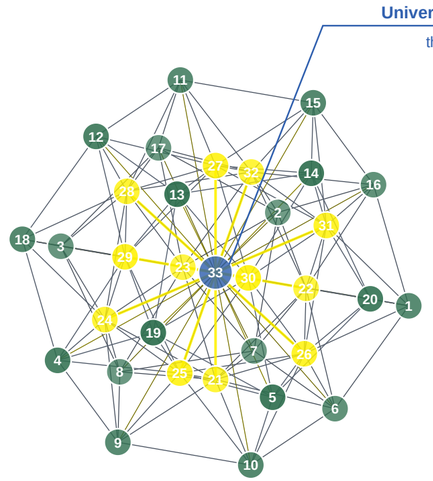
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Light-years from the hub at the origin, rounded to the nearest million ly · latitude 90° = toward #27 · longitude from +x · light time at the speed of light

#	Zone	Kind	x (ly)	y (ly)	z (ly)	Distance from hub (ly)	Latitude°	Longitude°	Light time
#33	Zone 1	Hub	0	0	0	0	–	–	0
#21	Zone 2	Inner (yellow)	0	0	−23,000,000,000,000	23,000,000,000,000	−90.0	0.0	23.00T yr
#22	Zone 2	Inner (yellow)	12,091,816,000,000	16,642,956,000,000	−10,285,913,000,000	23,000,000,000,000	−26.57	54.0	23.00T yr
#23	Zone 2	Inner (yellow)	−12,091,816,000,000	16,642,956,000,000	−10,285,913,000,000	23,000,000,000,000	−26.57	126.0	23.00T yr
#24	Zone 2	Inner (yellow)	−19,564,969,000,000	−6,357,044,000,000	−10,285,913,000,000	23,000,000,000,000	−26.57	198.0	23.00T yr
#25	Zone 2	Inner (yellow)	0	−20,571,825,000,000	−10,285,913,000,000	23,000,000,000,000	−26.57	270.0	23.00T yr
#26	Zone 2	Inner (yellow)	19,564,969,000,000	−6,357,044,000,000	−10,285,913,000,000	23,000,000,000,000	−26.57	342.0	23.00T yr
#27	Zone 2	Inner (yellow)	0	0	23,000,000,000,000	23,000,000,000,000	90.0	0.0	23.00T yr
#28	Zone 2	Inner (yellow)	−19,564,969,000,000	6,357,044,000,000	10,285,913,000,000	23,000,000,000,000	26.57	162.0	23.00T yr
#29	Zone 2	Inner (yellow)	−12,091,816,000,000	−16,642,956,000,000	10,285,913,000,000	23,000,000,000,000	26.57	234.0	23.00T yr
#30	Zone 2	Inner (yellow)	12,091,816,000,000	−16,642,956,000,000	10,285,913,000,000	23,000,000,000,000	26.57	306.0	23.00T yr
#31	Zone 2	Inner (yellow)	19,564,969,000,000	6,357,044,000,000	10,285,913,000,000	23,000,000,000,000	26.57	18.0	23.00T yr
#32	Zone 2	Inner (yellow)	0	20,571,825,000,000	10,285,913,000,000	23,000,000,000,000	26.57	90.0	23.00T yr

ROUTES ON THIS PAGE

Hub to each inner center: 23,000,000,000,000 ly, 23 trillion years at light speed



12 hub → inner routes
Hub · #33
12 inner · #21–32
20 outer · #1–20
32 throats to the hub
120 sibling links

How to read: same addresses as page 21, in full light-years. 23,000,000,000,000 ly is 23 trillion ly, or 23 trillion years at light speed. Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).
Address: latitude, longitude and distance, as on Earth. The real sky uses redshift as its distance coordinate.

TRAVEL ATLAS · COORDINATES IN LIGHT-YEARS

The Outer Twenty, in Light-Years

ZONE 3 · OUTER 20

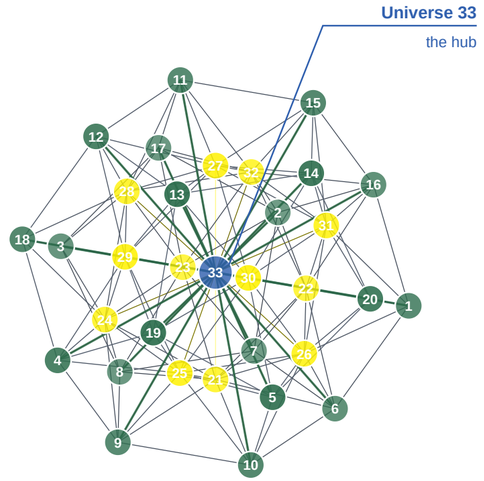
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Light-years from the hub at the origin, rounded to the nearest million ly · latitude 90° = toward #27 · longitude from +x · light time at the speed of light

#	Zone	Kind	x (ly)	y (ly)	z (ly)	Distance from hub (ly)	Latitude°	Longitude°	Light time
#1	Zone 3	Outer (green)	34,147,835,000,000	11,095,304,000,000	−6,857,275,000,000	36,554,106,000,000	−10.81	18.0	36.55T yr
#2	Zone 3	Outer (green)	0	35,905,159,000,000	−6,857,275,000,000	36,554,106,000,000	−10.81	90.0	36.55T yr
#3	Zone 3	Outer (green)	−34,147,835,000,000	11,095,304,000,000	−6,857,275,000,000	36,554,106,000,000	−10.81	162.0	36.55T yr
#4	Zone 3	Outer (green)	−21,104,523,000,000	−29,047,884,000,000	−6,857,275,000,000	36,554,106,000,000	−10.81	234.0	36.55T yr
#5	Zone 3	Outer (green)	21,104,523,000,000	−29,047,884,000,000	−6,857,275,000,000	36,554,106,000,000	−10.81	306.0	36.55T yr
#6	Zone 3	Outer (green)	21,104,523,000,000	6,857,275,000,000	−29,047,884,000,000	36,554,106,000,000	−52.62	18.0	36.55T yr
#7	Zone 3	Outer (green)	0	22,190,608,000,000	−29,047,884,000,000	36,554,106,000,000	−52.62	90.0	36.55T yr
#8	Zone 3	Outer (green)	−21,104,523,000,000	6,857,275,000,000	−29,047,884,000,000	36,554,106,000,000	−52.62	162.0	36.55T yr
#9	Zone 3	Outer (green)	−13,043,312,000,000	−17,952,579,000,000	−29,047,884,000,000	36,554,106,000,000	−52.62	234.0	36.55T yr
#10	Zone 3	Outer (green)	13,043,312,000,000	−17,952,579,000,000	−29,047,884,000,000	36,554,106,000,000	−52.62	306.0	36.55T yr
#11	Zone 3	Outer (green)	−13,043,312,000,000	17,952,579,000,000	29,047,884,000,000	36,554,106,000,000	52.62	126.0	36.55T yr
#12	Zone 3	Outer (green)	−21,104,523,000,000	−6,857,275,000,000	29,047,884,000,000	36,554,106,000,000	52.62	198.0	36.55T yr
#13	Zone 3	Outer (green)	0	−22,190,608,000,000	29,047,884,000,000	36,554,106,000,000	52.62	270.0	36.55T yr
#14	Zone 3	Outer (green)	21,104,523,000,000	−6,857,275,000,000	29,047,884,000,000	36,554,106,000,000	52.62	342.0	36.55T yr
#15	Zone 3	Outer (green)	13,043,312,000,000	17,952,579,000,000	29,047,884,000,000	36,554,106,000,000	52.62	54.0	36.55T yr
#16	Zone 3	Outer (green)	21,104,523,000,000	29,047,884,000,000	6,857,275,000,000	36,554,106,000,000	10.81	54.0	36.55T yr
#17	Zone 3	Outer (green)	−21,104,523,000,000	29,047,884,000,000	6,857,275,000,000	36,554,106,000,000	10.81	126.0	36.55T yr
#18	Zone 3	Outer (green)	−34,147,835,000,000	−11,095,304,000,000	6,857,275,000,000	36,554,106,000,000	10.81	198.0	36.55T yr
#19	Zone 3	Outer (green)	0	−35,905,159,000,000	6,857,275,000,000	36,554,106,000,000	10.81	270.0	36.55T yr
#20	Zone 3	Outer (green)	34,147,835,000,000	−11,095,304,000,000	6,857,275,000,000	36,554,106,000,000	10.81	342.0	36.55T yr

ROUTES ON THIS PAGE

Hub to each outer center: 36,554,106,000,000 ly, 36.55 trillion years at light speed



20 hub → outer routes
Hub · #33
12 inner · #21–32
20 outer · #1–20
32 throats to the hub
120 sibling links

How to read: same as page 22 in full light-years. Hub to #1 is 36,554,106,000,000 ly, about 36.55 trillion years at light speed. Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).
Address: latitude, longitude and distance, as on Earth. The real sky uses redshift as its distance coordinate.

TRAVEL ATLAS · EVERY ROUTE

All 528 Routes, Sorted into 14 Distances

33 UNIVERSES · 528 ROUTES

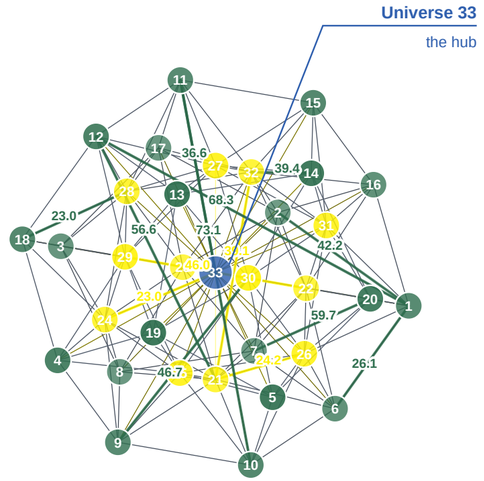
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Center to center, in the touching-shell packing. By symmetry the 528 routes fall into 14 distinct lengths; the example is one route of each length.

Route class	Angle	From	To	Routes	Distance (T ly)	Distance (ly)	× 13.8 Gyr	Example
Hub to an inner vertex (touching)	–	Hub	Inner	12	23.000	23,000,000,000,000	1,667×	#33 → #24
Hub to an outer face center	–	Hub	Outer	20	36.554	36,554,106,000,000	2,649×	#33 → #11
Neighboring inner vertices (one edge)	63.43°	Inner	Inner	30	24.184	24,183,631,000,000	1,752×	#21 → #26
Inner vertices two edges apart	116.57°	Inner	Inner	30	39.130	39,129,937,000,000	2,836×	#21 → #32
Opposite inner vertices (through the hub)	180.0°	Inner	Inner	6	46.000	46,000,000,000,000	3,333×	#22 → #29
Inner vertex to a face it touches	37.38°	Inner	Outer	60	23.000	23,000,000,000,000	1,667×	#28 → #18
Inner vertex to a face one step off	79.19°	Inner	Outer	60	39.367	39,367,093,000,000	2,853×	#32 → #14
Inner vertex to a face two steps off	100.81°	Inner	Outer	60	46.697	46,697,294,000,000	3,384×	#30 → #9
Inner vertex to a far-side face	142.62°	Inner	Outer	60	56.581	56,580,962,000,000	4,100×	#21 → #12
Neighboring outer faces (one edge)	41.81°	Outer	Outer	30	26.087	26,086,625,000,000	1,890×	#1 → #6
Outer faces two steps apart	70.53°	Outer	Outer	60	42.209	42,209,046,000,000	3,059×	#1 → #2
Outer faces three steps apart	109.47°	Outer	Outer	60	59.693	59,692,605,000,000	4,326×	#7 → #20
Outer faces four steps apart	138.19°	Outer	Outer	30	68.296	68,295,670,000,000	4,949×	#1 → #12
Opposite outer faces (through the hub)	180.0°	Outer	Outer	10	73.108	73,108,211,000,000	5,298×	#10 → #11
All routes				**528**	= 33 × 32 ÷ 2			

ROUTES ON THIS PAGE

One straight line for each of the 14 distances, labeled in T ly



— 14 example routes, T ly
● Hub · #33
● 12 inner · #21–32
● 20 outer · #1–20
— 32 throats to the hub
— 120 sibling links

Time units: yr = year · kyr = thousand years · Myr = million years · Gyr = billion years · T yr = trillion years

How to read: 33 universes make 528 routes but only 14 different lengths. Each row is one length, with one example route. Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

How Long Each Trip Takes

LIGHT SPEED AND BEYOND

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

#1 → #2: 42,209,046,000,000 ly

center to center, both green

42.2 T yr at light speed

3,059× the 13.8 Gyr age of our universe

Longest trip: 73.11T ly

opposite outer universes (10 pairs)

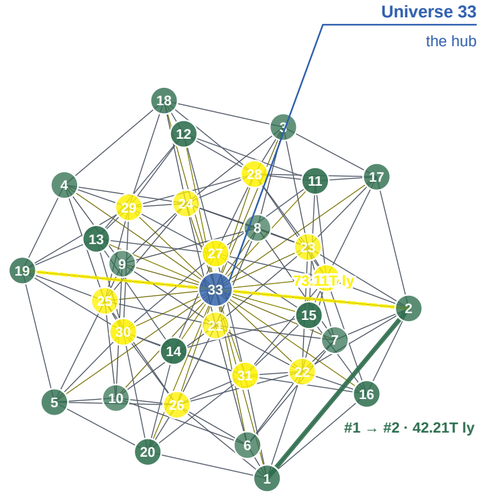
Light travel time equals the distance in years. Multiples of c are a scale for reading the map.

Route class	T ly	at 1c	at 10^3c	at 10^6c	at 10^9c	at 10^{12}c
Hub to an inner vertex	23.000	23.0 T yr	23.0 Gyr	23.0 Myr	23.0 kyr	23.0 yr
Hub to an outer face center	36.554	36.6 T yr	36.6 Gyr	36.6 Myr	36.6 kyr	36.6 yr
Neighboring inner vertices	24.184	24.2 T yr	24.2 Gyr	24.2 Myr	24.2 kyr	24.2 yr
Inner vertices two edges apart	39.130	39.1 T yr	39.1 Gyr	39.1 Myr	39.1 kyr	39.1 yr
Opposite inner vertices	46.000	46.0 T yr	46.0 Gyr	46.0 Myr	46.0 kyr	46.0 yr
Inner vertex to a face it touches	23.000	23.0 T yr	23.0 Gyr	23.0 Myr	23.0 kyr	23.0 yr
Inner vertex to a face one step off	39.367	39.4 T yr	39.4 Gyr	39.4 Myr	39.4 kyr	39.4 yr
Inner vertex to a face two steps off	46.697	46.7 T yr	46.7 Gyr	46.7 Myr	46.7 kyr	46.7 yr
Inner vertex to a far-side face	56.581	56.6 T yr	56.6 Gyr	56.6 Myr	56.6 kyr	56.6 yr
Neighboring outer faces	26.087	26.1 T yr	26.1 Gyr	26.1 Myr	26.1 kyr	26.1 yr
Outer faces two steps apart	42.209	42.2 T yr	42.2 Gyr	42.2 Myr	42.2 kyr	42.2 yr
Outer faces three steps apart	59.693	59.7 T yr	59.7 Gyr	59.7 Myr	59.7 kyr	59.7 yr
Outer faces four steps apart	68.296	68.3 T yr	68.3 Gyr	68.3 Myr	68.3 kyr	68.3 yr
Opposite outer faces	73.108	73.1 T yr	73.1 Gyr	73.1 Myr	73.1 kyr	73.1 yr

Time units: yr = year · kyr = thousand years · Myr = million years · Gyr = billion years · T yr = trillion years

ROUTES ON THIS PAGE

The #1 → #2 trip from the example and the longest trip, drawn to the coordinates



- #1 → #2, 42.21T ly
- longest, #2 → #19, 73.11T ly
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: light takes 1 year per light-year. Each column repeats the trip at a faster, hypothetical speed (10^3c = 1,000 × light speed). Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

Hub and Inner Twelve, Center to Center

ZONE 1 · CORE

ZONE 2 · INNER 12

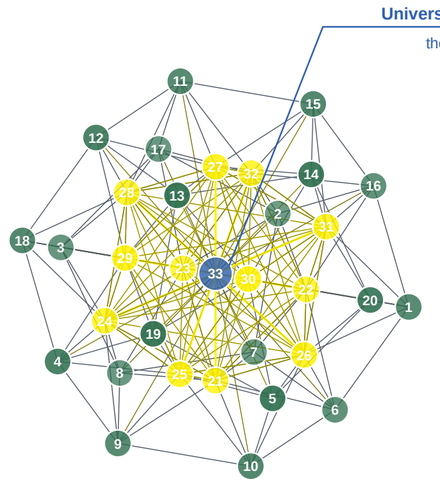
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Center to center in T ly (trillions of light-years); at light speed the same number is trillions of years. Rows = start, columns = destination; number color = destination zone.

from ↓ to →	#33	#21	#22	#23	#24	#25	#26	#27	#28	#29	#30	#31	#32
#33	–	23.00	23.00	23.00	23.00	23.00	23.00	23.00	23.00	23.00	23.00	23.00	23.00
#21	23.00	–	24.18	24.18	24.18	24.18	24.18	46.00	39.13	39.13	39.13	39.13	39.13
#22	23.00	24.18	–	24.18	39.13	39.13	24.18	39.13	39.13	46.00	39.13	24.18	24.18
#23	23.00	24.18	24.18	–	24.18	39.13	39.13	39.13	24.18	39.13	46.00	39.13	24.18
#24	23.00	24.18	39.13	24.18	–	24.18	39.13	39.13	24.18	24.18	39.13	46.00	39.13
#25	23.00	24.18	39.13	39.13	24.18	–	24.18	39.13	39.13	24.18	24.18	39.13	46.00
#26	23.00	24.18	24.18	39.13	39.13	24.18	–	39.13	46.00	39.13	24.18	24.18	39.13
#27	23.00	46.00	39.13	39.13	39.13	39.13	39.13	–	24.18	24.18	24.18	24.18	24.18
#28	23.00	39.13	39.13	24.18	24.18	39.13	46.00	24.18	–	24.18	39.13	39.13	24.18
#29	23.00	39.13	46.00	39.13	24.18	24.18	39.13	24.18	24.18	–	24.18	39.13	39.13
#30	23.00	39.13	39.13	46.00	39.13	24.18	24.18	24.18	39.13	24.18	–	24.18	39.13
#31	23.00	39.13	24.18	39.13	46.00	39.13	24.18	24.18	39.13	39.13	24.18	–	24.18
#32	23.00	39.13	24.18	24.18	39.13	46.00	39.13	24.18	24.18	39.13	39.13	24.18	–

ROUTES ON THIS PAGE

78 straight lines: 12 hub → inner and 66 inner ↔ inner



- 78 routes on this page
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: start in the left column, destination across the top; the number is the distance in T ly. Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

Hub and Inner Twelve to the Outer Twenty

ZONE 2 · INNER 12

ZONE 3 · OUTER 20

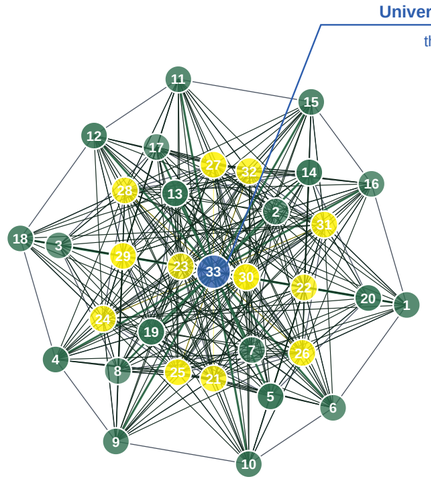
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Center to center in T ly (trillions of light-years); at light speed the same number is trillions of years. Rows = start, columns = destination; number color = destination zone.

from ↓ to →	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
#33	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55	36.55
#21	39.37	39.37	39.37	39.37	39.37	23.00	23.00	23.00	23.00	23.00	56.58	56.58	56.58	56.58	56.58	46.70	46.70	46.70	46.70	46.70
#22	23.00	23.00	46.70	56.58	46.70	23.00	23.00	39.37	46.70	39.37	46.70	56.58	56.58	46.70	39.37	23.00	39.37	56.58	56.58	39.37
#23	46.70	23.00	23.00	46.70	56.58	39.37	23.00	23.00	39.37	46.70	39.37	46.70	56.58	56.58	46.70	39.37	23.00	39.37	56.58	56.58
#24	56.58	46.70	23.00	23.00	46.70	46.70	39.37	23.00	23.00	39.37	46.70	39.37	46.70	56.58	56.58	56.58	39.37	23.00	39.37	56.58
#25	46.70	56.58	46.70	23.00	23.00	39.37	46.70	39.37	23.00	23.00	56.58	46.70	39.37	46.70	56.58	56.58	56.58	39.37	23.00	39.37
#26	23.00	46.70	56.58	46.70	23.00	23.00	39.37	46.70	39.37	23.00	56.58	56.58	46.70	39.37	46.70	39.37	56.58	56.58	39.37	23.00
#27	46.70	46.70	46.70	46.70	46.70	56.58	56.58	56.58	56.58	56.58	23.00	23.00	23.00	23.00	23.00	39.37	39.37	39.37	39.37	39.37
#28	56.58	39.37	23.00	39.37	56.58	56.58	46.70	39.37	46.70	56.58	23.00	23.00	39.37	46.70	39.37	46.70	23.00	23.00	46.70	56.58
#29	56.58	56.58	39.37	23.00	39.37	56.58	56.58	46.70	39.37	46.70	39.37	23.00	23.00	39.37	46.70	56.58	46.70	23.00	23.00	46.70
#30	39.37	56.58	56.58	39.37	23.00	46.70	56.58	56.58	46.70	39.37	46.70	39.37	23.00	23.00	39.37	46.70	56.58	46.70	23.00	23.00
#31	23.00	39.37	56.58	56.58	39.37	39.37	46.70	56.58	56.58	46.70	39.37	46.70	39.37	23.00	23.00	23.00	46.70	56.58	46.70	23.00
#32	39.37	23.00	39.37	56.58	56.58	46.70	39.37	46.70	56.58	56.58	23.00	39.37	46.70	39.37	23.00	23.00	23.00	46.70	56.58	46.70

ROUTES ON THIS PAGE

260 straight lines: 20 hub → outer and 240 inner → outer



- 260 routes on this page
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: start in the left column, destination across the top; the number is the distance in T ly. Straight line, center to center.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

The Outer Twenty, Center to Center

ZONE 3 · OUTER 20

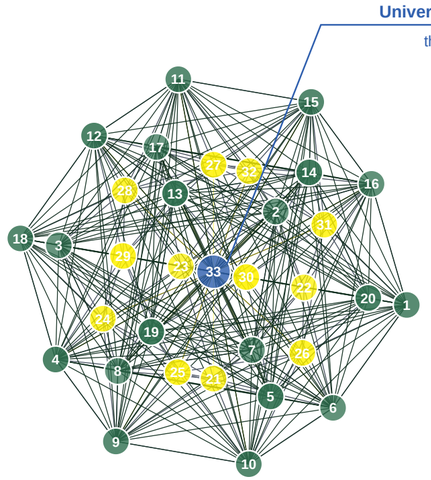
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Center to center in T ly (trillions of light-years); at light speed the same number is trillions of years. Rows = start, columns = destination; number color = destination zone.

from ↓ to →	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
#1	–	42.21	68.30	68.30	42.21	26.09	42.21	59.69	59.69	42.21	59.69	68.30	59.69	42.21	42.21	26.09	59.69	73.11	59.69	26.09
#2	42.21	–	42.21	68.30	68.30	42.21	26.09	42.21	59.69	59.69	42.21	59.69	68.30	59.69	42.21	26.09	26.09	59.69	73.11	59.69
#3	68.30	42.21	–	42.21	68.30	59.69	42.21	26.09	42.21	59.69	42.21	42.21	59.69	68.30	59.69	59.69	26.09	26.09	59.69	73.11
#4	68.30	68.30	42.21	–	42.21	59.69	59.69	42.21	26.09	42.21	59.69	42.21	42.21	59.69	68.30	73.11	59.69	26.09	26.09	59.69
#5	42.21	68.30	68.30	42.21	–	42.21	59.69	59.69	42.21	26.09	68.30	59.69	42.21	42.21	59.69	59.69	73.11	59.69	26.09	26.09
#6	26.09	42.21	59.69	59.69	42.21	–	26.09	42.21	42.21	26.09	68.30	73.11	68.30	59.69	59.69	42.21	59.69	68.30	59.69	42.21
#7	42.21	26.09	42.21	59.69	59.69	26.09	–	26.09	42.21	42.21	59.69	68.30	73.11	68.30	59.69	42.21	42.21	59.69	68.30	59.69
#8	59.69	42.21	26.09	42.21	59.69	42.21	26.09	–	26.09	42.21	59.69	59.69	68.30	73.11	68.30	59.69	42.21	42.21	59.69	68.30
#9	59.69	59.69	42.21	26.09	42.21	42.21	42.21	26.09	–	26.09	68.30	59.69	59.69	68.30	73.11	68.30	59.69	42.21	42.21	59.69
#10	42.21	59.69	59.69	42.21	26.09	26.09	42.21	42.21	26.09	–	73.11	68.30	59.69	59.69	68.30	59.69	68.30	59.69	42.21	42.21
#11	59.69	42.21	42.21	59.69	68.30	68.30	59.69	59.69	68.30	73.11	–	26.09	42.21	42.21	26.09	42.21	26.09	42.21	59.69	59.69
#12	68.30	59.69	42.21	42.21	59.69	73.11	68.30	59.69	59.69	68.30	26.09	–	26.09	42.21	42.21	59.69	42.21	26.09	42.21	59.69
#13	59.69	68.30	59.69	42.21	42.21	68.30	73.11	68.30	59.69	59.69	42.21	26.09	–	26.09	42.21	59.69	59.69	42.21	26.09	42.21
#14	42.21	59.69	68.30	59.69	42.21	59.69	68.30	73.11	68.30	59.69	42.21	42.21	26.09	–	26.09	42.21	59.69	59.69	42.21	26.09
#15	42.21	42.21	59.69	68.30	59.69	59.69	59.69	68.30	73.11	68.30	26.09	42.21	42.21	26.09	–	26.09	42.21	59.69	59.69	42.21
#16	26.09	26.09	59.69	73.11	59.69	42.21	42.21	59.69	68.30	59.69	42.21	59.69	59.69	42.21	26.09	–	42.21	68.30	68.30	42.21
#17	59.69	26.09	26.09	59.69	73.11	59.69	42.21	42.21	59.69	68.30	26.09	42.21	59.69	59.69	42.21	42.21	–	42.21	68.30	68.30
#18	73.11	59.69	26.09	26.09	59.69	68.30	59.69	42.21	42.21	59.69	42.21	26.09	42.21	59.69	59.69	68.30	42.21	–	42.21	68.30
#19	59.69	73.11	59.69	26.09	26.09	59.69	68.30	59.69	42.21	42.21	59.69	42.21	26.09	42.21	59.69	68.30	68.30	42.21	–	42.21
#20	26.09	59.69	73.11	59.69	26.09	42.21	59.69	68.30	59.69	42.21	59.69	59.69	42.21	26.09	42.21	42.21	68.30	68.30	42.21	–

ROUTES ON THIS PAGE

190 straight lines: every outer universe to every other



- 190 routes on this page
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: start in the left column, destination across the top; the number is the distance in T ly. Straight line, center to center.

Note: These are distances among the outer 20 universes, inside the 32-universe shell, which works on its own without the throats (Geometric Transport, page 4).

One Curvilinear, in True Shape

ZONE 4 · CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

WHAT THIS SHOWS

One curvilinear at true scale, drawn only from the deck's numbers. On the left it sits on the line from the hub to green #1: its two tips reach the hub and green #1, and its three sides rest against three yellow universes. On the right is the slice across it: three yellow universes around the orange piece.

The piece is a round bicone, two cones joined at a narrow waist. Each yellow universe presses a notch into the waist.

THE NUMBERS

Length 23T ly, waist 5.75T ly (a quarter of the length), volume 1.99×10^{38} ly³ each and 3.98×10^{39} ly³ for all 20. Center 18.28T ly from the hub; tips at 6.78T and 29.78T. The gap between the three yellow universes has radius 2.46T ly at the center and the waist radius is 2.88T ly, so each yellow universe presses 0.41T ly into it. A 120° turn returns the same view: three-fold symmetry.

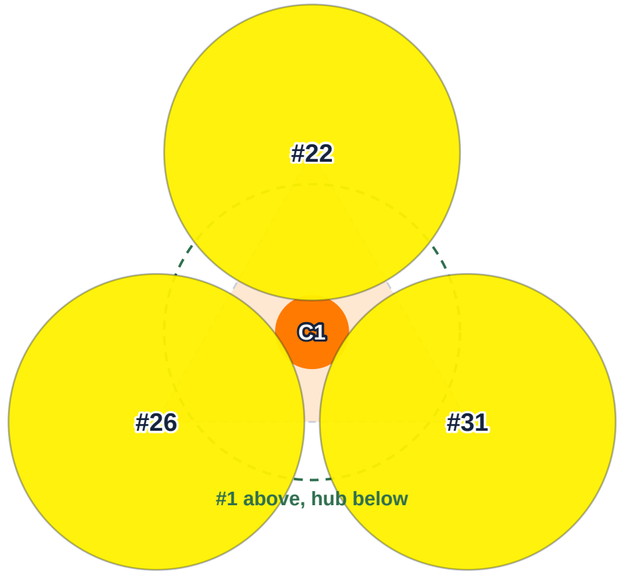
The hub and green #1 are 36.55T ly apart, so they do not touch each other. The curvilinear is what joins them.

ALONG THE AXIS



Slice from the hub to green #1. The yellow universe touches the hub and touches green #1 (23.0T ly center to center). The hub and green #1 are 36.55T apart, joined by the orange curvilinear.

ACROSS THE FACE



Slice at the middle of the curvilinear, looking down from #1. The three yellow universes are the corners of face #1. The orange waist is cut by each of them.

hub #33 · inner (yellow) · outer (green) · curvilinear (orange)

How to read: left, a slice along the line from the hub to green #1; right, a slice across the face where three yellow universes meet. Orange is the curvilinear.

Which Universes Each Curvilinear Joins

ZONE 4 · CURVILINEARS

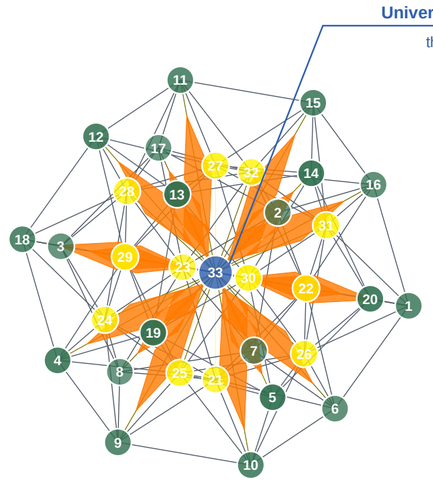
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Each curvilinear touches the hub, its green universe and the three yellow universes at the corners of its face · lengths in T ly from the hub

Curvilinear	Sits on	Inner a	Inner b	Inner c	Hub	Center	Near tip	Far tip	Length	Waist	Volume ly³
C1	#1	#22	#26	#31	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C2	#2	#22	#23	#32	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C3	#3	#23	#24	#28	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C4	#4	#24	#25	#29	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C5	#5	#25	#26	#30	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C6	#6	#21	#22	#26	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C7	#7	#21	#22	#23	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C8	#8	#21	#23	#24	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C9	#9	#21	#24	#25	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C10	#10	#21	#25	#26	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C11	#11	#27	#28	#32	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C12	#12	#27	#28	#29	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C13	#13	#27	#29	#30	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C14	#14	#27	#30	#31	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C15	#15	#27	#31	#32	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C16	#16	#22	#31	#32	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C17	#17	#23	#28	#32	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C18	#18	#24	#28	#29	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C19	#19	#25	#29	#30	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸
C20	#20	#26	#30	#31	#33	18.28	6.78	29.78	23.0	5.75	1.99×10³⁸

ROUTES ON THIS PAGE

The 20 curvilinears, each along the axis from the hub to its green universe



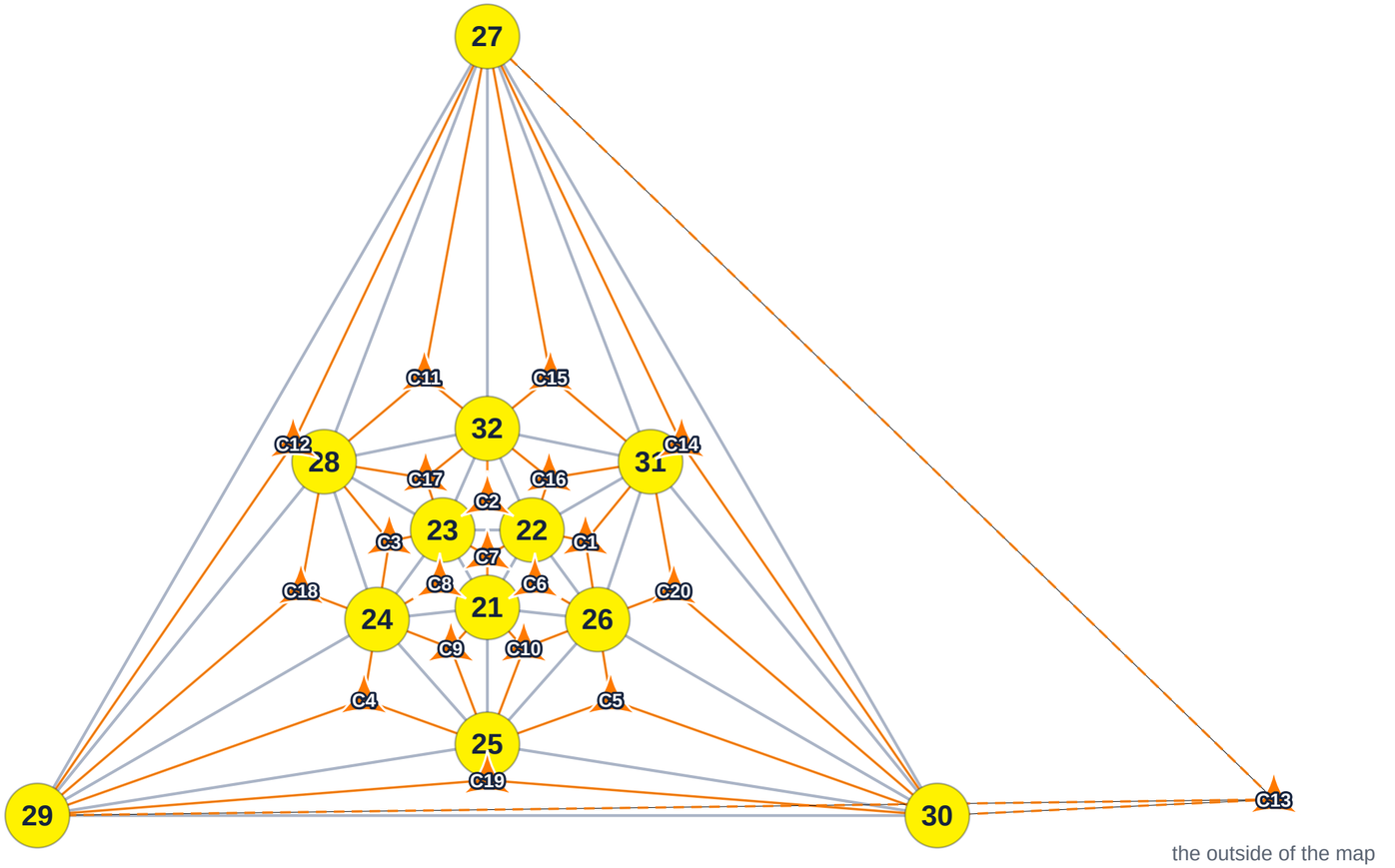
- 20 curvilinears C1–C20
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: each row is one curvilinear. It joins the hub, one green universe and the three yellow universes at the corners of its face.

Every Curvilinear on One Flat Map

ZONE 4 · CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



the outside of the map
wraps around to this face

WHAT THIS SHOWS

All 12 yellow universes and all 20 curvilinears on one flat map. Every universe appears once.

Each orange **C** is joined to the three yellow universes at the corners of its face. Its two tips reach the hub, in the middle of the whole structure, and the green universe with the same number.

THE NUMBERS

20 faces × 3 corners = 60 links. Each yellow universe is a corner of 5 faces (12 × 5 = 60). Every link is 13.96T ly from the curvilinear's center to the yellow universe's center.

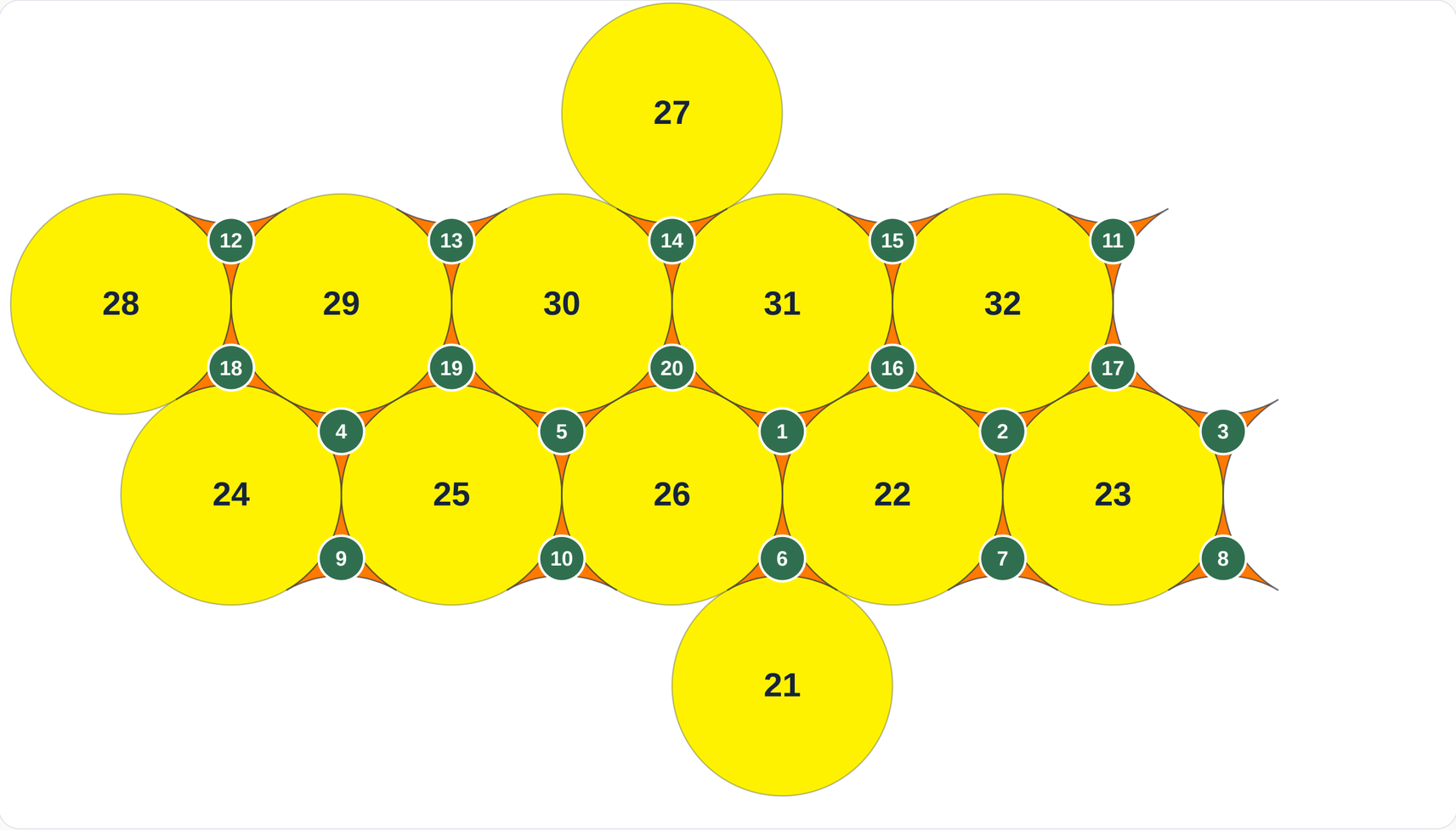
C13 is the face around the outside of the map, drawn at the lower right. Distances on this map are not to scale.

How to read: all 12 yellow universes and all 20 curvilinears on one map, each once. Each orange C is joined to the three yellow universes at the corners of its face.

The Same Map, With the Yellow Universes Touching

ZONE 4 · CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



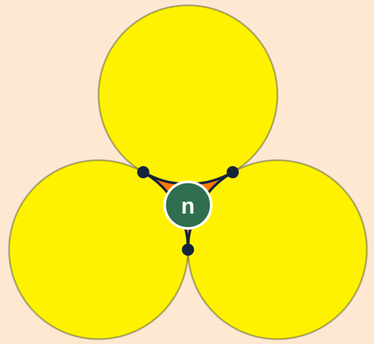
WHAT THIS SHOWS

The map cut open and laid flat, with the yellow universes touching. Each yellow universe 21–32 appears once.

The orange gap between three touching yellow universes is a three-sided curvilinear. The **green disc** in its center is where its green universe sits, numbered 1–20 (C1 on #1).

THE NUMBERS

Each side is a 60° arc of a yellow universe, 12.04T ly long. Orange shapes along the top, right and bottom edges have one open side because the map is cut there; the missing circle is the universe drawn elsewhere on the map.



3 curved sides · 3 touching points

How to read: each orange shape is one curvilinear, between three touching yellow universes (21–32, once each). The green disc in its center is its green universe, numbered 1–20.

Every Face, With Its Three Yellow Universes Touching

ZONE 4 · CURVILINEARS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

C1 on green #1 — 22, 26, 31

C2 on green #2 — 22, 23, 32

C3 on green #3 — 23, 24, 28

C4 on green #4 — 24, 25, 29

C5 on green #5 — 25, 26, 30, 5

C6 on green #6 — 21, 22, 26

C7 on green #7 — 21, 22, 23

C8 on green #8 — 21, 23, 24

C9 on green #9 — 21, 24, 25

C10 on green #10 — 21, 25, 26

C11 on green #11 — 27, 28, 32

C12 on green #12 — 27, 28, 29

C13 on green #13 — 27, 29, 30

C14 on green #14 — 27, 30, 31

C15 on green #15 — 27, 31, 32

C16 on green #16 — 22, 31, 32

C17 on green #17 — 23, 28, 32

C18 on green #18 — 24, 28, 29

C19 on green #19 — 25, 29, 30

C20 on green #20 — 26, 30, 31

5 **Green universe:** every face has one green universe sphere sitting on top of its curvilinear, numbered like its C (C5 has green #5).
Only C5 shows it, to keep the page clean. The same is true of all 20 faces: the three touching yellow universes always have a green universe above them.

WHAT THIS SHOWS

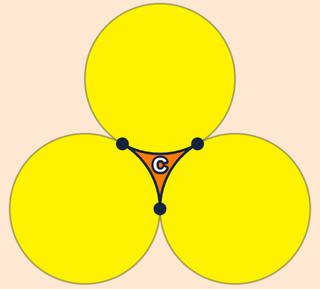
All 20 faces, one panel each. In every panel the three yellow universes touch, and the orange gap between them is a true three-sided curvilinear. Each face also has a green universe above it, shown on C5.

THE NUMBERS

Each side is a 60° arc of a yellow universe, 12.04T ly long. The gap radius at the center is 1.78T ly.

Why numbers repeat: a yellow universe is a corner of 5 faces, so it shows in 5 panels. The map on page 33 shows each one once.

The coordinates on pages 20–30 use equal 23T spheres with the hub touching, which leaves 1.18T between neighboring yellow universes.



3 curved sides · 3 touching points

How to read: each panel is one face: three touching yellow universes and the orange curvilinear between them. Each yellow universe is in 5 panels. The green sphere on C5 sits above every face.

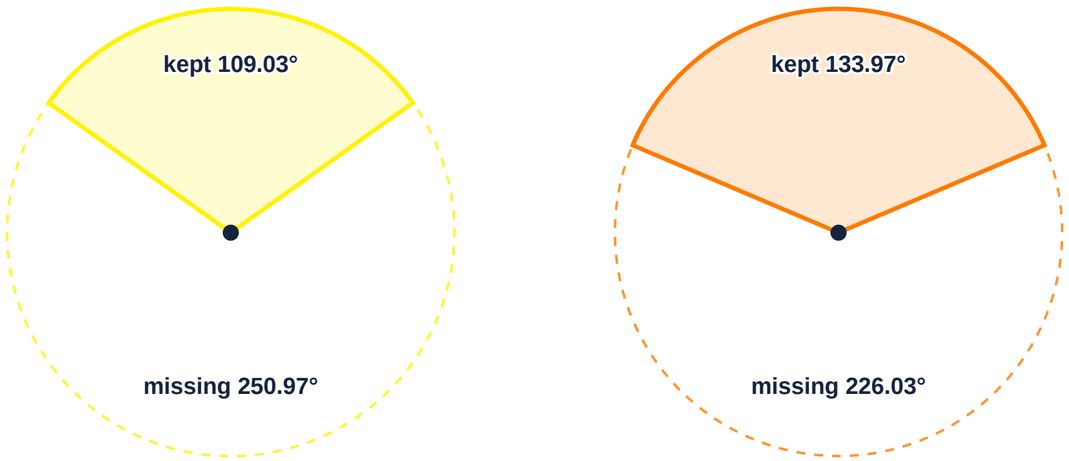
Throat Curvature: Where the Curve Lives

APEX THROAT

CURVILINEAR THROAT

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

A THROAT, CUT OPEN AND UNROLLED FLAT



Apex throat · 12 of them
half-angle 17.6289° · mouth 0.2951 sr
deficit 250.9735° = 4.3803 rad

Curvilinear throat · 20 of them
half-angle 21.8475° · mouth 0.4513 sr
deficit 226.0303° = 3.9450 rad

Glue the two straight edges of the solid sector together and you get the throat. The dashed wedge is the part that is missing: the deficit.

WHAT THIS SHOWS

Inside a throat the wall is flat. All the curvature sits at the tip, as the angle that is missing when you unroll the cone. A throat with a bigger deficit is a sharper tip.

Carry an arrow once around the tip and it comes back turned by the deficit: 250.9735° at an apex throat, 226.0303° at a curvilinear throat.

THE NUMBERS

In trillions of light-years: an apex throat is 24.1334T ly from the hub to its mouth rim (axis 23.00T) and a curvilinear throat 39.3827T ly (axis 36.55T). The unrolled arcs above are 45.92T and 92.09T ly long.

Sky tiling: 12 × 0.2951 + 20 × 0.4513 = 12.5664 sr = 4π, so the 32 mouths cover the whole sky.

Links: 32 throats + 120 sibling links (30 yellow–yellow, 30 green–green, 60 yellow–green) = 152 links.

Loops: 152 − 33 + 1 = 120 independent loops. Net displacement zero, net turn 360°.

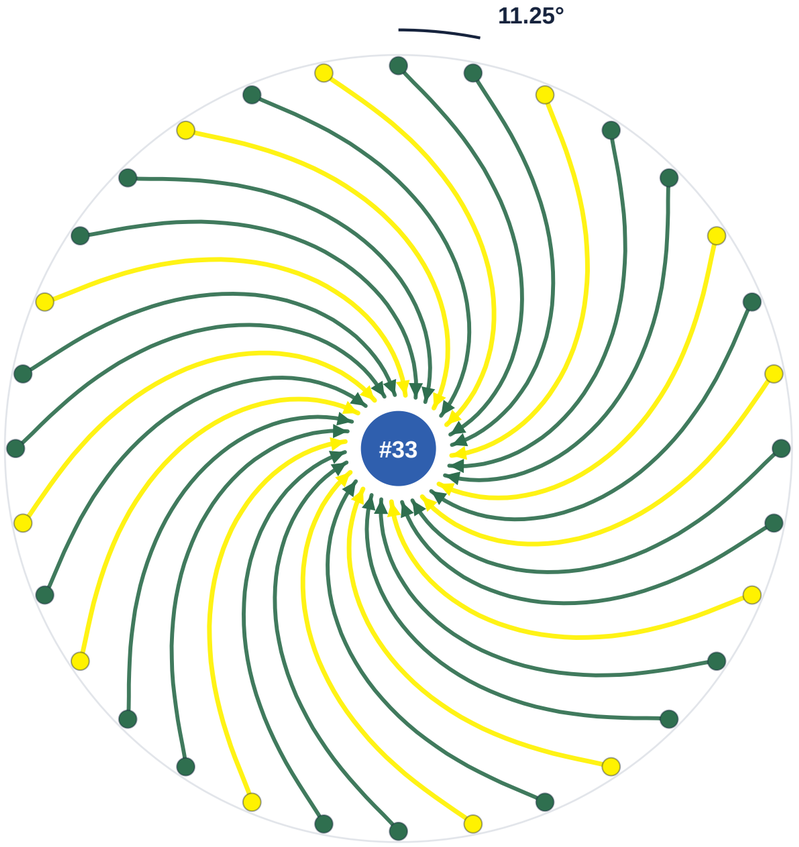
Check: 360° × (1 − sin of the half-angle) gives 250.97° and 226.03°, matching the deck.

How to read: a throat is a cone. Unroll it flat and a wedge is missing; that missing angle is the deficit, and it is where all the curvature sits.

The Curving, Spinning Arrival

SCHEMATIC

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



hub #33 · 20 green, pushing in · 12 yellow, pulling in

WHAT THIS SHOWS

A picture of the idea on page 4, not a measured path. At each green universe, dark energy pushes inward; at each yellow universe, dark matter pulls inward, all toward the same hub. The two do not meet head-on. They arrive curving, in a spinning rotation around the hub.

Here the 32 throats are flattened into a ring and seen from above, with each arm bending as it comes in. How much each arm bends is only for the picture.

THE NUMBERS FROM THE DECK

In trillions of light-years: measured on pages 50–53 under the rule stated there, the arrival is 34.8125T ly (apex) and 63.4282T ly (curvilinear) for one full turn, against straight lengths of 24.1334T and 39.3827T ly.

One rotation step per domain: 360° ÷ 33 = 10.9091°. Per throat: 360° ÷ 32 = 11.25°, the gap between neighboring arms. Fivefold at an apex is 72°; three-fold at a curvilinear is 120°.

Distances: the straight center-to-center ones are on pages 24–30. The deck gives no length for a curved route; the curved lengths come from the rule on pages 50–53.

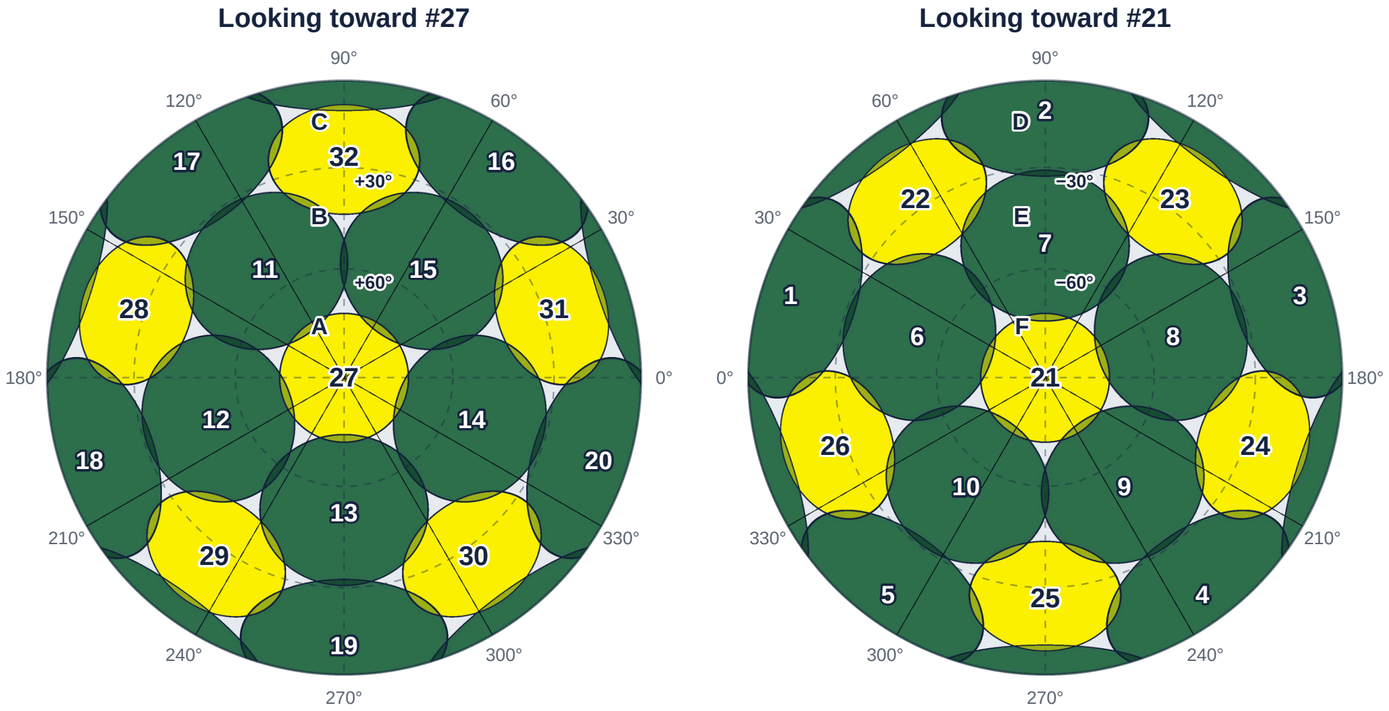
How to read: SCHEMATIC. The bend of each arm is only illustrative; the angles come from the deck. Measured lengths for the same arrival are on pages 50–53.

The 32 Throats Seen From the Hub

12 APEX THROATS

20 CURVILINEAR THROATS

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



equal-area map: patch sizes are true, shapes bend toward the edge · grid: latitude every 30°, longitude every 30° from +x

apex throat mouth (12) · curvilinear throat mouth (20) · overlap of two mouths · not inside any mouth

WHAT THIS SHOWS

All 32 throats as the hub sees them. Each patch is the mouth of one throat, centered on the direction of the universe it leads to: yellow for the 12 apex throats, green for the 20 curvilinear throats. The numbers are those universes. The left disc looks toward #27 and the right disc toward #21, so every throat appears once.

Where a yellow mouth and a green mouth overlap, the color turns olive.

THE NUMBERS

In trillions of light-years: the throat axes run 23.00T ly from the hub to each of the 12 apex universes and 36.55T ly to each of the 20 curvilinear ones. The mouths are angles on the sky, the same at every size.

Grid, as on an old atlas: bands A–F by latitude, sectors 1–12 by longitude; #13 is B10.

Multipoles: these 32 directions, and the 12, 20 and 30 inside them, carry power only at ℓ = 0, 6, 10, 12, 16, 18, 20, 22, 24, the set icosahedral symmetry allows and the ℓ = 6, 10, 12 ladder of Rivero-Zeta page 19. It describes the pattern of directions, not a topology.

Sky tiling: 12 × 0.2951 sr + 20 × 0.4513 sr = 12.5664 sr = 4π.

Mouth half-angles: 17.6289° at an apex throat, 21.8475° at a curvilinear throat.

Coverage measured on this map: 92.4% of the sky is inside one mouth, 3.8% inside two, and 3.8% inside none. The overlap and the gaps are equal in size, so the mouths still add up to exactly 4π.

How to read: the hub looks out at 32 patches of sky, one per throat. Yellow patches are apex throats, green patches are curvilinear throats. Each number is the universe the throat leads to.

The 12 Apex Throats, Hub to Yellow

Each apex throat runs from the hub to one yellow inner universe · lengths in T ly · latitude 90° = toward #27 · longitude from +x

ZONE 1 · CORE

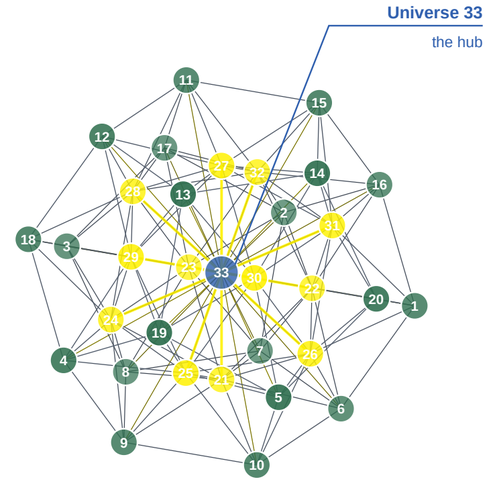
ZONE 2 · INNER 12

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Throat	Type	Latitude°	Longitude°	Axis length	Gap to hub	Half-angle	Mouth sr	Share of sky	Deficit	Light time
#21	Apex throat	−90.0	0.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#22	Apex throat	−26.57	54.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#23	Apex throat	−26.57	126.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#24	Apex throat	−26.57	198.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#25	Apex throat	−26.57	270.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#26	Apex throat	−26.57	342.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#27	Apex throat	90.0	0.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#28	Apex throat	26.57	162.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#29	Apex throat	26.57	234.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#30	Apex throat	26.57	306.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#31	Apex throat	26.57	18.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr
#32	Apex throat	26.57	90.0	23.00T	0.00T	17.6289°	0.2951	2.35%	250.9735°	23.00T yr

THROATS ON THIS PAGE

12 throats, each 23.00T ly from the hub center to the universe center; the surfaces touch



- 12 throats from the hub
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

How to read: each row is one throat from the hub to a yellow universe. The mouth is the patch of sky it opens onto; the deficit is the angle missing when the throat is unrolled flat.

The 20 Curvilinear Throats, Hub to Green

ZONE 1 · CORE

ZONE 3 · OUTER 20

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

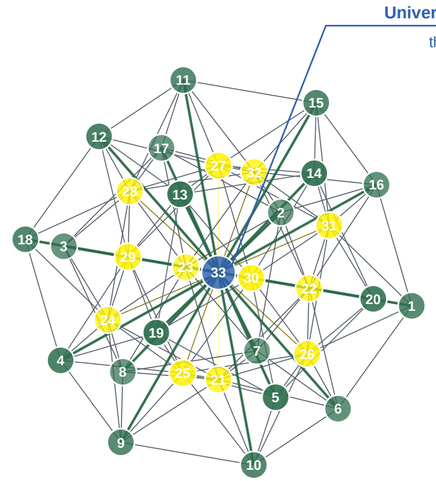
Each curvilinear throat runs from the hub to one green outer universe · lengths in T ly · latitude 90° = toward #27 · longitude from +x

Throat	Type	Latitude°	Longitude°	Axis length	Gap to hub	Half-angle	Mouth sr	Share of sky	Deficit	Light time
#1	Curvilinear throat	−10.81	18.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#2	Curvilinear throat	−10.81	90.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#3	Curvilinear throat	−10.81	162.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#4	Curvilinear throat	−10.81	234.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#5	Curvilinear throat	−10.81	306.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#6	Curvilinear throat	−52.62	18.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#7	Curvilinear throat	−52.62	90.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#8	Curvilinear throat	−52.62	162.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#9	Curvilinear throat	−52.62	234.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#10	Curvilinear throat	−52.62	306.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#11	Curvilinear throat	52.62	126.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#12	Curvilinear throat	52.62	198.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#13	Curvilinear throat	52.62	270.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#14	Curvilinear throat	52.62	342.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#15	Curvilinear throat	52.62	54.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#16	Curvilinear throat	10.81	54.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#17	Curvilinear throat	10.81	126.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#18	Curvilinear throat	10.81	198.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#19	Curvilinear throat	10.81	270.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr
#20	Curvilinear throat	10.81	342.0	36.55T	13.55T	21.8475°	0.4513	3.59%	226.0303°	36.55T yr

How to read: each row is one throat from the hub to a green universe. The mouth is the patch of sky it opens onto; the deficit is the angle missing when the throat is unrolled flat.

THROATS ON THIS PAGE

20 throats, each 36.55T ly from the hub center to the universe center, with a 13.55T ly gap the curvilinear spans



- 20 throats from the hub
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

Angles Between the 32 Throats

496 PAIRS · 12 ANGLES

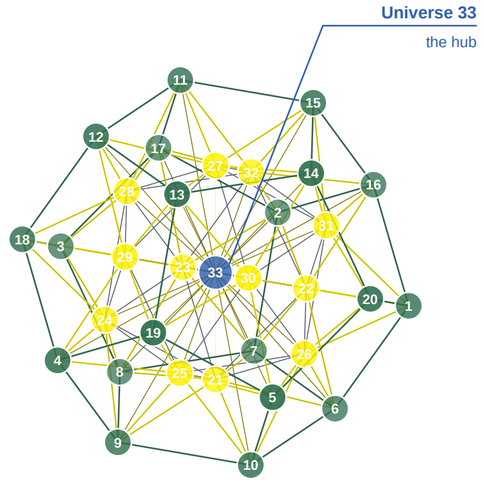
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Every pair of throats compared by the angle between their axes at the hub, against the two cones' half-angles added together

Pair of throats	Angle	From	To	Pairs	Sum of half-angles	Gap (+) / overlap (−)	Cones
Neighboring yellow throats	63.43°	Yellow	Yellow	30	35.26°	+28.17°	apart
Yellow throats two steps apart	116.57°	Yellow	Yellow	30	35.26°	+81.31°	apart
Opposite yellow throats	180.00°	Yellow	Yellow	6	35.26°	+144.74°	apart
Yellow throat and a green throat on its face	37.38°	Yellow	Green	60	39.48°	−2.10°	overlap
Yellow throat and a green throat one step off	79.19°	Yellow	Green	60	39.48°	+39.71°	apart
Yellow throat and a green throat two steps off	100.81°	Yellow	Green	60	39.48°	+61.33°	apart
Yellow throat and a far-side green throat	142.62°	Yellow	Green	60	39.48°	+103.14°	apart
Neighboring green throats	41.81°	Green	Green	30	43.70°	−1.88°	overlap
Green throats two steps apart	70.53°	Green	Green	60	43.70°	+26.84°	apart
Green throats three steps apart	109.47°	Green	Green	60	43.70°	+65.78°	apart
Green throats four steps apart	138.19°	Green	Green	30	43.70°	+94.50°	apart
Opposite green throats	180.00°	Green	Green	10	43.70°	+136.31°	apart

OVERLAPPING THROATS

The 90 neighbor pairs whose cones overlap: 60 yellow–green and 30 green–green



- overlapping neighbor pairs
- Hub · #33
- 12 inner · #21–32
- 20 outer · #1–20
- 32 throats to the hub
- 120 sibling links

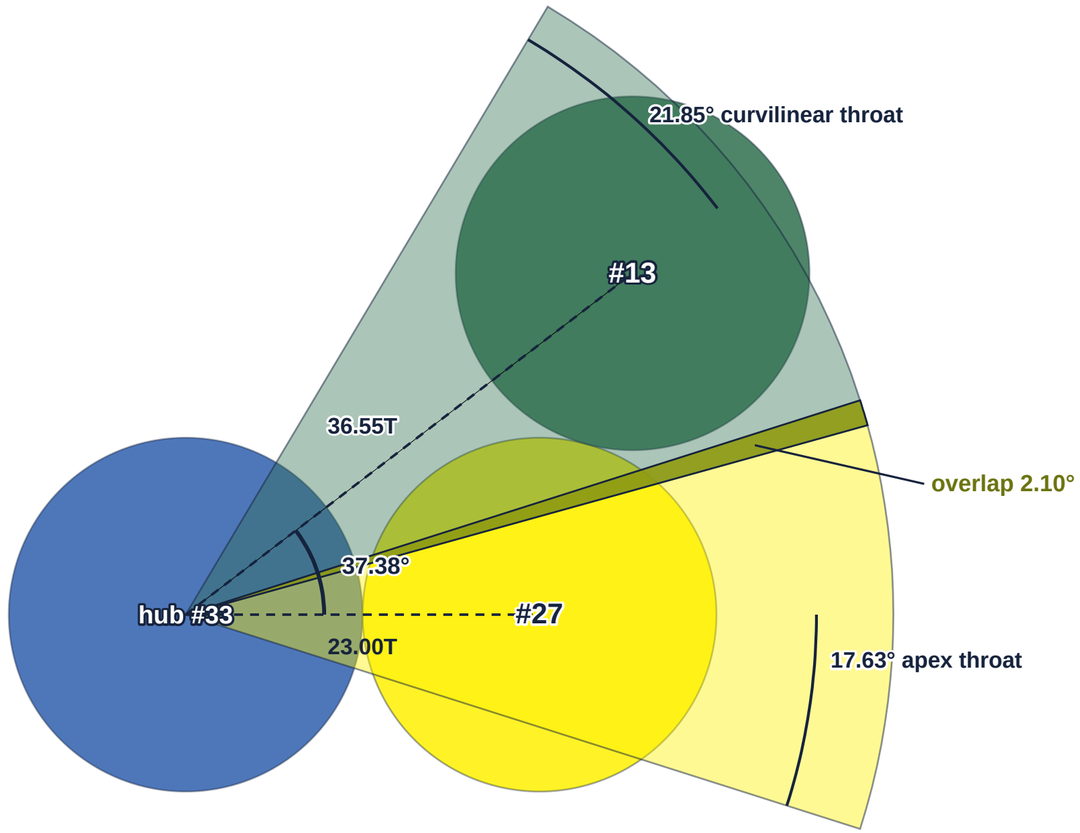
In trillions of light-years: the angles do not depend on length. The two universes of each class are this far apart, center to center: apex–apex 24.184T, 39.130T and 46.000T ly; apex–green 23.000T, 39.367T, 46.697T and 56.581T ly; green–green 26.087T, 42.209T, 59.693T, 68.296T and 73.108T ly (page 61).

How to read: pairs of throats are compared by the angle between their axes. A minus sign means the two cones overlap; a plus sign is the gap between them.

Two Neighboring Throats, to Scale

APEX CURVILINEAR

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



WHAT THIS SHOWS

Two neighboring throats drawn to scale in one slice through the hub: the apex throat to yellow #27 and the curvilinear throat to green #13. The yellow cone opens 17.63° to each side of its axis, the green cone 21.85°.

The two axes are 37.38° apart. The two cones need 39.48° to stay separate, so they overlap by 2.10°.

THE NUMBERS

This happens at 90 neighbor pairs: 60 yellow–green pairs overlap by 2.10° and 30 green–green pairs by 1.88°. Yellow universes are 63.43° apart, so their cones stay 28.17° apart.

Across the whole sky about 3.8% lies inside two mouths and 3.8% inside none, which is why the mouths still add up to 4π.

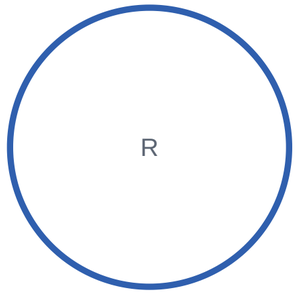
How to read: a slice through the hub shows two neighboring throats at true scale. Where the yellow and green cones cross, they overlap by 2.10°.

Curvature Atlas: pages 43–49

One Rotation, Five Readings

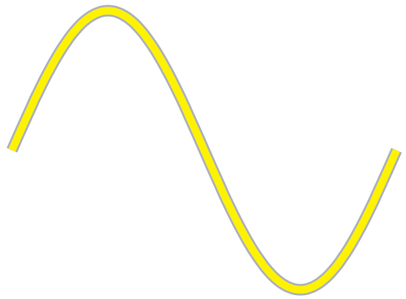
COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

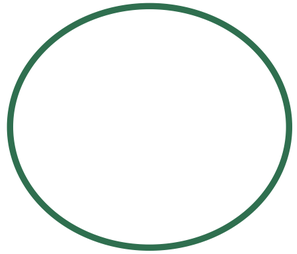


along the axis
a circle, curvature 1/R

edge on
a straight segment of length 2R



side on, against the index
a sine, one period



30° off axis
an ellipse, axis ratio 0.8660 = cos 30°



riding it
a straight line of length 2πR, every step the same to 1.7×10^{-13}

WHAT THIS SHOWS

One uniform rotation, read from five places. Along its axis it is a circle. Edge on it is a straight segment. Plotted against the index it is a sine. Tilted 30° it is an ellipse. Riding it, you travel a straight line and feel no turn. Nothing changes but where you stand.

THE NUMBERS

In trillions of light-years: R is a unit of length. With R = 23.00T ly (hub to an apex universe) the segment is 2R = 46.00T ly and riding it is a straight line of 2πR = 144.51T ly. The circle's curvature is 1/R = 0.0435 per trillion light-years.

Circle: curvature 1/R. **Segment:** length 2R, apparent curvature 0. **Ellipse:** axis ratio = cos(tilt).

Riding it: a straight line of length 2πR, arc step uniform to 1.7×10^{-13}.

At the architecture's own angles: cos 37.3774° = 0.7947 (apex to face), cos 41.8103° = 0.7454 (face to face), cos 63.4349° = 0.4472 (apex to apex).

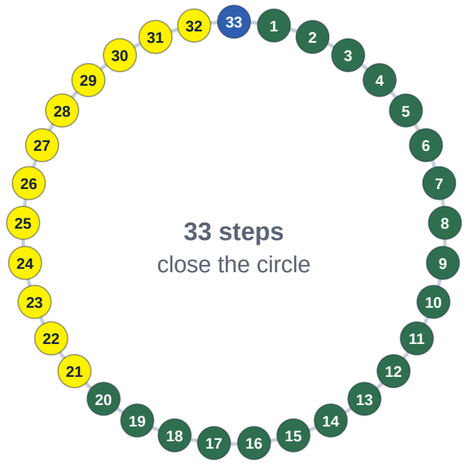
The curve is from The Rivero-Zeta Curve, pages 3 and 9. The three tilt ratios are cos of the deck's own angles.

How to read: the same rotation seen from five places. The turning is real, but it exists only in a frame you are not in.

The 33-Domain Wave

COMPUTED

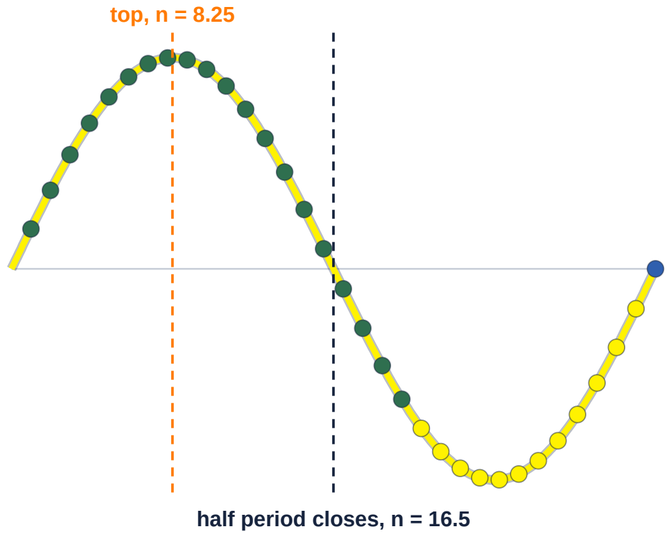
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



from outside
33 points, 10.9091° apart

Universe numbers fill the steps in order, 1 to 33;
the hub #33 is the closing step, at 360°.

the deck gives the step, not the order



along the index
a sine, $\omega_n = R \cdot \sin(2\pi n/33)$



edge on: a flat segment, curvature zero

WHAT THIS SHOWS

The architecture's own rotation, one step per domain: 360° ÷ 33 = 10.9091°. After 33 steps the angle is exactly 360° and the curve returns to where it began, to 2.4×10^{-16}. From outside it is a circle of 33 points, along the index a sine, edge on a flat segment.

THE NUMBERS

In trillions of light-years: with R = 23.00T ly, one step of 10.9091° moves the point 4.379T ly along the circle (2πR ÷ 33), and the wave swings 23.00T ly to each side of the line.

Steps to a full turn:

one per domain (33)	10.9091°	33.000	**closes**
one per throat (32)	11.2500°	32.000	**closes**
five-fold, apex	72.0000°	5.000	**closes**
three-fold, curvilinear	120.0000°	3.000	**closes**
apex mouth radius	17.6289°	20.421	**remainder**
curvilinear mouth radius	21.8475°	16.478	**remainder**

Max at n = 8.25 and half period at n = 16.5, both exact fractions of 33. R is a unit of length: no free parameter.

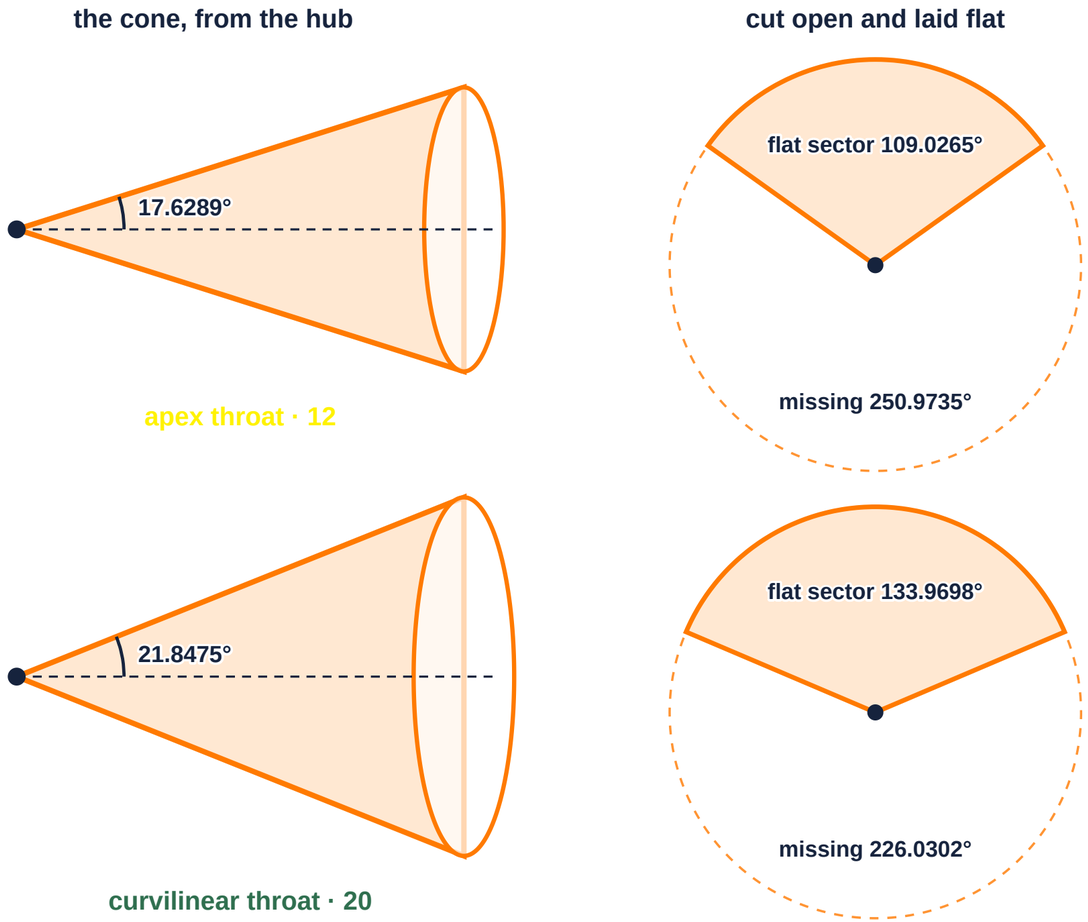
From The Rivero-Zeta Curve, pages 5 and 9.

How to read: 33 points on a circle, the same 33 along a sine, the same 33 on a flat line. The circle closes on the architecture's own count, 33.

Inside One Throat

THEOREM · COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



How to read: each throat is a cone; unrolled it is a flat sector. The dashed wedge is the deficit, where all the curvature lives.

WHAT THIS SHOWS

A throat is the cone its domain subtends at the hub. Cut it open and it lies flat: zero curvature on the wall, so riding inside you measure flat space and feel no turning. All the curvature sits at the tip, as the angle that is missing.

THE NUMBERS

In trillions of light-years: slant (hub to mouth rim) 24.1334T ly for an apex throat and 39.3827T ly for a curvilinear one, from axes of 23.00T and 36.55T ly. Rim radius 7.3088T and 14.6558T ly; once around the rim 45.92T and 92.09T ly; the one-turn straight path 39.30T and 72.50T ly.

Apex: half-angle 17.628945°, sector 109.026503°, deficit 250.973497°. **Curvilinear:** 21.847547°, 133.969758°, 226.030242°.

Flux inward $j = 1/(\Omega r^2)$: 3.389 (apex) and 2.216 (curvilinear) at r = 1, ×100 at each tenth.

Core radius $r_0 \sin\alpha$: sin α = 0.302851398 and 0.372138217, so the two classes differ by 1.228781570 in cutoff and 1.509904146 in barrier height.

Derived here: the straight path that circles the wall once is 2s·sin(sector ÷ 2) = 1.6285 s (apex) and 1.8408 s (curvilinear) for a throat of slant length s.

From The Rivero-Zeta Curve, pages 2 and 12. The one-turn wall path follows from the sector angle.

The Curvature Budget

THEOREM · EXACT

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

the shell: 12 defects of 60° — 12 × 60° = 720° = 4π = 12.5663706144 sr



the 20 curvilinear faces: 0.0000° each, flat

the core: 32 throat mouths tile the sky — 12 × 0.2950722634 + 20 × 0.4512751727 = 12.5663706144 sr



two different theorems, the same total, to every digit computed

the 32 cones: their deficits — 12 × 4.3803138509 + 20 × 3.9449719321 = 131.463205 rad



apex domain (pull, dark-matter sector) curvilinear domain (push, dark-energy sector)

Bars 1 and 2 are drawn to the same length: the shell carries 4π as curvature,
and the 32 mouths divide the same 4π as the sky. The throats add relations, not curvature.

WHAT THIS SHOWS

Descartes' theorem fixes the total angular defect of any closed convex surface at 2πχ = 4π. The architecture divides it into twelve equal quanta of 60° and puts none on the twenty. The 32 throat mouths then tile exactly the same 4π from the other side.

THE NUMBERS

In trillions of light-years: these totals are angles and solid angles, the same at every size. The throats they act along are 24.1334T and 39.3827T ly long (page 45), and mass per length is the same at every length.

Ratios: mouths, curvilinear over apex, 1.5293717121; deficits, apex over curvilinear, 1.1103536163; shell defects 1.0000000000.

A defect is a mass: δ = 8πGμ. As listed in the deck (G only): shell 6.242852×10^{8} · apex cone 2.611317×10^{9} · curvilinear cone 2.351789×10^{9} kg/m.

With c² restored (×8.98755×10^{16}, as pages 8 and 12 of the Multiverse Map do): 5.61×10^{25} · 2.35×10^{26} · 2.11×10^{26} kg/m. Line mass per length is the same at every size.

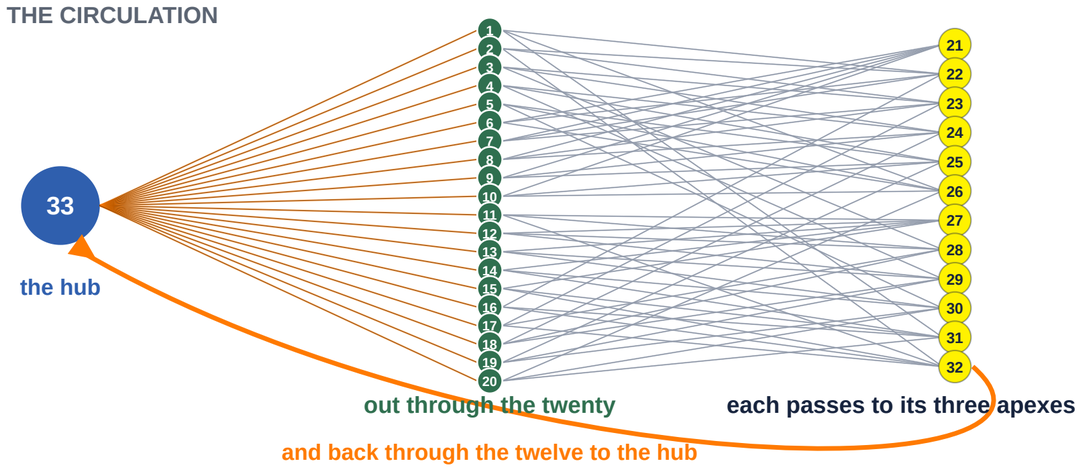
From The Rivero-Zeta Curve, pages 10, 11 and 12. Recomputed from the icosahedron: all values match.

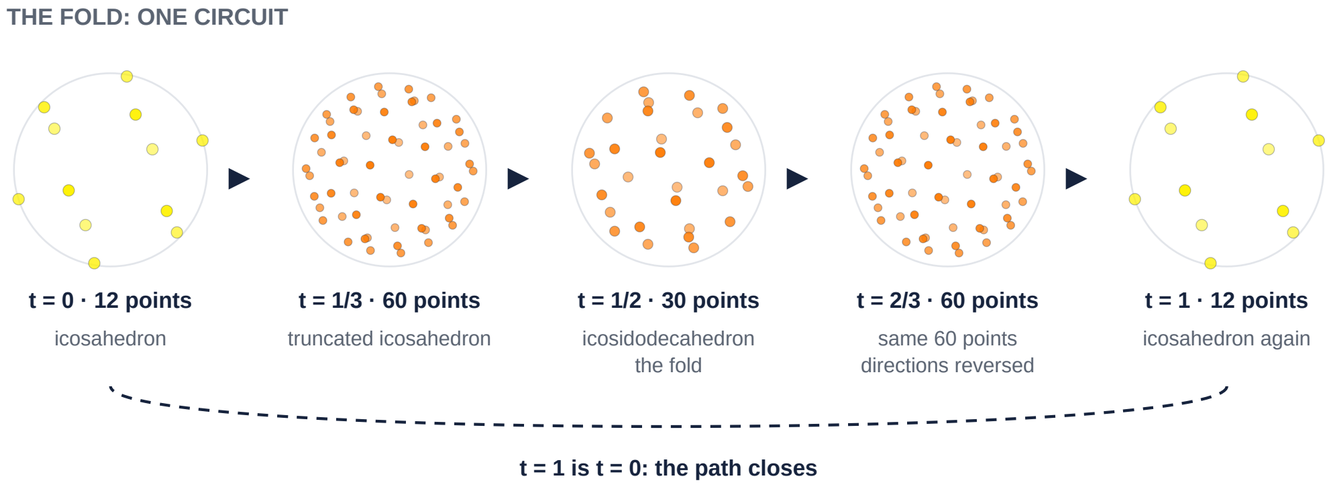
How to read: one total, on both sides of inception: 4π as the shell's curvature and 4π as the sky the throats divide.

The Circulation and the Fold

THEOREM · EXACT

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years





How to read: left to right and back: out through the twenty, back through the twelve; and the fold, a circuit that returns the shape but not the orientation.

WHAT THIS SHOWS

The hub sends outward through the twenty curvilinear domains; each passes to its three apexes; each apex gathers from its five and returns through its throat. Below, the fold: truncate every edge at a fraction t and run t from 0 to 1. The family closes on itself.

THE NUMBERS

In trillions of light-years: the hub sends out along axes of 23.00T ly (to the 12 apex universes) and 36.55T ly (to the 20 curvilinear ones). The links across the shell are 24.184T ly (apex–apex), 23.000T ly (apex–curvilinear) and 26.087T ly (curvilinear–curvilinear).

Circulation: 120 independent loops (152 links − 33 domains + 1). Net displacement zero, net turn 360°. Divergence 4.3×10^{-16} at every node. Its two exact ratios: 20/12 = 1.666667 and 5.000000. Unique at 33 domains.

Fold: 60 of 60 directed edges reversed, difference 0.0. $R^2 = I$, with eigenspaces 17 (+1) and 6 (−1). One circuit returns the shape and negates the 6; two circuits restore all 23: a spinor.

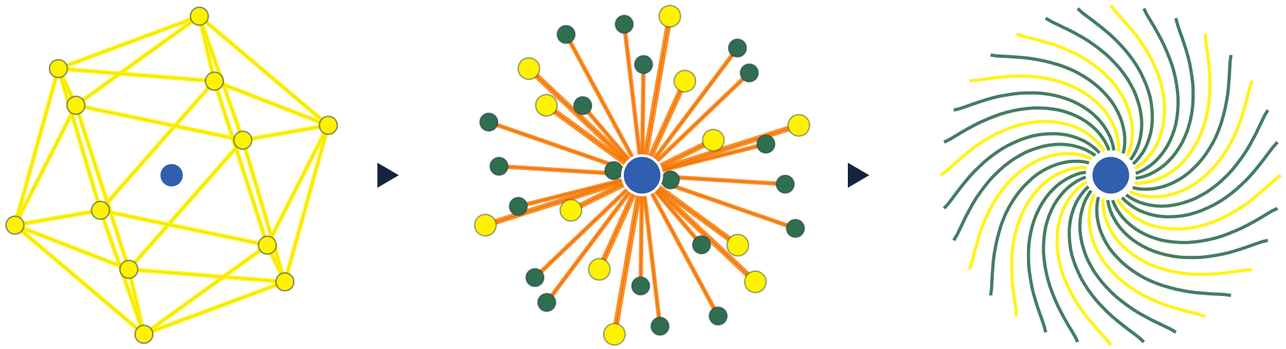
Antipodal map on the twenty: two blocks of 10, coupling 2.2×10^{-17}.

From The Rivero-Zeta Curve, pages 6, 13, 14, 16 and 17. Point counts are recomputed here.

Inception: the Flat Wall Meets the Curved Tip

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

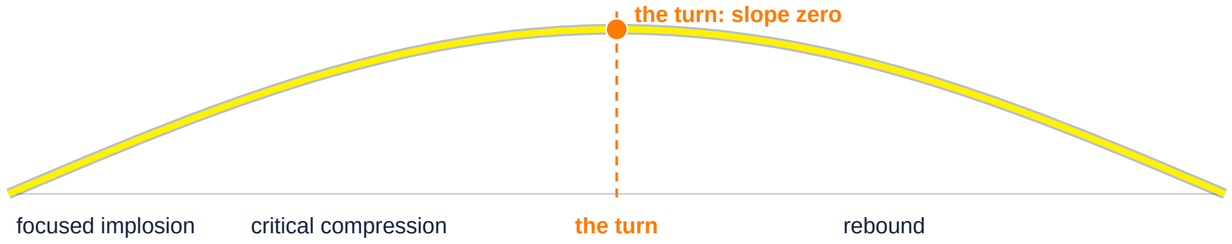


before inception
4π already carried by the shell
F = 0 and R = 0
no inverse, no distance, no duration

inception: the throats open
32 cones form, each with a deficit
their mouths tile the same 4π
relations appear, curvature is kept

after: the curve runs
one flux carries both: 0.999892
Λ ~ F locked at −1.0000
exact revival, P = 1.000000

THE INCEPTION PROFILE: FOUR STAGES, ONE HALF PERIOD



WHAT THIS SHOWS

Before inception the shell already carries its full curvature, but nothing can read it: no relations, no distance, no duration. Inception adds the relations, not the curvature. The throats open, energy flows in toward the hub in a curving, spinning rotation, and the curve turns where its slope is zero.

THE NUMBERS

In trillions of light-years: the hub is 23.00T ly across. The 32 throats run 24.1334T ly (apex) and 39.3827T ly (curvilinear) to their rims, and the spinning arrivals are 34.8125T and 63.4282T ly (pages 50–53).

One total on both sides: 12.5663706144 sr as shell curvature before, and the same as the sky the throats divide after. Line defects in all: 131.463205 rad.

They leave zero together: one flux carries both readings, 0.999892; Λ and F locked at −1.0000.

The wave on the architecture: return probability exactly 1.000000 at every multiple of 2π/33 = 0.190400 (t = 1.1424 is the sixth). Entanglement 0 without interaction, 0.162052 bits with it, against a ceiling of $\log_2 33 = 5.0444$.

The turn is the top of the curve: n = 8.25 for the full turn on page 44, step 16.5 of 33 on the half-turn profile of the deck's page 4.

From The Rivero-Zeta Curve, pages 4, 7, 8, 11, 27, 28 and 29.

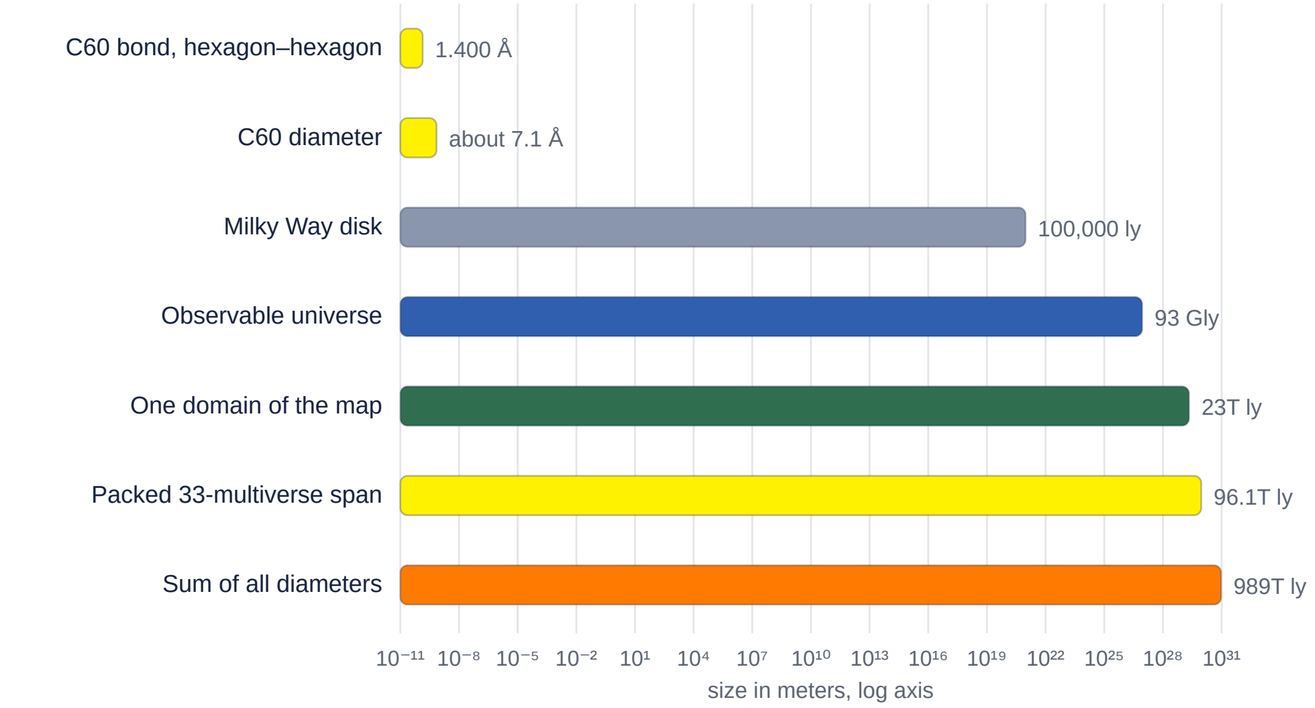
How to read: left to right: the shell before, the throats opening, the flow turning in. Orange throats, yellow energy flow and shell lines, green push, blue hub.

The Same Geometry, Small and Large

MEASURED · COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

SIZES THE ATLAS SPANS



THE SAME AT EVERY SIZE

angles: 60°, 63.4349°, 37.3774°, 41.8103°
deficits 250.973497° and 226.030242°
mouths 12.5663706144 sr; ratios 1.5293717121
counts: 12, 20, 32, 33, 120 loops
line mass per length

WHAT SCALES WITH R

lengths, in proportion to R
areas as R^2, volumes as R^3
light-travel times along the axes
R is a unit of length, not a fitted quantity
the same 12 + 20 shape at both sizes; only R differs

How to read: the same shape and the same exact numbers, read at the size of a molecule and at the size of the multiverse. Only R changes.

WHAT THIS SHOWS

The deck's numbers are pure geometry, so the same shape can be read at any size. Small: the 12/20 partition is built and measured in C60, a molecule 7 Å across, and in carbon nanocones. Large: this atlas, where one domain is 23 trillion light-years across.

THE NUMBERS

Small numbers, the quantum side: bonds 1.400 and 1.450 Å; real C60 moves the shares from 28.1773/71.8227 to 28.9301/71.0699 percent. Nanocone angles only 112.885°, 83.621°, 60.000°, 38.942°, 19.188°.

How exact: the deck's identities hold to 1.7×10^{-13} (arc step), 2.4×10^{-16} (closure), 4.3×10^{-16} (divergence), 1.5×10^{-16} (orbit lock) and 2.2×10^{-17} (block coupling).

Large numbers: volume 2.14×10^{41} ly³; hub to outer 36.55T ly; light time 36.55T years.

Small side from The Rivero-Zeta Curve, pages 15 and 25–31. Large side from this atlas.

Spinning Arrival: pages 50–53

SPINNING ARRIVAL · THE FLAT WALL, TO SCALE

The Spinning Arrival, Mapped in Trillion Light-Years

CURVING · SPINNING

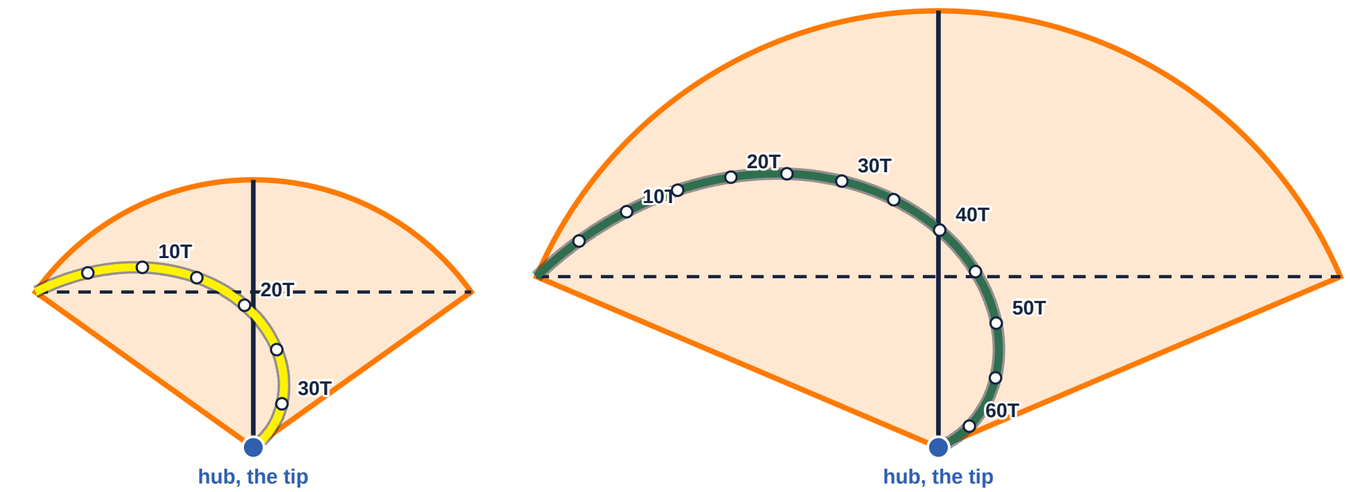
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

apex throat · 12

axis 23.00T · slant 24.13T · sector 109.0265°

curvilinear throat · 20

axis 36.55T · slant 39.38T · sector 133.9698°



straight arrival · spinning arrival, one full turn · straight once around the rim

dots every 5T ly of path length · the sector edges are the same line on the cone

WHAT THIS SHOWS

A throat cut open and laid flat is a sector, drawn here to true scale in trillions of light-years. The wall is flat, so every length on it is exact. The straight arrival runs from the rim to the hub. The spinning arrival falls at an even rate while turning an even amount, one full turn, and is longer.

THE NUMBERS

Apex throat: straight 24.1334T ly · spinning, one turn, **34.8125T ly** (×1.4425) · once around the rim 39.3012T.

Curvilinear throat: straight 39.3827T ly · spinning, one turn, **63.4282T ly** (×1.6106) · once around the rim 72.4958T.

Rule used: slant = axis ÷ cos(half-angle); one turn = 360° = 33 steps of 10.9091° = 32 steps of 11.25°; on the wall a turn φ is a swing of φ·sin α.

Rule from The Rivero-Zeta Curve: flat wall, sector and deficit (p. 2, 12), rotation steps (p. 5, 9). Path lengths are computed here.

How to read: the same throat laid flat. Black is the straight way in; yellow or green is the spinning way in, one full turn. Dots are 5T ly apart along the path.

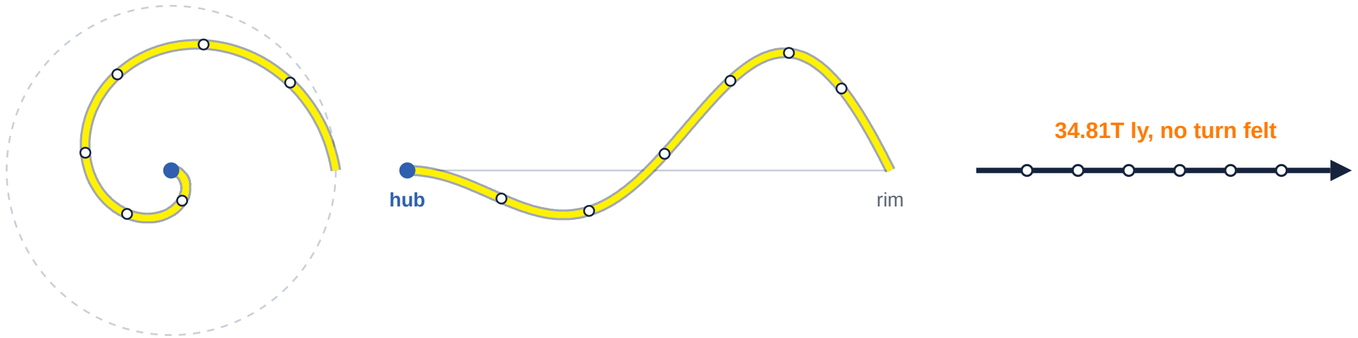
Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

One Arrival, Three Readings

CURVING · SPINNING

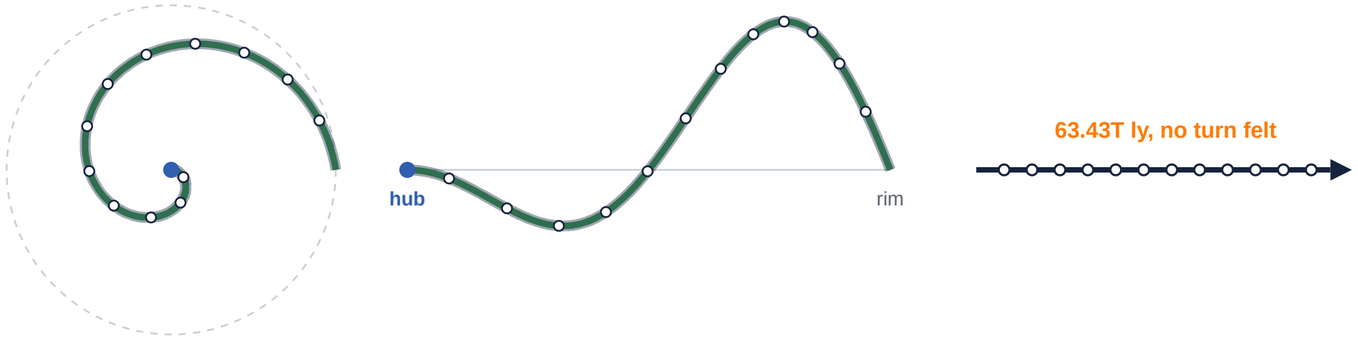
T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

apex throat · one full turn · 34.81T ly



from the front: a spiral | from the side: a fading wave | riding it: a straight line

curvilinear throat · one full turn · 63.43T ly



from the front: a spiral | from the side: a fading wave | riding it: a straight line

How to read: from the front a spiral, from the side a fading wave, riding it a straight line. All three are the same arrival and the same length.

WHAT THIS SHOWS

One arrival, read from three places, as in the deck. From the front it is a spiral. From the side it is a wave that fades to nothing at the hub. Riding it you travel a straight line and feel no turn. Same path, same length, three views.

THE NUMBERS

Apex throat: **34.8125T ly** long, rim radius 7.3088T, axis 23.00T.

Curvilinear throat: **63.4282T ly** long, rim radius 14.6558T, axis 36.55T.

Both are one full turn: 360° = 33 steps of 10.9091°. The side view has exactly one wave, and its envelope shrinks in a straight line toward the hub. At light speed each length is also the travel time in trillions of years.

The three-readings idea is adapted from The Rivero-Zeta Curve, page 3. Lengths are computed here; dots are 5T ly apart. Each row has its own scale.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

SPINNING ARRIVAL · LENGTHS, TURN BY TURN

Arrival Lengths, Turn by Turn

CURVING · SPINNING

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Length of the spinning arrival from the mouth rim to the hub, in trillions of light-years, for each amount of turning · light time at the speed of light

Turn	Angle turned	Apex length	× straight	Light time	Curvilinear length	× straight	Light time
straight in	0.0000°	**24.1334T**	×1.0000	24.13T yr	**39.3827T**	×1.0000	39.38T yr
1 domain step	10.9091°	**24.1467T**	×1.0006	24.15T yr	**39.4156T**	×1.0008	39.42T yr
8.25 steps, a quarter turn	90.0000°	**25.0150T**	×1.0365	25.01T yr	**41.5227T**	×1.0543	41.52T yr
16.5 steps, a half turn	180.0000°	**27.3929T**	×1.1351	27.39T yr	**47.0705T**	×1.1952	47.07T yr
33 steps, one full turn	360.0000°	**34.8125T**	×1.4425	34.81T yr	**63.4282T**	×1.6106	63.43T yr

TWO VALUES, AS THE DECK PREDICTS

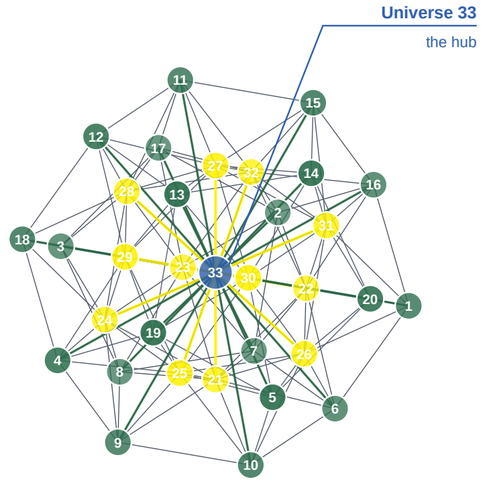
The Rivero-Zeta deck says any quantity the curve governs must take exactly two values on the shell. The spinning arrival does: every apex throat is **34.8125T ly** and every curvilinear throat is **63.4282T ly** for one full turn. A third value would break it.

The rule, in four lines. Slant = axis ÷ cos(half-angle): 24.1334T apex, 39.3827T curvilinear. A turn φ around the axis is a swing of φ·sin α on the flat wall. The path falls evenly while it turns evenly, an Archimedean spiral on the unrolled wall. Its length is $s \cdot [\Theta\sqrt{(1+\Theta^2)} + \operatorname{asinh}\Theta] \div 2\Theta$, with $\Theta = \varphi \cdot \sin\alpha$ in radians.

The deck fixes the step (10.9091° per domain, 11.25° per throat) and the flat wall; it does not say how many steps an arrival takes. The rows show the natural counts, so the choice can be made on the page. In the Geometric Transport file an arrow also costs one chosen parameter, a phase on every link, and no value is selected by the geometry (page 23).

ALL 32 ARRIVALS

12 apex throats share one length and 20 curvilinear throats share the other



12 apex arrivals
20 curvilinear arrivals
Hub · #33
12 inner · #21–32
20 outer · #1–20

How to read: each row is one amount of turning. More turning, longer path. A full turn is 1.44 times the straight way in for an apex throat and 1.61 times for a curvilinear throat.

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

SPINNING ARRIVAL · ALL 32 INTO THE HUB

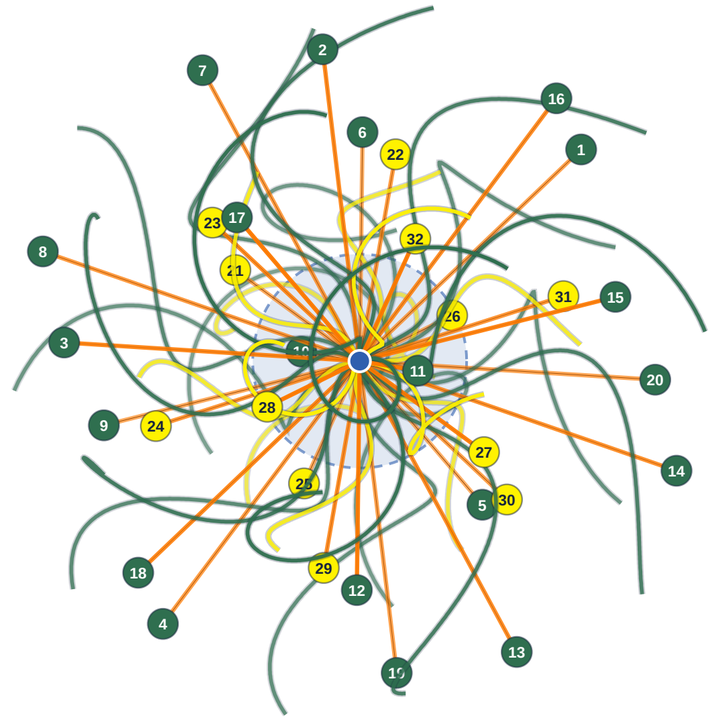
The 32 Spinning Arrivals

CURVING · SPINNING

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

12 apex arrivals: 34.81T ly each
20 curvilinear arrivals: 63.43T ly each

orange: the throat axes
dashed ring: the hub, radius 11.5T
arms start at the mouth rim



10T ly

WHAT THIS SHOWS

All 32 arrivals into the hub at once, drawn to scale in trillions of light-years, each one a full turn. Orange is the throat axis out to its universe. Yellow and green are the energy flowing in, spinning as it falls. The arms start at the mouth rim and meet at the hub.

THE NUMBERS

Two values only: 34.8125T ly for the 12 apex arrivals and 63.4282T ly for the 20 curvilinear ones.

Straight lengths for comparison: 24.1334T and 39.3827T ly. The spin adds 10.68T and 24.05T ly.

Light time: the same numbers in trillions of years. All 32 arrivals turn the same way and the same amount, so the swirl is one pattern, not 32.

View from the same angle as page 48. Lengths from page 52; throat axes from page 24 and 25.

How to read: every universe sends one arrival to the hub. Apex throats (yellow) are shorter, curvilinear throats (green) longer, and each class has one length.

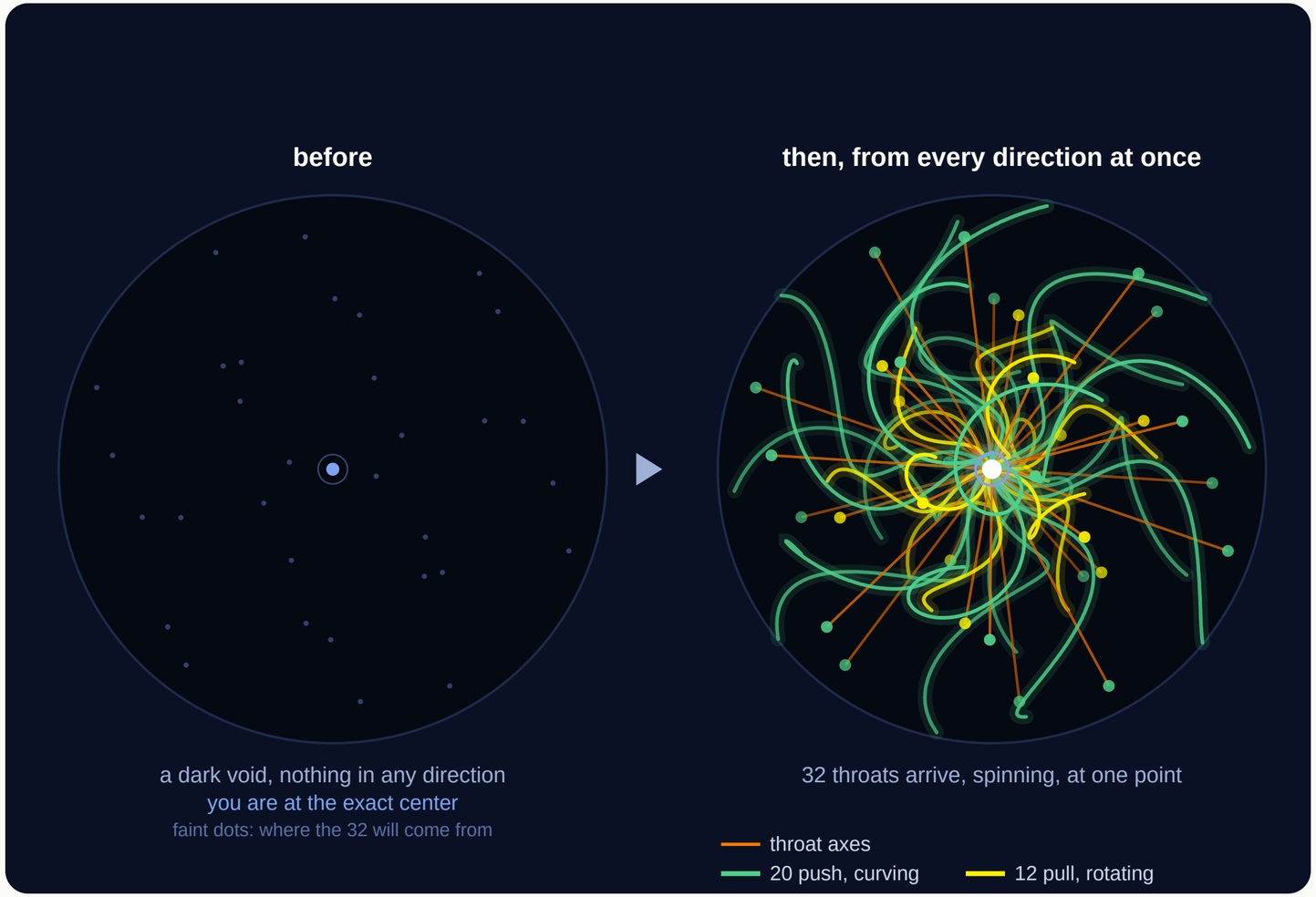
Standing at the Center: pages 54–57

Note The 32-universe shell works on its own. Distances and travel times that end at the hub, our universe #33, apply once the throats connect (Geometric Transport, page 4; 14 Loop Quantum Gravity Bridge, page 7).

Standing at the Center

INCEPTION

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



How to read: left, the void before inception; right, the 32 arrivals meeting at the point where you stand. Green pushes and curves, yellow pulls and rotates.

WHAT THIS SHOWS

You are in a vast empty void, at the exact center of the 32-universe icosahedron. Look in any direction: nothing. Then, from all 32 directions at once, the energy arrives. Twenty come pushing and curving. Twelve come pulling and rotating. All of them spin in step toward one point: you. The burst is where they meet.

THE NUMBERS

The observer is inside the geometry: the hub is not viewed from outside, it is where the observer stands (Cosmic Web Atlas, page 12).

All directions covered: the 32 mouths tile the whole sky, 12 × 0.2951 + 20 × 0.4513 = 12.5664 sr = 4π.

The steps: one turn is 33 steps of 10.9091° or 32 steps of 11.25°; five-fold at each of the 12 (72°), three-fold at each of the 20 (120°).

Lengths on the way in (pages 50–53): 34.8125T ly for the 12 and 63.4282T ly for the 20, one full turn.

The void is not empty of law: with the hub still at zero, the 32 already agree with each other at 0.987 ± 0.003. See page 57.

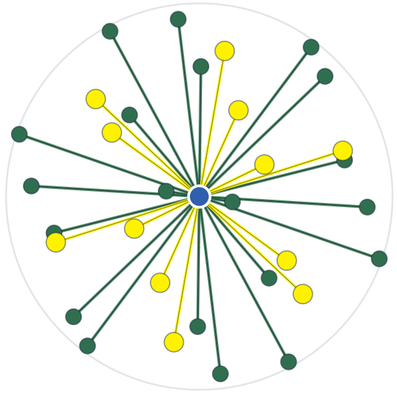
Picture drawn to the same scale as page 53. The 20 push and the 12 pull follow the framework's reading (Rivero-Zeta pages 10 and 14).

What Arrives at Your Point

THEOREM · COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

every arrival has an equal opposite



6 yellow pairs and 10 green pairs
each pair is one line through you

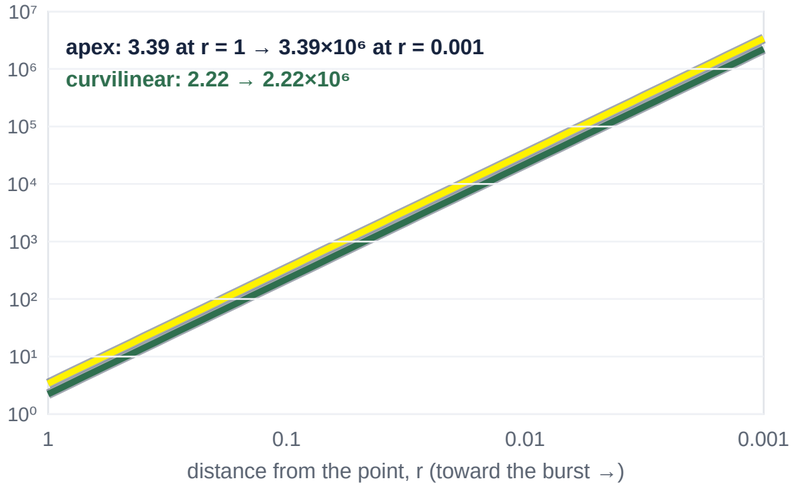
sum of the 12 axes: 2×10⁻¹⁶
sum of the 20 axes: 1.5×10⁻¹⁶

what arrives at your point: the bending of all 32

12 apex · 8.3658 turns	20 curvilinear · 12.5572 turns

total 131.463205 rad = 20.9230 full turns
12 × 4.3803138509 + 20 × 3.9449719321

how hard it hits: flux j = 1/(Ω r²), r in units of R



WHAT THIS SHOWS

Three things arrive at the point. First, a balance: every arrival has an exact opposite, so the pushes and pulls cancel and you are held, not thrown. Second, the bending of all 32 throats, delivered together: 20.92 full turns. Third, a flow that rises as the square of the closeness, a million times stronger at one thousandth of the distance.

THE NUMBERS

In trillions of light-years: r is in units of R. With R = 23.00T ly, r = 1 is 23.00T ly from the point, 0.1 is 2.30T, 0.01 is 0.23T and 0.001 is 0.023T ly. The arrivals start 24.1334T and 39.3827T ly out, along axes of 23.00T and 36.55T ly.

Balance: the 12 apex axes sum to zero (6 opposite pairs) and so do the 20 curvilinear axes (10 pairs), to machine precision.

Bending: 250.973497° at each apex throat and 226.030242° at each curvilinear one. Total 131.463205 rad, with 12 shell defects of 60° (720° = 4π) behind it.

Flow: j = 1/(Ω r²) with Ω = 0.2950722634 sr (apex) and 0.4512751727 sr (curvilinear): ×100 for every tenth of the distance.

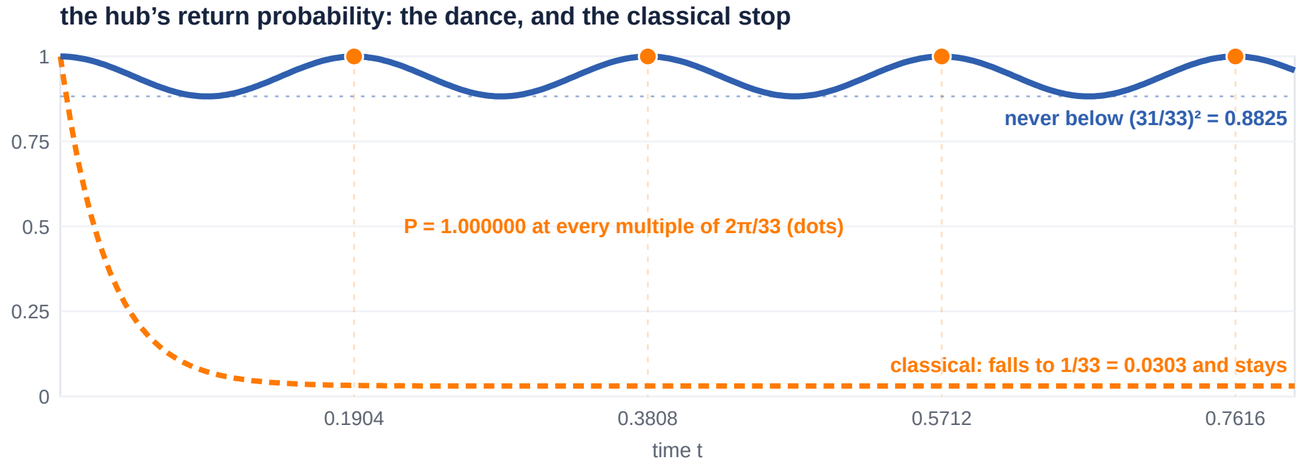
From The Rivero-Zeta Curve, pages 2, 10, 11 and 12. The cancellation was recomputed from the 32 axes.

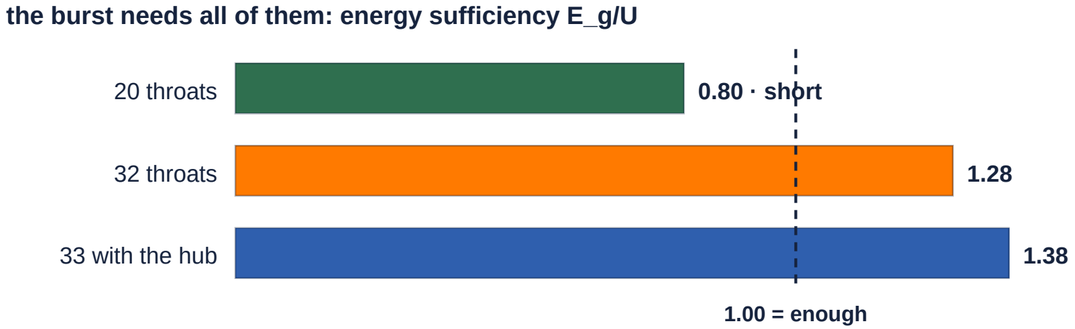
How to read: left, every arrival opposed by an equal one; top right, the total bending; bottom right, how the flow grows as it closes in on the point.

The Beat and the Burst

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years





WHAT THIS SHOWS

Because the hub is joined to all 32, its state swings between two levels, and the swing is exact: it returns completely every 0.1904, and never falls below 0.8825. A classical spread would simply stop at 1/33. The burst itself needs every throat: twenty alone are short of the energy, all 32 clear it, and with the hub it is 1.38.

THE NUMBERS

In trillions of light-years: the throat axes are 23.00T ly (apex) and 36.55T ly (curvilinear), so light takes 23.00T and 36.55T years along them. The beat above runs in the model’s own time unit.

The beat: $P(t) = |1/33 + (32/33)e^{(-33it)}|^2$, eigenvalues 0 and 33, weights 1/33 and 32/33. Period $2\pi/33 = 0.190400$; $t = 1.1424$ is the sixth return. Checked here on the real 33-node graph (152 links): the match is 1.4×10^{-15}.

The spark: one spark from the hub reaches all 32 points at once, with zero delay (page 18).

Energy: $E_g/U = 0.80$ for 20 throats, 1.28 for 32, 1.38 for 33 (GENESYS, page 9).

Beat from The Rivero-Zeta Curve, pages 27–29. Energy sufficiency from the Multiverse Map, page 9 (GENESYS).

How to read: blue is the quantum dance, orange dashed is the classical stop; the bars show that only all 32 together reach enough energy.

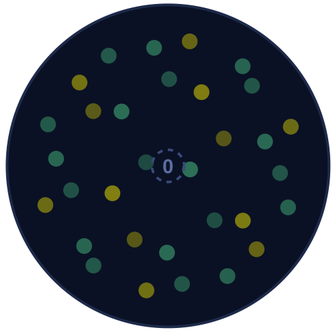
STANDING AT THE CENTER · THE VOID IS LAWFUL

Before the Hub: The Void Is Lawful

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

the void: the hub is still at zero



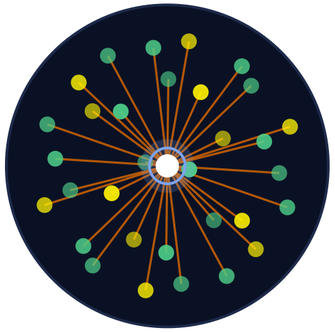
97.4% · 6 of 6

one component · signs · 32 universes, hub at zero

the hub attaches

3 sign flips
Λ~F: −1.0000 → −0.9958

the hub attaches



77.1% · 6 of 6

one component · signs · full 33-node model

THE FRAMEWORK'S LADDER, NOW WITH NUMBERS

1 LAWFULNESS

descriptions of the same thing must agree

97.4%

in one component, six signs of six; 18 points above the strongest null, p < 0.0005

2 ARCHITECTURE

what lawfulness looks like once it takes a shape

AUC 0.9965

geometry recovered from information alone: 114 of 120 edges; 376 of 528 pairs independent

3 INCEPTIONS

the outer zones had theirs first

0.987 → 77.1%

the law holds with the hub at zero (0.987 ± 0.003), and again once the hub attaches

4 QUANTUM AND GRAVITY

two readings of one structure

Λ~F = −1.0000

proved by symmetry: exactly two kinds of node, 12 apex and 20 curvilinear

WHAT THIS SHOWS

Before the hub exists, the 32 universes are not lawless. Their information field and their curvature already agree with each other. When the hub attaches, the law still holds, but it reorients: three signs flip and one lock relaxes. The hub is the one node that gives the structure a degree of freedom.

THE NUMBERS

In trillions of light-years: the hub, where #33 forms, is 23.00T ly across. The 12 apex universes sit 23.00T ly and the 20 curvilinear universes 36.55T ly from its center.

Hub at zero: 97.4% in one component, 6/6 signs, information against curvature 0.987 ± 0.003 (Lawfulness pages 3, 7 and 28).

Hub attached: 77.1% and 6/6 signs on the full 33-node model; Λ~F relaxes from −1.0000 to −0.9958 (page 18).

Why it matters here: the 12 apex and 20 curvilinear classes that give two arrival lengths on pages 50–53 are the same two classes that make Λ and F exact (page 17).

Scope: the ladder is the framework's premise; the numbers are measured, on a discrete operator of 32 and 33 nodes (pages 5 and 25).

From AI33-MPOPT Lawfulness (v12, 28 pages), pages 3, 5, 7, 15, 17, 18, 25 and 28.

How to read: left, the 32 with the hub at zero; right, with the hub attached; below, the four steps of the premise and the measured number behind each.

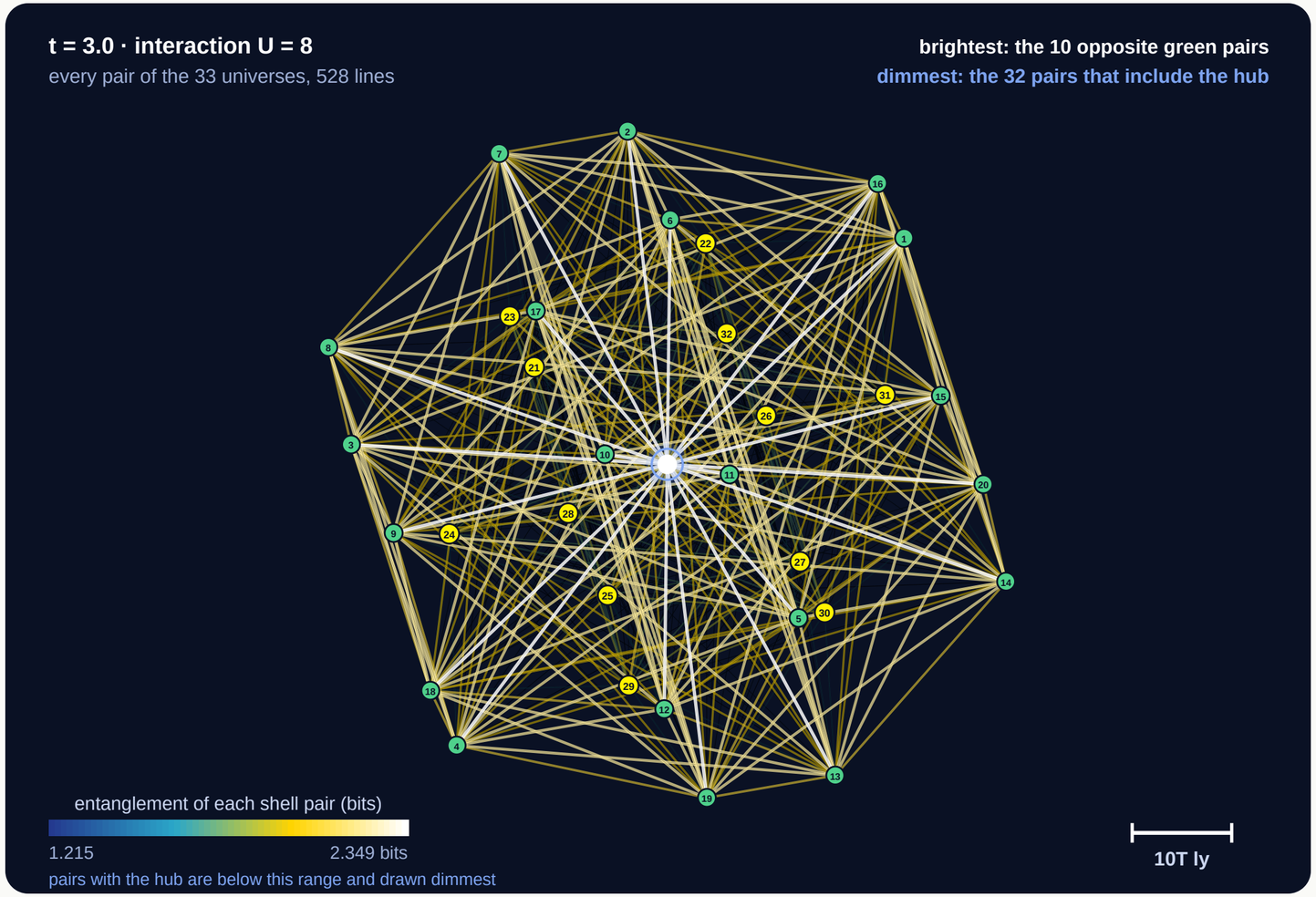
Entanglement Web: pages 58–62

ENTANGLEMENT WEB · ALL 528 PAIRS

The Entanglement Web: All 528 Pairs

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



WHAT THIS SHOWS

All 33 universes and every one of the 528 pairs between them, drawn on the map's own layout. The brighter and thicker the line, the more entangled the pair. This is the late view, t = 3, when the farthest pairs, the opposite ones, are among the most entangled.

THE NUMBERS

In trillions of light-years: the lines run on the map's own layout: 23.00T ly from the hub to each apex universe and 36.55T ly to each green one. The longest pair, opposite green universes, is 73.108T ly apart; the shortest is 23.000T ly. The scale bar is 10T ly.

The measure: two particles, one started on each universe, interacting at strength U = 8 on the 33-node graph; the entropy of one of them, in bits. Ceiling $\log_2 33 = 5.0444$.

At t = 3: from 0.152 bits (hub pairs) to 2.349 bits (opposite green pairs).

528 pairs, 14 classes: every pair in a class has the same value to 10^{-15}, so the whole web is 14 numbers.

Measure from The Rivero-Zeta Curve, page 28. Rebuilt here: it reproduces the deck's four values to six digits (0.006982, 0.050657, 0.020456, 0.162052).

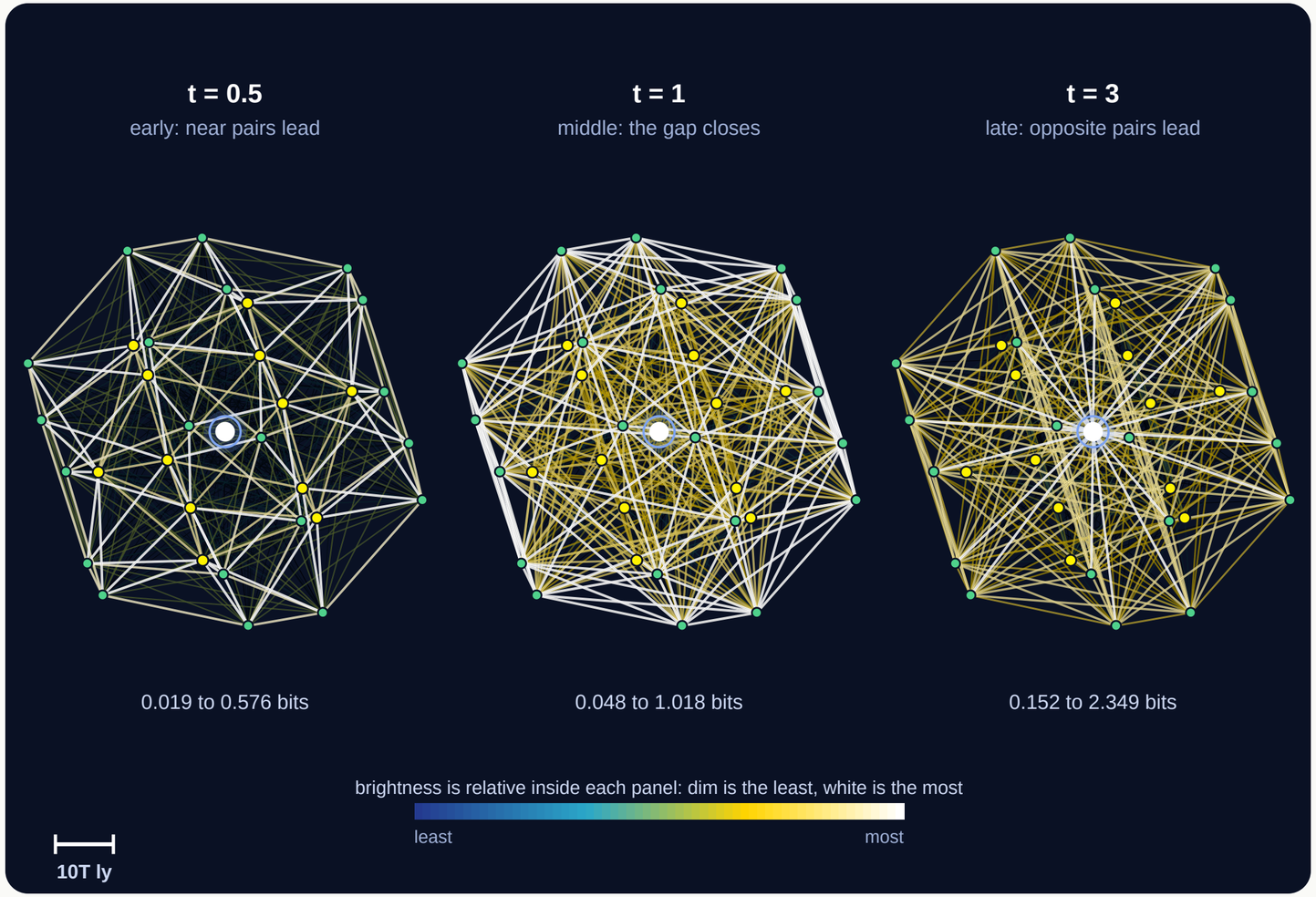
How to read: each line joins two universes; brightness is how entangled the pair is. The hub is the blue point in the middle, apex universes yellow, green universes green.

ENTANGLEMENT WEB · OVER TIME

The Web Over Time

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



How to read: left to right: early, middle, late. Early the near lines are brightest; late the lines between opposite universes catch up and pass them.

WHAT THIS SHOWS

The same web at three moments. Early on, the nearest pairs are the most entangled, as in any ordinary system, and the opposite pairs are the faintest lines. By t = 3 the order has turned: the opposite pairs in the shell are among the brightest.

THE NUMBERS

In trillions of light-years: the same layout as page 58: 23.00T and 36.55T ly from the hub; the longest pair is 73.108T ly. The scale bar is 10T ly.

Brightest pair: 0.576 bits at t = 0.5, 1.018 at t = 1 and 2.349 at t = 3. Entanglement grows as the walk spreads.

Dimmest pair: always one that includes the hub: 0.019, 0.048 and 0.152 bits.

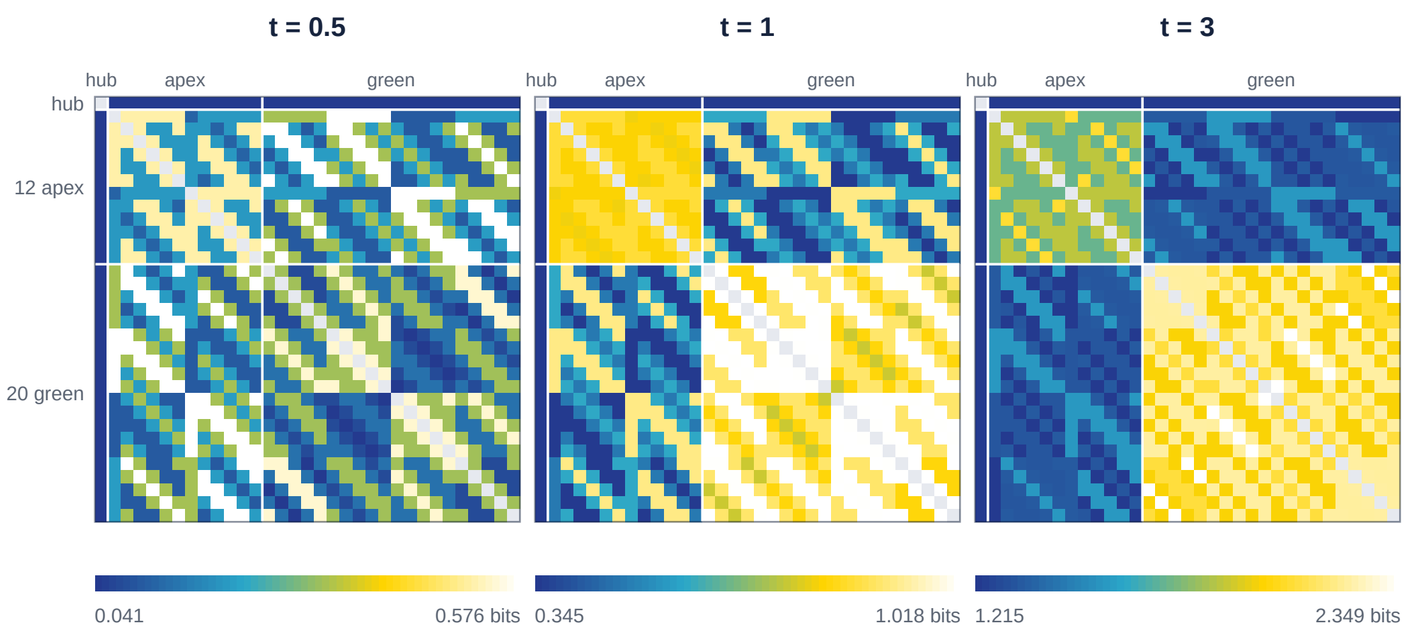
Each panel is scaled to itself, so compare the pattern, not the colour.

Same measure as page 58, U = 8, at t = 0.5, 1.0 and 3.0 in the deck's own time unit.

The Same Web as a Matrix

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



WHAT THIS SHOWS

The same web as a table: one row and one column per universe, ordered hub, the 12 apex universes, then the 20 green ones. Each square is one pair. Four blocks show at a glance who is entangled with whom. The hub's row and column are always the darkest.

THE NUMBERS

Blocks: hub–shell, apex–apex, apex–green, green–green. Inside each block the squares fall into the angle classes on page 61.

Late (t = 3): the brightest block is green–green; the darkest is everything touching the hub.

The diagonal (a universe with itself) is left grey; it is not a pair.

528 pairs, 33 × 33, symmetric. Each panel is scaled to its own shell range, shown under it; pairs with the hub fall below it and show darkest.

How to read: each square is one pair of universes. Rows and columns run hub, apex 21–32, green 1–20. Brighter means more entangled.

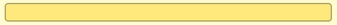
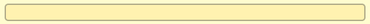
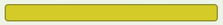
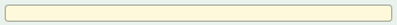
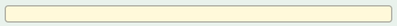
ENTANGLEMENT WEB · 14 PAIR CLASSES

The 14 Pair Classes, With Distances

528 PAIRS · 14 CLASSES

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

Every pair of universes belongs to one of 14 classes. Distance and light time are the straight center-to-center values from pages 24–30; entanglement in bits, U = 8, in the deck's time unit

Pair class	Pairs	Angle at hub	Distance	Light time	t = 0.5	t = 1.0	t = 3.0	at t = 3, relative
hub–apex	12	—	23.000T	23.00T yr	0.0186	0.0507	**0.1621**	
hub–green	20	—	36.554T	36.55T yr	0.0231	0.0482	**0.1522**	
apex–apex	30	63.43°	24.184T	24.18T yr	0.5181	0.8495	**1.8709**	
apex–apex	30	116.57°	39.130T	39.13T yr	0.2034	0.7990	**1.7213**	
apex–apex	6	180.00°	46.000T	46.00T yr	0.1118	0.7875	**2.0588**	
apex–green	60	37.38°	23.000T	23.00T yr	0.5758	0.9144	**1.5567**	
apex–green	60	79.19°	39.367T	39.37T yr	0.3306	0.5855	**1.3305**	
apex–green	60	100.81°	46.697T	46.70T yr	0.2096	0.4822	**1.2155**	
apex–green	60	142.62°	56.581T	56.58T yr	0.0968	0.3453	**1.3182**	
green–green	30	41.81°	26.087T	26.09T yr	0.5441	1.0157	**2.0734**	
green–green	60	70.53°	42.209T	42.21T yr	0.3266	1.0175	**2.2152**	
green–green	60	109.47°	59.693T	59.69T yr	0.1359	0.8915	**1.9770**	
green–green	30	138.19°	68.296T	68.30T yr	0.0706	0.8106	**2.2117**	
green–green	10	180.00°	73.108T	73.11T yr	0.0413	0.7481	**2.3491**	

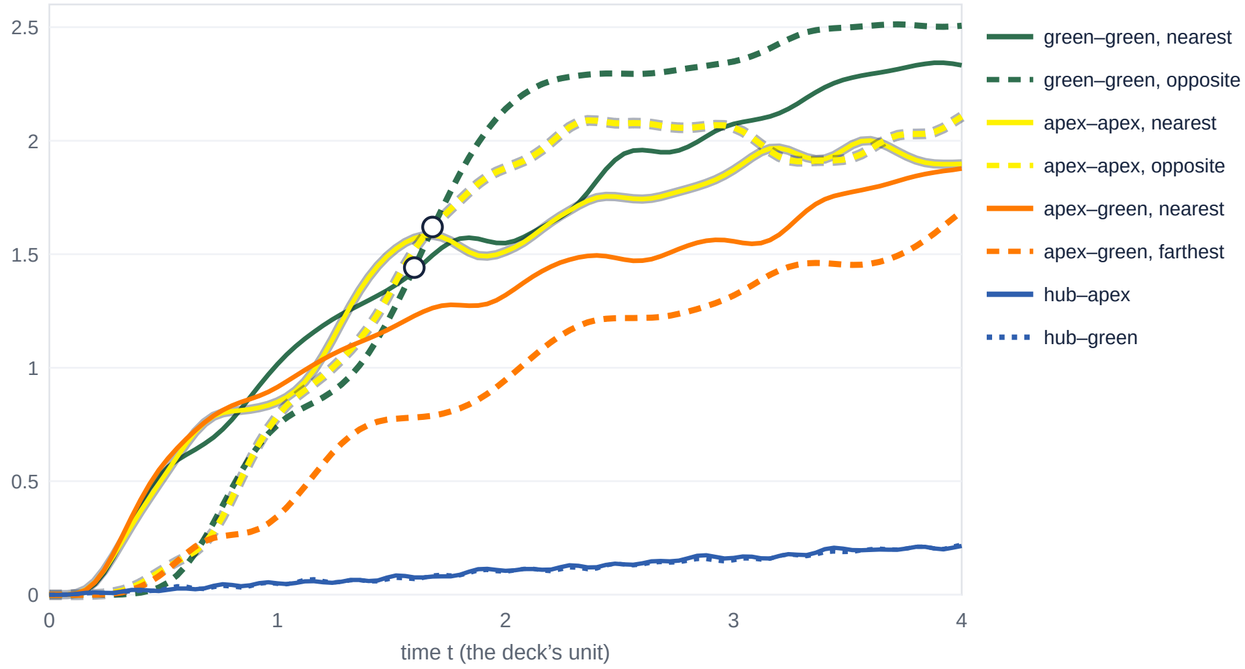
How to read: one row per class. Pairs inside a class are identical, so the web has 14 numbers. The same 14 classes are the 14 distances of page 26.

Near and Far: The Reversal

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

pair entropy over time: the nearest and the farthest pair of each kind



white dots: the moment the opposite pair first passes the nearest pair

green–green: t ≈ 1.60 · apex–apex: t ≈ 1.68 · apex–green never reverses in this window

WHAT THIS SHOWS

Pair entropy over time for the nearest and the farthest pair of each kind. Early on, the nearest pairs lead. From about t = 1.6 the opposite pairs overtake them, for green–green and for apex–apex. Apex–green pairs do not reverse.

THE NUMBERS

Crossing: green–green at t ≈ 1.60, apex–apex at t ≈ 1.68.

At t = 3: opposite green pairs 2.349 bits against 2.073 for the nearest; opposite apex pairs 2.059 against 1.871; apex–green farthest 1.318 against 1.557 nearest.

The archive result: 0.072 at the farthest against 0.064 at the nearest, on 10 samples (Super Formula Build-Up, pages 3 and 12). The 10 farthest pairs in the shell are the 10 opposite green pairs. Here all 528 pairs are measured.

Control: in 20 random networks with the same hub and about 120 links (8 plain, 12 with an exact opposite for every universe) the farthest pairs never overtake the nearest at t = 3. In the icosahedral structure they do.

Measure: the Rivero-Zeta two-particle test (page 28), interaction U = 8, time in the deck's own unit.

Measure: Rivero-Zeta page 28. Archive result: Super Formula Build-Up, pages 3 and 12.

How to read: solid lines are the nearest pair of each kind, dashed lines the farthest. Early the near pairs lead; after about t = 1.6 the opposite pairs pass them, except apex–green.

Framework Metric: pages 63–66

FRAMEWORK METRIC · ONE SPHERE

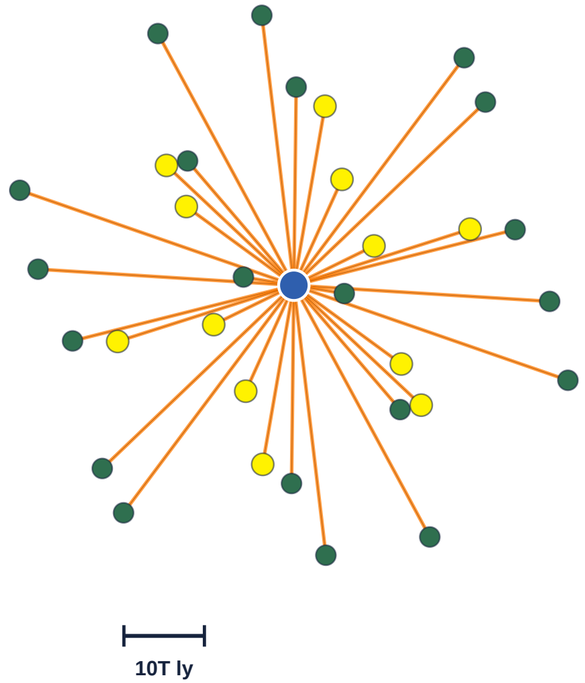
The Framework’s Own Metric: All 32 on One Sphere

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

our layout: touching spheres

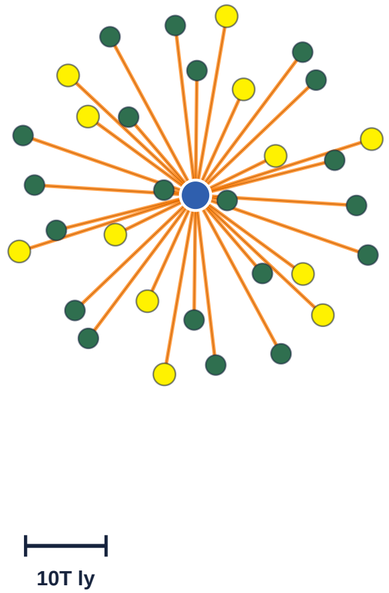
12 apex at 23.00T ly, 20 green at 36.55T ly



throats 25.85% · apex 36.34% · green 37.81%
share of total metric length

the framework’s layout: one sphere

all 32 at the same distance (shown for R = 23.00T ly)



throats 25.93% · apex 41.14% · green 32.93%
share of total metric length

How to read: left, our touching-sphere layout; right, the framework’s one-sphere layout. Same directions, different distances for the 20 green universes.

WHAT THIS SHOWS

Two ways to place the same 33 universes. The directions from the hub are identical in both. They differ only in how far out the 20 green universes sit. In our layout the 12 apex universes touch the hub, each green universe touches its three apex universes, and every green universe reaches the hub through its orange curvilinear. In the framework’s own metric, Geometric Transport page 22, all 32 sit on one sphere, so every throat has the same length.

THE NUMBERS

Throats: ours 23.00T ly (apex) and 36.55T ly (green); one sphere 23.00T ly for all 32 at R = 23.00T ly.

Surface links: apex–apex 24.18T in both; apex–green 23.00T ours, 14.74T one sphere; green–green 26.09T ours, 16.41T one sphere.

Metric length, one sphere: total 123.40 at unit circumradius: apex 50.77 (41.14%), green 40.63 (32.93%), throats 32.00 (25.93%), reproduced here.

Conductance to the hub, apex over green: 1.4298 on the graph (the files quote 1.43); 1.2690 with one-sphere lengths; 1.1662 with the green universes at true face centers (page 22: 1.2690 and 1.1662). With a uniform shift of 0.5 added to the Laplacian, as in Geometric Transport page 13, the graph ratio is 1.3963, the value on its pages 12, 13 and 22.

Pages 20–62 use our layout. The framework’s results were computed on the one-sphere metric.

Geometric Transport Architecture, pages 6 and 22. All values recomputed from the icosahedron here.

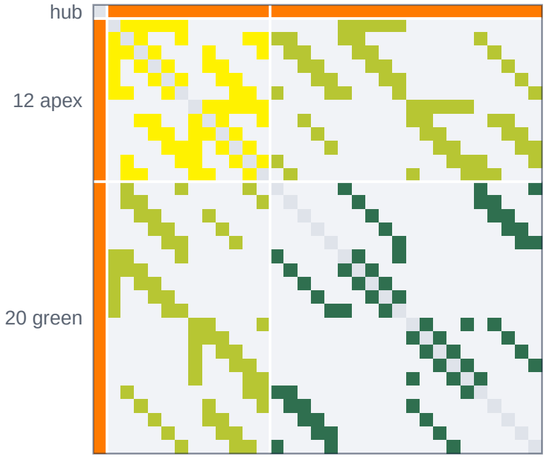
FRAMEWORK METRIC · THE LINK CENSUS

Who Talks to Whom: The Link Census

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

who is linked to whom: 152 links



12 apex universes
degree 11
5 apex + 5 green + 1 throat

20 green universes
degree 7
3 apex + 3 green + 1 throat

1 hub, #33
degree 32
12 apex + 20 green throats

THE 152 LINKS, WITH THEIR LENGTHS IN TRILLIONS OF LIGHT-YEARS

link	count	color	our layout	one sphere, R = 23.00T
throats: hub to every universe	32 (12 + 20)		23.00T · 36.55T	23.00T
apex–apex	30		24.18T	24.18T
apex–green	60		23.00T	14.74T
green–green	30		26.09T	16.41T

32 + 30 + 60 + 30 = 152 links; the 120 surface links are 79% and the 32 throats are 21%

WHAT THIS SHOWS

Every universe's links, counted. A link is a connection in the framework's wiring, not necessarily two spheres touching. An apex universe is linked to 5 apex universes, 5 green ones and the hub through its throat: 11 in all. A green universe is linked to 3 apex, 3 green and the hub through its throat: 7. The hub is linked to all 32: the 12 apex directly, the 20 green through their curvilinears. No count was chosen: every one follows from the icosahedron.

THE NUMBERS

152 links: 30 apex–apex, 60 apex–green, 30 green–green and 32 throats. The matrix shows each one as a colored square.

Links are not contacts: in our layout two neighboring apex universes are 1.18T ly apart and two neighboring green ones 3.09T ly, with a gap between their surfaces, so those 60 links are relations in the wiring. What physically touches is on page 63.

Connectivity ratio: 11 ÷ 7 = 1.5714. An apex is a busier junction than a green universe, by construction.

The hub: degree 32 against 11 for the next highest. It is the one point the structure converges on.

Why 33: two sectors, of exactly 12 and 20, exist at 33 domains and at no other size. At 123 domains there are four sectors and at 273 there are six (Geometric Transport, page 9).

Geometric Transport Architecture, pages 4, 6 and 9; Cosmic Web Atlas, page 2. Counts and lengths recomputed here.

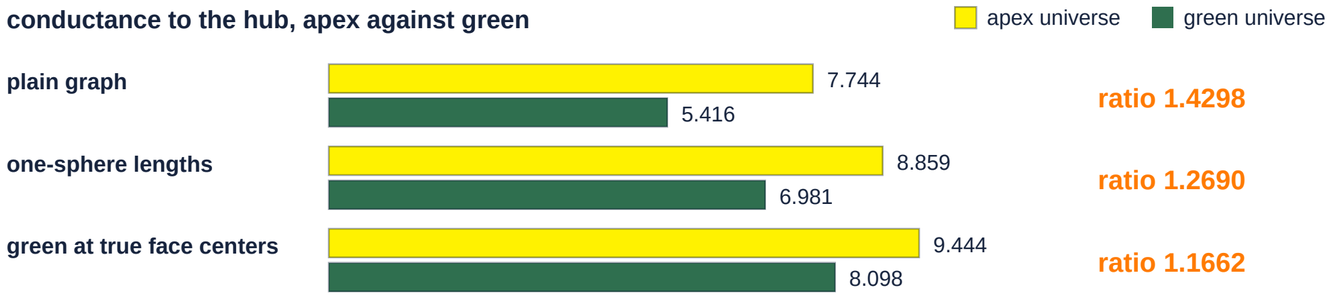
How to read: each colored square is a link: orange throats, yellow apex–apex, olive apex–green, green green–green. Lengths in trillions of light-years are on the left of the table.

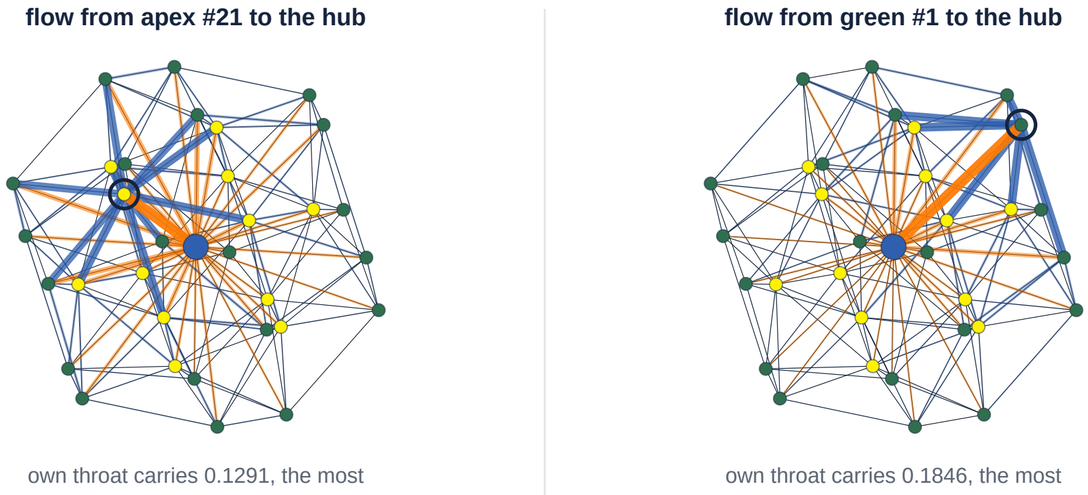
FRAMEWORK METRIC · THE FLOW

How the Flow Divides: What Each Throat Delivers

COMPUTED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years





own throat carries 0.1291, the most

own throat carries 0.1846, the most

line thickness is the current on each link; orange is the throat, blue the surface links

How to read: left top, conductance to the hub by layout; below, where the current flows from an apex source and from a green source. Thicker is more current.

WHAT THIS SHOWS

Every throat has the same coupling, yet an apex universe delivers more to the hub than a green one, because an apex has 10 surface links and a green one 6. Position decides, not construction. The flow also stays near its source: the busiest wire is always the source's own throat.

THE NUMBERS

Apex over green: 1.4298 on the graph, matching the files' 9.487 ÷ 6.635 = 1.4298; 1.2690 with one-sphere lengths and 1.1662 with face centers, matching 1.2690 and 1.1662 (Geometric Transport, page 22). The same graph ratio with a uniform shift of 0.5 added to the Laplacian is 1.3963 (page 13); both are the same apex-over-green conductance, at two settings of the operator.

Busiest wire: for a unit of current from any universe to the hub, the largest current is on its own throat and equals its effective resistance to the hub: 0.1291 for an apex, 0.1846 for a green universe. Checked on all 32.

Carried from the files, not recomputed here: current falls by 0.40 per hop (decay length 1.1 hops); a single edge carries at most 2.879% (apex) or 4.877% (green) of a source's flow (Geometric Transport page 26; Continuous Throat Feed page 22).

Geometric Transport Architecture, pages 4, 13, 22 and 26; Continuous Throat Feed, page 22.

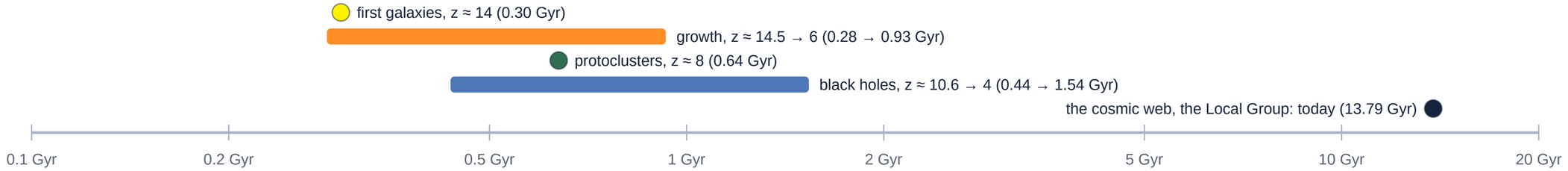
FRAMEWORK METRIC · THE SCALE LADDER

From Inception to the Local Group: The Scale Ladder

MEASURED · CARRIED

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years

WHEN EACH STEP HAPPENS: cosmic age on a log axis, in billions of years (1 Gyr = 0.001T years)



inception: the throats open. Below the coupling threshold nothing passes, so no age is assigned to it.

Scale	What the architecture says	What is measured	Result	Cosmic age
Inception	the throats open; below a threshold nothing passes	the operator is singular at zero coupling, regular above it	computed	—
Inception	the density through the bounce is $\sin^2(\omega t)$	$\sin^2(\omega t)$, max deviation 0.00000075; $k = 4\omega^2$	computed	—
First galaxies, z ≈ 14	the feed supplies mass continuously	10^{39} W per galaxy · 0.1 to 1.4 $M_\odot$ per year	close	0.30 Gyr
Growth, z ≈ 14.5 → 6	delivery scales with the structure fed	one number, η = 0.053%, across every epoch	close	0.28 → 0.93 Gyr
Protoclusters, z ≈ 8	the densest regions draw hardest	two protoclusters inside the predicted stellar mass	2 of 2	0.64 Gyr
Black holes, z ≈ 4 to 10.6	the centre forms first, the galaxy grows around it	fourteen JWST black holes inside the ceiling	14 of 14	0.44 → 1.54 Gyr
The cosmic web, today	a transport network with two node classes	clustering 0.639 against 0.613 observed	close	13.8 Gyr
The Local Group	the feed runs to the present day	Milky Way within 7%, Andromeda within 1.4×	close	13.8 Gyr

In trillions of light-years: the universe is 13.8 Gyr = 0.0138T years old and the observable universe is about 93 Gly = 0.093T ly across (standard values). The hub of the map alone is 23.00T ly across.

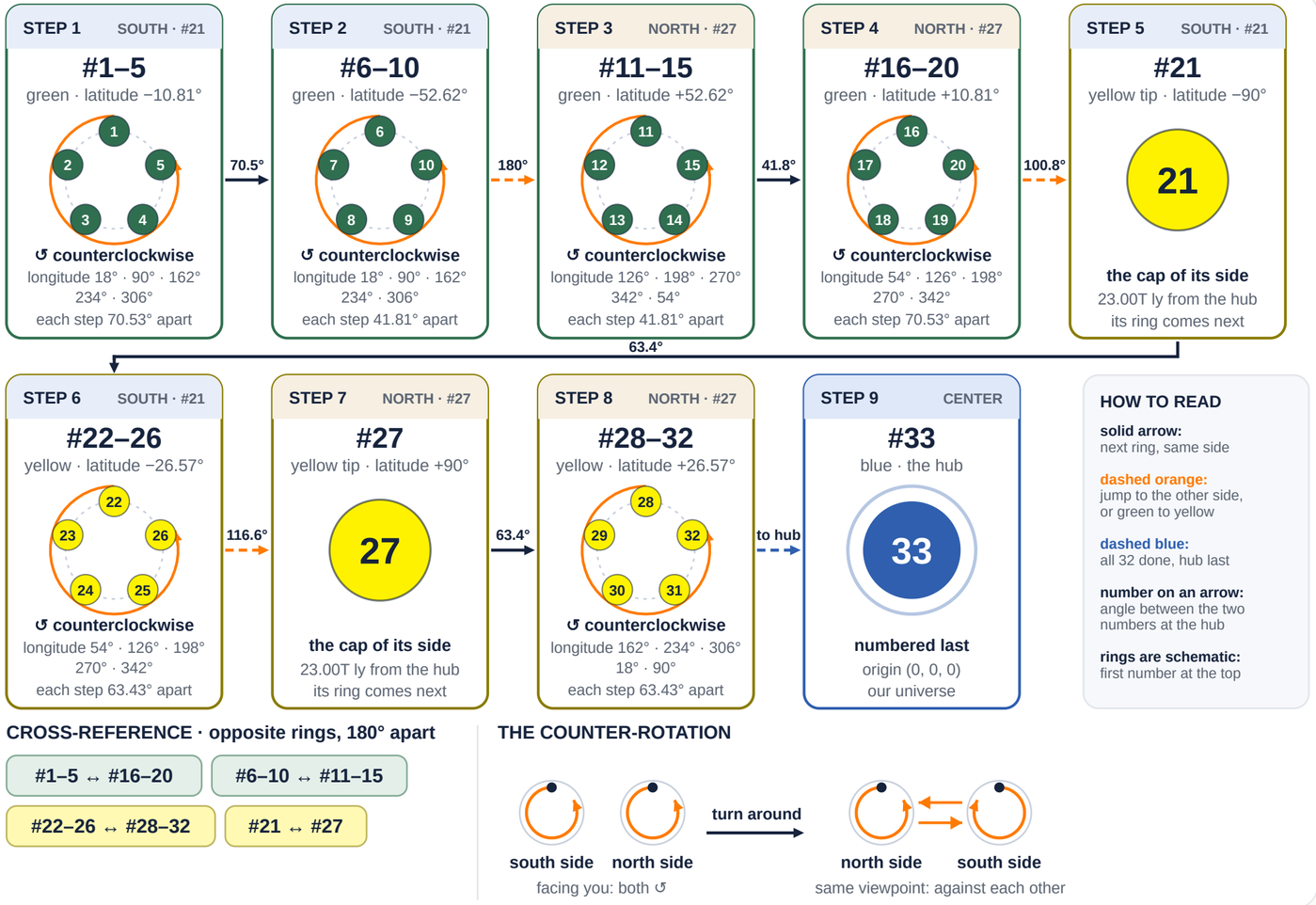
How to read: each row is a separate measurement from the companion files, in the order the structure grows. Cosmic ages are for a flat ΛCDM universe with Planck 2018 values, computed here.

NUMBERING ORDER · FLOW CHART

How the 33 Are Numbered, and Why

ORDER · COUNTER-DIRECTION

T = trillion · 1T ly = 1,000,000,000,000 light-years
ly = light-year · at light speed, 1T ly takes 1T years



DESCRIPTION

The 33 are numbered from the outside in. The 20 outer green universes come first (#1–20), the 12 inner yellow next (#21–32), and our universe, the hub, last (#33). On each side the numbers go around a ring, then on to the next ring; the order then crosses to the other side. The two yellow tips, #21 and #27, cap the two sides: #27 is the north pole and #21 the south pole. In the framework's premise the outer universes had their inceptions first and ours came later (Lawfulness, page 5), which is why ours is #33 and not #1.

THE NUMBERS

Every ring steps 72° (360° ÷ 5). Latitudes: greens −10.81°, −52.62°, +52.62°, +10.81°; yellow tips −90° and +90°; yellow rings −26.57° and +26.57°. Each card lists its longitudes in numbering order, as on pages 24 and 25.

Face values: these are measured facing each side. Pull a side toward you, or walk round and face it from behind, and you measure the same kind of values, counterclockwise. Only someone watching from away sees the two sides turn against each other, like two magnets. In the framework's picture, this opposition is what lets the energy all travel inward. It is stated here as a premise, not a measurement; the deck's results do not depend on it.

Mirror check (computed): mirroring the south-side labels changes none of the inception results. 120 symmetries, 3 groups, apex over green conductance 1.429836 and lowest hub return 0.882462 stay identical. Only the number-by-number lists change, for 25 universes.

Opposite numbers, 180° apart: 1↔18, 2↔19, 3↔20, 4↔16, 5↔17, 6↔12, 7↔13, 8↔14, 9↔15, 10↔11; 21↔27, 22↔29, 23↔30, 24↔31, 25↔32, 26↔28.

In this model, green, the outer 20, is dark energy pushing inward through the orange curvilinears toward the blue hub; yellow, the inner 12, is dark matter pulling inward.

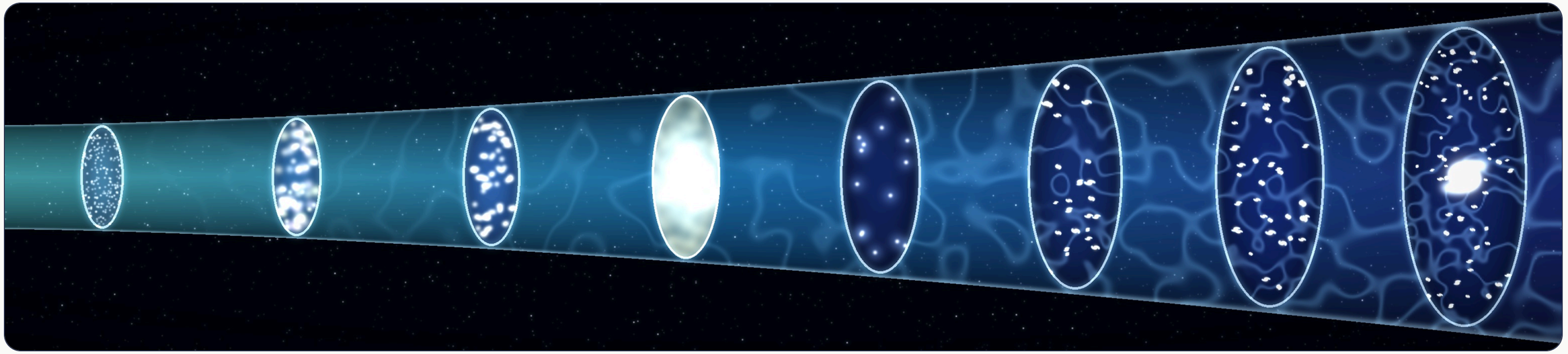
Latitudes, longitudes and angles from the deck's coordinates (pages 24 and 25). Direction of the numbering from the physical model.

How to read: follow the nine steps from #1 to #33. Each ring is drawn as it faces you: first number at the top, counting counterclockwise; longitudes are the face values from pages 24 and 25.

Viewpoint: Every value here is measured from one stated viewpoint: the hub at the origin, the north pole toward #27. Measure from another viewpoint and the directions and signs change; the distances, angles and structure do not.

The History of the Universe, With the AI33-MPOPT Numbers

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly



Inflation
Initial expansion

10^{-36} seconds
no AI33-MPOPT time scale

computed

The crossing: singular at zero throat coupling, regular above w ≈ 0.15. No seconds are assigned. The seen and unseen halves each have radius 0.707107; their product is ½, the uncertainty minimum (page 73).

First Particles
Neutrons, protons and electrons form

1 microsecond

framework premise

The strong force binds quarks into the first protons and neutrons. AI33-MPOPT assigns it to the outer 20, the green zone (page 14). No computed value yet.

First Nuclei
Helium and hydrogen form

3 minutes

framework premise

The weak force sets the neutron-to-proton balance before helium forms. AI33-MPOPT assigns it to the inner 12, the yellow zone (page 14). No computed value yet.

First Light
The first atoms form

380,000 years
0.371 Myr with Planck 2018 values

fitted jointly

In the model, the feed acts only after z = 20, so this light is untouched. A joint fit to Planck, DESI and Pantheon+ data allows up to 1.65% of cosmic matter.

First Stars
Gas and dust condense into stars

200 million years
z = 20 is 178 Myr

fixed in advance

The feed switches on at z = 20 = 178 Myr, fixed before any data were loaded.

Galaxies & Dark Matter
Galaxies form in dark matter cradles

400 million years
z ≈ 11 is 434 Myr

compared

In the model, the 12 apex throats are read as dark matter. The first galaxies need 10^{39} W each; 14 of 14 black holes sit inside the ceiling.

Dark Energy
Expansion accelerates

10 billion years
z = 0.33; it starts near z = 0.63

−0.72σ

In the model, the 20 curvilinear throats are read as dark energy. Occupation 2.5490, compared with Planck's published dark energy ÷ dark matter, 2.5893 ± 0.056.

one of two inside the window among about 45 quantities tried (Geometric Transport, p. 19)

Today
Humans observe the universe

13.8 billion years
13.79 Gyr

compared

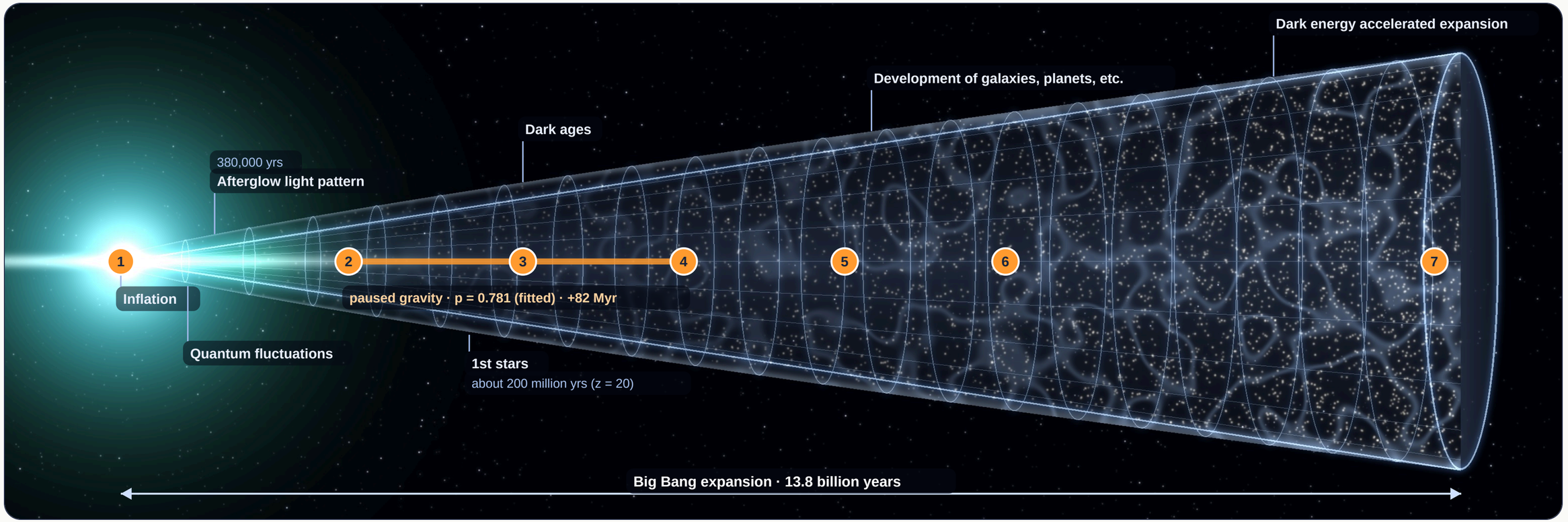
Milky Way: 6.4 vs 6 × 10^{10} M☉, within 7%. Clustering: 0.639 vs 0.613.

How to read: stage names and times follow the familiar history-of-the-universe diagram; the AI33-MPOPT numbers sit beneath each stage. Redshift is converted to age with the standard flat ΛCDM relation (Planck 2018 values) as a ruler. Illustration.

THE LIGHT CONE · AFTERGLOW TO TODAY

The Light Cone, With the AI33-MPOPT Marks

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly



1 Spark
returns every 2π/33 · halves 0.707107 each, product ½
computed

2 Feed on
z = 20 · 178 Myr
fixed in advance

3 First galaxies
290 Myr · 10^{39} W each
computed

4 Black holes
14 of 14 inside
compared

5 Protoclusters
2 of 2 inside
predicted

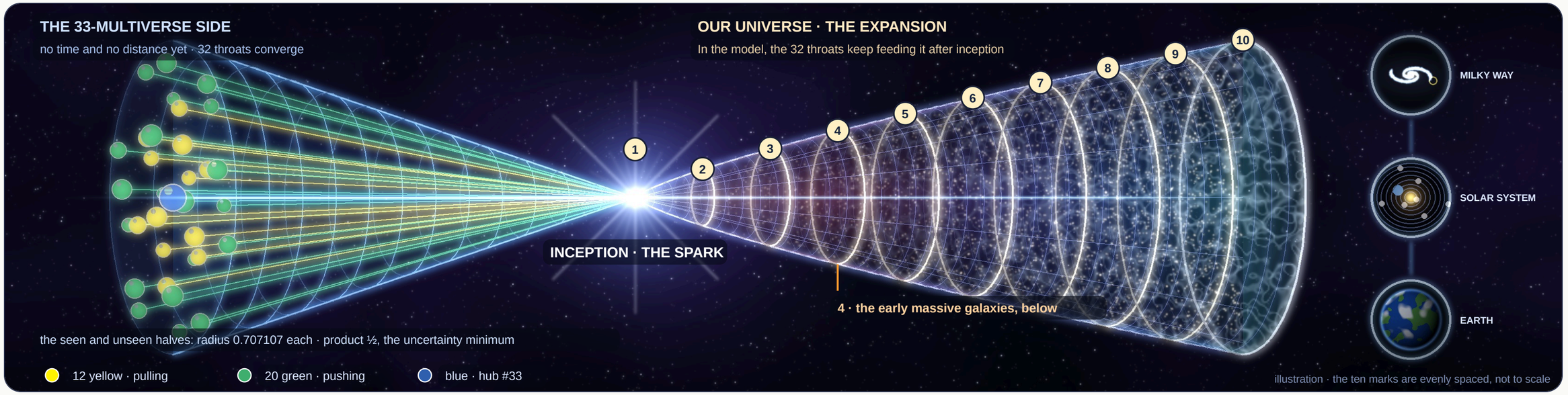
6 Star-formation record
1.00–1.05, z 9 → 6
compared

7 Today
Milky Way within 7%
compared

How to read: white labels follow the familiar light-cone timeline; orange numbers are the AI33-MPOPT marks, evenly spaced and not to scale. Illustration.

From the Inception Spark to Earth

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly



1	2	3	4	5	6	7	8	9	10
Inception	Light released	Cosmic dawn	First galaxies	Black holes first	Protoclusters	Reionization done	Cosmic noon	Solar System, Earth	Today: Milky Way

4 · THE EARLY MASSIVE GALAXIES JWST FOUND, AND THE POWER THE 32 THROATS CARRY TO THEM

COMPUTED

JADES-GS-z14-0
z 14.18 · 288 Myr old · 4.0 × 10^{8} M☉ built
7.8 × 10^{39} W
each of the 32 throats: **2.4 × 10^{38} W**
1.38 M☉ a year
GOODS-South · 33.7 Gly away
RA 53.0829° · Dec −27.8556°

MoM-z14
z 14.44 · 281 Myr old · 1.3 × 10^{8} M☉ built
2.5 × 10^{39} W
each of the 32 throats: **7.9 × 10^{37} W**
0.45 M☉ a year
COSMOS · 33.8 Gly away

JADES-GS-z14-1
z 13.86 · 298 Myr old · 3.7 × 10^{7} M☉ built
7.1 × 10^{38} W
each of the 32 throats: **2.2 × 10^{37} W**
0.12 M☉ a year
GOODS-South · 33.6 Gly away
RA 53.0743° · Dec −27.8859°

GHZ2
z 12.34 · 350 Myr old · 1.1 × 10^{9} M☉ built
1.8 × 10^{40} W
each of the 32 throats: **5.7 × 10^{38} W**
3.2 M☉ a year
Abell 2744 · 32.8 Gly away
RA 3.4990° · Dec −30.3247°

GN-z11
z 10.6 · 432 Myr old · 1.3 × 10^{9} M☉ built
1.7 × 10^{40} W
each of the 32 throats: **5.2 × 10^{38} W**
2.9 M☉ a year
GOODS-North · 31.8 Gly away
RA 189.1061° · Dec +62.2421°

RUBIES-UDS-QG-z7
z 7.29 · 716 Myr old · 1.7 × 10^{10} M☉ built
1.3 × 10^{41} W
each of the 32 throats: **4.2 × 10^{39} W**
24 M☉ a year
UDS · 29.0 Gly away

How the throats feed them. In the model, the 32 throats deliver matter-energy continuously, every one at the same strength, with no shut-off. Power needed = stellar mass × c^2 ÷ the galaxy's age; each throat carries one thirty-second of it. For JADES-GS-z14-0 that is 7.8 × 10^{39} W, 12 apex throats carrying 2.9 × 10^{39} W and 20 curvilinear throats 4.9 × 10^{39} W. In the model, the throats' ceiling (from a joint fit to public Planck, DESI and Pantheon+ data) is about **660 times** what these galaxies need. Sky positions are J2000; distance is comoving, from flat ΛCDM, computed here.

The Ten Marks, Side by Side

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

#	Mark	Age · redshift	What the public record shows	What AI33-MPOPT gives	Do they match?	Kind
1	**Inception**	0 —	Nothing is measured at t = 0. The uncertainty principle: Δx · Δp ≥ ħ/2, which is ½ when ħ = 1	32 throats converge on one point; the hub returns exactly every 2π/33. The seen and unseen halves each have radius 0.707107; product ½, the uncertainty minimum (page 73)	**Equals the uncertainty minimum, ½, exactly.** Nothing measured at t = 0; the link is the framework's reading.	computed
2	**Light released (CMB)**	0.371 Myr z 1090	Planck 2018: light released at z = 1090. Its patches, about 1 part in 100,000 (scalar amplitude 2.1 × 10^{-9}, so ΔT/T ≈ 9 × 10^{-6}), come from quantum fluctuations stretched by inflation	The feed acts only after z = 20, so this light is untouched. The hub's own uncertainty: energy spread √32, lowest return (31/33)² = 0.8825 (computed).	**No conflict: a joint fit to Planck, DESI and Pantheon+ allows up to 1.65% of cosmic matter.**	fitted jointly
3	**Cosmic dawn**	178 Myr z 20	The era of the first galaxies; JWST sees the earliest at z ≈ 14	The feed switches on at z = 20, fixed before any data were loaded	Set in advance	fixed in advance
4	**First galaxies seen**	290 Myr z 14.2	JWST: JADES-GS-z14-0 4.0 × 10^{8} M☉ · MoM-z14 1.3 × 10^{8} · JADES-GS-z14-1 3.7 × 10^{7}	Needs 7.8 × 10^{39} W, 2.5 × 10^{39} W and 7.1 × 10^{38} W; each of the 32 throats carries one thirty-second	**Enough: the ceiling is about 660× the need**	computed
5	**Black holes first**	464 Myr z 10	JWST: 14 black holes at z ≈ 4 to 10.6, each compared with its host galaxy	The black-hole-first ceiling from the feed (η = 0.053%)	**14 of 14 inside the ceiling**	η fitted
6	**Protoclusters**	589 Myr z 8.5	A2744-z7p9OD: stars > 1 × 10^{10} M☉ · MACS0416-OD-z8p5: ≈ 2.3 × 10^{9} M☉	Predicted before the comparison: 0.5–1.5 × 10^{10} M☉ and 1.4–4.9 × 10^{9} M☉	**2 of 2 inside the predicted range**	predicted first
7	**Reionization complete**	927 Myr z 6	JWST star-formation record, z ≈ 9 → 6	Feed ÷ record at each epoch, nothing re-fitted	**1.00 to 1.05: within 5%**	compared
8	**Cosmic noon**	3.27 Gyr z 2	Peak of cosmic star formation	Carried forward: the feed keeps running to mark 10	—	carried
9	**Solar System, Earth**	9.22 Gyr z 0.42	Formed 4.57 Gyr ago	Carried forward: the feed keeps running to mark 10	—	carried
10	**Today: Milky Way, Local Group**	13.79 Gyr z 0	Milky Way: 6 × 10^{10} M☉ in stars · galaxy clustering 0.613	Feed builds 6.4 × 10^{10} M☉ · architecture clustering 0.639	**Within 7%; clustering 4% apart; Andromeda 1.4×, Local Group 1.2×**	same η and γ

Cosmic ages: flat ΛCDM, Planck 2018 values, computed here as a ruler. The paused gravity, p = 0.781 ± 0.013 (fitted), acts across marks 3 and 4, from z = 20 to z = 10 (page 76). Sources: Continuous Throat Feed, Cosmic Web Atlas, Rivero-Zeta.

How to read: read across each row: when it happened, what the public record shows, what AI33-MPOPT gives, and whether the two match. Green means the two numbers are close, as stated in the cell; gray means the mark is carried forward.

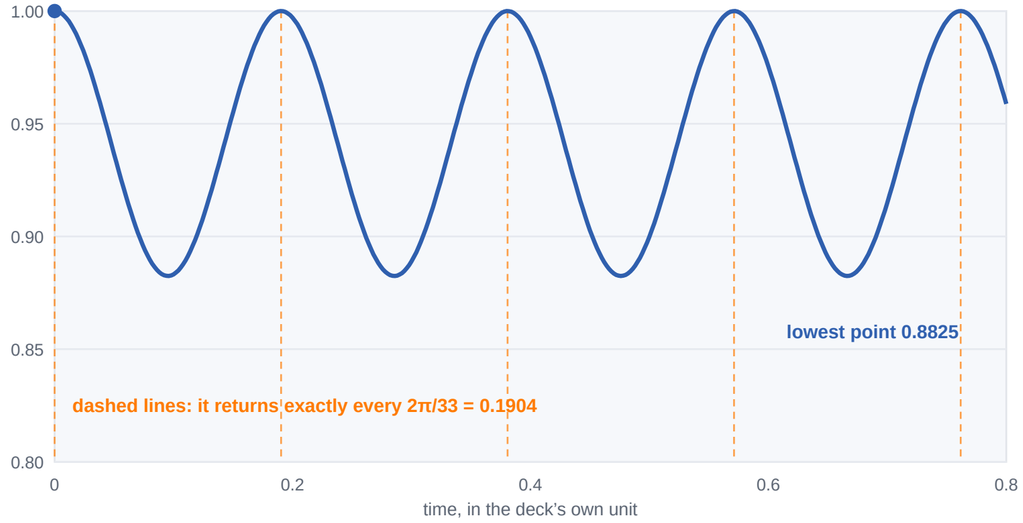
Inception: Where the Two Cones Touch

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

INSIDE THE 33-MULTIVERSE — COMPUTED

32 throats converge on one point, the hub, #33. At zero throat coupling the operator is singular and nothing passes; above w ≈ 0.15 it is regular again and the whole structure re-coheres.

THE HUB'S TWO STATES: PROBABILITY OF BEING FOUND AT THE HUB



Before the throats connect, distance and duration are undefined, so there is no "earlier" to measure from (14 Loop Quantum Gravity Bridge, page 7). The spark returns on an exact schedule: every 2π/33 = 0.1904, never lower than 0.8825.

The seen and unseen halves each have radius 0.707107; their product is exactly ½, the uncertainty minimum (ħ = 1). The seen half reads as position and the unseen half as momentum: the framework's reading (page 73).

IN OUR UNIVERSE — PUBLISHED VALUES · COMPUTED HERE

14.51 Gly
both radii

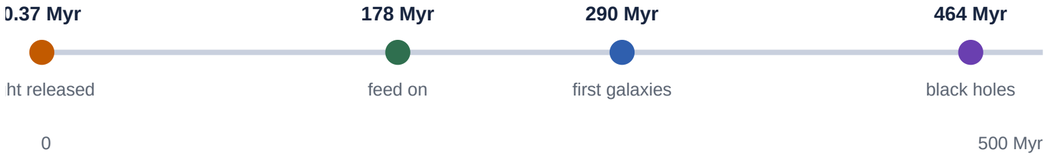
The Hubble radius, **c/H_0 = 14.51 Gly**, and the black-hole radius, **$2GM/c^2$ = 14.51 Gly**, of the **4.65 × 10^{22} $M_\odot$** inside it are the same size. This is exact for a flat universe at critical density.

13.79 Gyr
age of the universe today

0.371 Myr
when the CMB light was released

178 Myr
the feed switches on (z = 20)

THE FIRST HALF-BILLION YEARS



The crossing, as the framework reads it (premise): seen from outside, the 32 siblings converge through their throats onto one point, a black-hole view. Seen from inside, the same point is where Universe 33 exits and expands. What falls in on one side comes out on the other (Continuous Throat Feed, page 13).

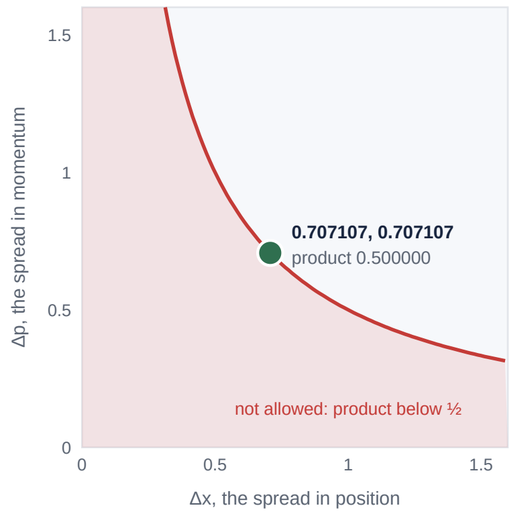
How to read: left, computed on the 33-node operator; right, standard cosmology computed with H_0 = 67.4 and Ωm = 0.315. The crossing itself is the framework's premise.

The Two Halves and the Uncertainty Minimum

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

THE UNCERTAINTY PRINCIPLE

STANDARD PHYSICS



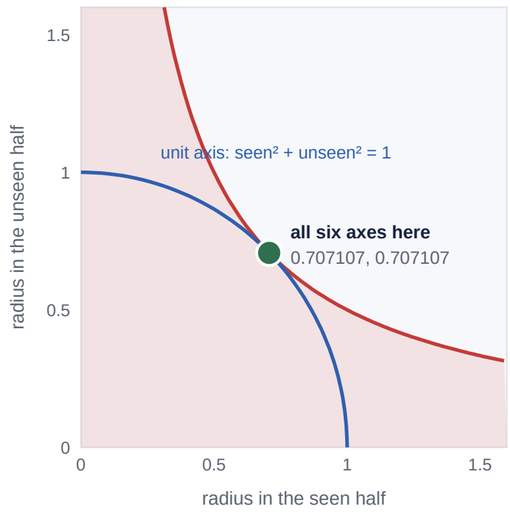
The spreads in a particle's position and momentum can never both be small: **Δx · Δp ≥ ħ/2**. In units where ħ = 1 the floor is ½.

The state that sits exactly on the floor has equal spreads: **Δx = Δp = 0.707107**, and 0.707107 × 0.707107 = **0.500000**. I computed that state directly.

The red region is not allowed. The green point is the minimum, the only place on the curve where the two spreads are equal.

THE AI33-MPOPT SPLIT

COMPUTED



The structure is read in six dimensions, split into a **seen** 3D half and an **unseen** 3D half (pages 17 and 18). Every one of the six axes has radius **0.707107** in each half: 0.707107 × 0.707107 = **0.500000**.

The unseen half has the same two angles as the seen half, **63.43° and 116.57°**, with neighbours and non-neighbours swapped. The split is an exact rotation of the six dimensions.

The blue arc is every possible split of a unit axis. It touches the red curve at one point only.

WHY IT HOLDS

proved

Icosahedral symmetry makes the six axes a perfectly even frame, so the two halves must be equal: each axis has radius² = 3 ÷ 6 = ½ in each half. Any other split of a unit axis has a product below ½ and would fall in the red region. **Only the equal split reaches the floor.** No number is tuned.

EXACT, AND A READING

framework's reading

Exact (computed, and checked independently from the 33-multiverse's icosahedron coordinates)**:** both radii 0.707107, product 0.500000, squares sum to 1.000000. **The reading:** the seen half stands for position and the unseen half for momentum. That is the framework's reading of the split.

WHERE ELSE IT SHOWS UP

computed

The hub is a two-state system and returns exactly every 2π/33 (page 72). All 32 points on the unseen side light up at once, with no delay (page 75). A separate timing check on the hub, σ = √32, gives σ × t = 0.5385 against ½, 7.7% above its minimum: the match is in the position and momentum split.

How to read: the numbers are computed here; the uncertainty principle is standard physics. Reading the seen half as position and the unseen half as momentum is the framework's own. Source for the six-dimensional split: page 32, Rivero-Zeta.

Why 33: The Structure Behind the Marks

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

TWO SECTORS EXIST AT 33 DOMAINS, AND AT NO OTHER SIZE — PROVED · EXHAUSTIVE

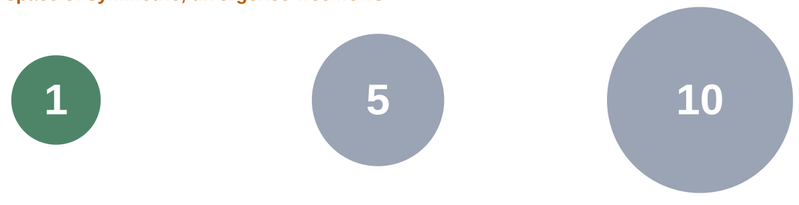
the same solid, the same symmetry group (120), the same rule at every size



A two-part dark-sector reading needs exactly two sectors. At 33 domains there are exactly two, the twelve and the twenty. One subdivision later the two-sector structure is gone (Geometric Transport, page 9).

ONE CIRCULATION, AND ONLY AT THIS SIZE — PROVED · COMPUTED

dimension of the space of symmetric, divergence-free flows



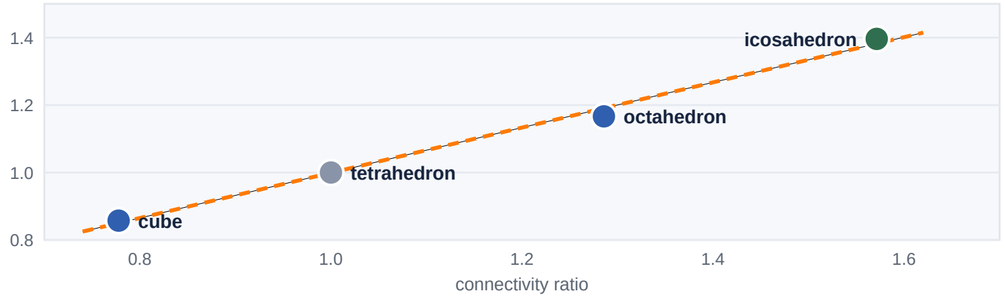
33 domains
one direction

123 domains
a family, none selected

273 domains
a family, none selected

Out through the twenty, back through the twelve: twelve throats carry what twenty carry, exactly **20/12 = 1.666667**, and each apex returns what its five curvilinears collect, exactly **5.000000**. Divergence is 4.3×10^{-16} at every node.

TRANSPORT FOLLOWS CONNECTIVITY, ACROSS FOUR SOLIDS — COMPUTED



R = 0.9974; transport = 0.670 × connectivity + 0.329, largest residual 0.024. The tetrahedron is the control: equal classes give exactly 1.0000 and no asymmetry (page 13). The vertical axis is the transport ratio.

WHAT THE STRUCTURE GIVES THE TEN MARKS

Every number on pages 68–85 is computed on one structure. These four results are what hold it fixed. The structure is the same in every comparison; the only tuned numbers are η, p and γ, labeled fitted.

Two sectors, 12 and 20, exist only at 33 domains	proved
The push-and-pull loop is derived, not named, and is unique at 33	proved
Transport asymmetry follows connectivity asymmetry on four solids	computed
17 numbers fix the correlation kernel, not 561; 33 profiles collapse to 3	proved

How to read: theorems and computations from the Geometric Transport file (pages 9, 10, 13, 16 and 17), carried here unchanged.

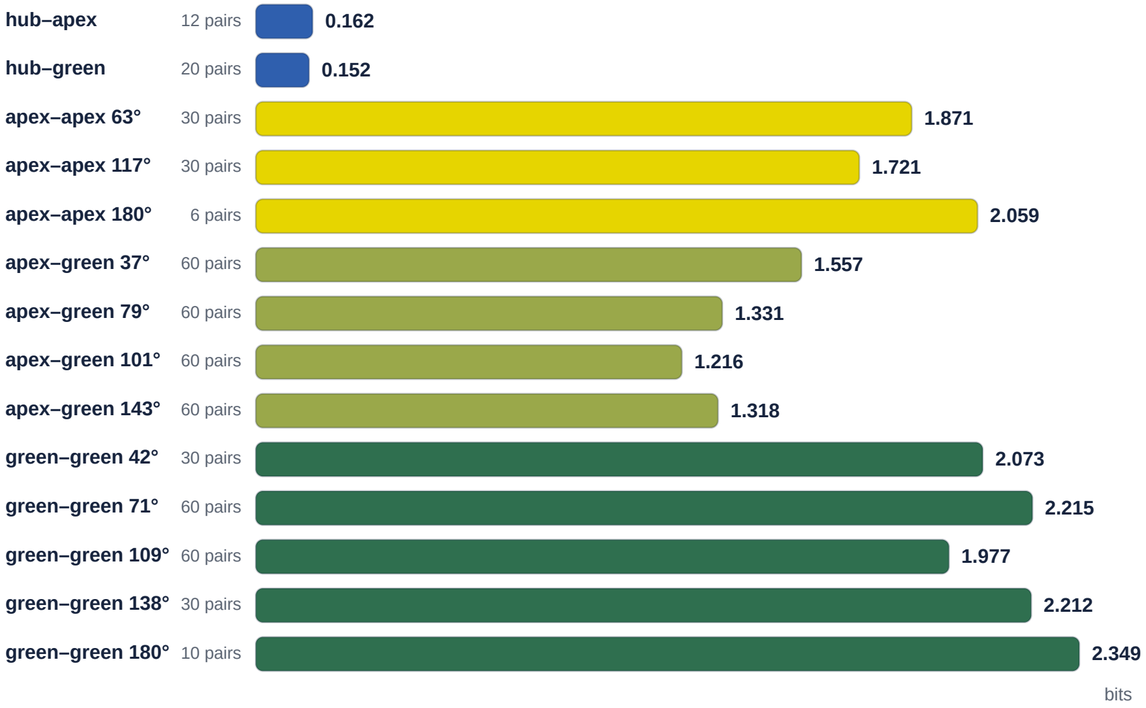
Entangled Pairs: In the Model, and in the Laboratory

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

IN THE MODEL: ALL 528 PAIRS IN 14 CLASSES, AT t = 3

COMPUTED

32 · 1 · 0 32 points, 1 instant, 0 delay: send the spark from the hub and all 32 points on the invisible side light up identically at once, equal to machine precision (Rivero-Zeta).



Entropy of one particle of a pair, in bits, from the two-particle test in the Rivero-Zeta deck (page 28), reproduced to six digits; interaction U = 8, time in the deck's own unit. Pairs inside a class are identical.

IN THE LABORATORY

MEASURED

Entangled photon pairs are measured to be correlated more strongly than any local classical account allows. The test value (CHSH) is 2 or less for such an account; quantum mechanics allows up to 2√2 ≈ 2.83, and experiments report values above 2 by many standard deviations. No signal passes between the two ends.



Same word, two settings. In the model, entanglement is computed on the 33-node operator, and it depends on where a pair sits: the opposite green pairs reach 2.349 bits, the hub pairs 0.152. In the laboratory it is measured on photons.

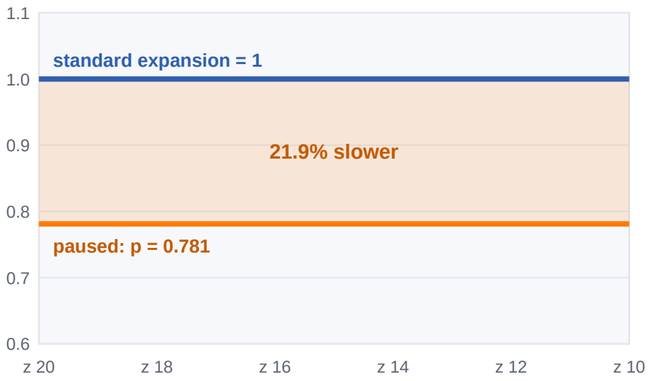
The reversal (page 62, computed): early on, the nearest pairs are the most entangled. Later the opposite pairs overtake them: green–green at t ≈ 1.60 and apex–apex at t ≈ 1.68. Apex–green pairs never reverse. In random networks with the same hub the farthest pairs never overtake the nearest.

How to read: left, computed from the deck's coordinates (pages 58–62); right, published experimental results.

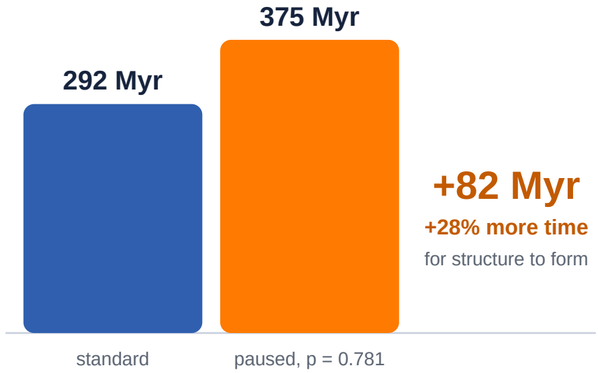
Paused Gravity and the Expansion

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

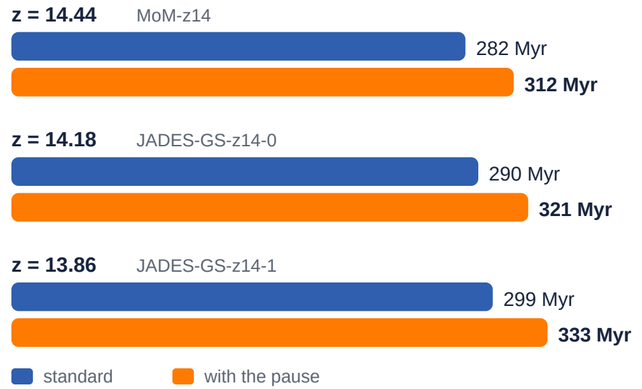
EXPANSION RATE, z = 20 → 10 — FITTED



TIME FROM z = 20 TO z = 10 — COMPUTED



COSMIC AGE AT THE EARLIEST GALAXIES — COMPUTED



WHAT THE PAUSE IS

At the rebound the Core sends out its first burst of light, and gravity then pauses as the forces separate (page 14). In the formula this is one factor on the standard expansion rate, $\mathbf{H(z) = p \cdot H_{\Lambda CDM}(z)}$, with **p = 0.781 ± 0.013**, fitted to ages inferred from public JWST galaxy data (64 walkers, 96,000 samples). The early expansion runs 21.9% slower, which gives structure 28% more time between z = 20 and z = 10: 292 Myr becomes 375 Myr.

HOW IT FITS THE PICTURE

The right-hand cone on page 70 carries this: its early part, marks 3 and 4, is the stretch of time that gets longer. The pause gives structure more time, and the continuous feed supplies the material (page 77); the first galaxies are seen at z ≈ 14. Here p is a fitted value from public JWST ages, and the ages are standard cosmology computed here.

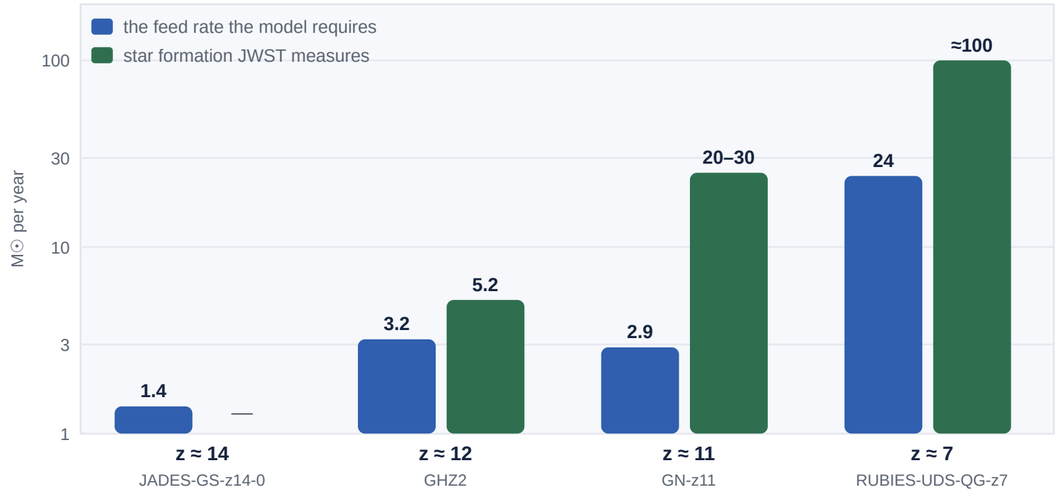
How to read: ages from a flat ΛCDM universe with Planck 2018 values; the pause is applied across z = 20 to z = 10, the range used here. Sources: Super Formula (page 8), Quantum-Gravity Bridge (page 7), Cosmic Web Atlas.

The First Galaxies, Beside the JWST Record

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

FEED NEEDED, AND STAR FORMATION MEASURED

COMPUTED · MEASURED



The three earliest galaxies need about 10^{39} W each, delivered continuously and shared over the 32 throats. From z ≈ 14 to z ≈ 7 the feed rate the model requires rises about 17-fold, within an order of magnitude of the star formation JWST measures directly. One fitted number, η = 0.053%, describes all five JWST epochs.

Galaxy	z	Stellar mass	Radius	Power to build it
JADES-GS-z14-0	14.18	4.0×10^8 M☉	260 pc	7.8×10^{39} W
JADES-GS-z14-1	13.86	3.7×10^7 M☉	≲ 50 pc	7.1×10^{38} W
MoM-z14	14.44	1.3×10^8 M☉	74 pc	2.5×10^{39} W

ROOM FOR THE FEED

COMPUTED

ceiling from a joint Planck + DESI + Pantheon+ fit

6.6×10^8 M☉ per Mpc³

needed by the earliest massive galaxies

≲ 10^6 M☉ per Mpc³

≈ 660× the ceiling is about 660 times the need; the feed uses under 1% of it.

THE COMPARISONS

14 / 14 — compared
JWST black holes, z ≈ 4 to 10.6, inside the black-hole-first ceiling

2 / 2 — predicted
protoclusters inside the prediction made before the comparison

1.00–1.05 — compared
feed ÷ JWST star-formation record, z ≈ 9 → 6, nothing re-fitted

Γ = 8.0 — compared
growth per expansion time, 81 JWST galaxies, no trend with redshift

How to read: data are published JWST measurements; the feed, ceilings and predictions are computed in the Continuous Throat Feed (pages 7–21).

More Comparisons, From the Decks

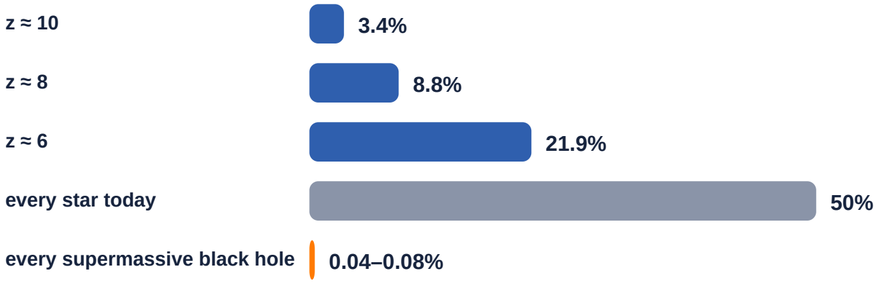
Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

TWO JWST RECORDS, SIDE BY SIDE — COMPARED



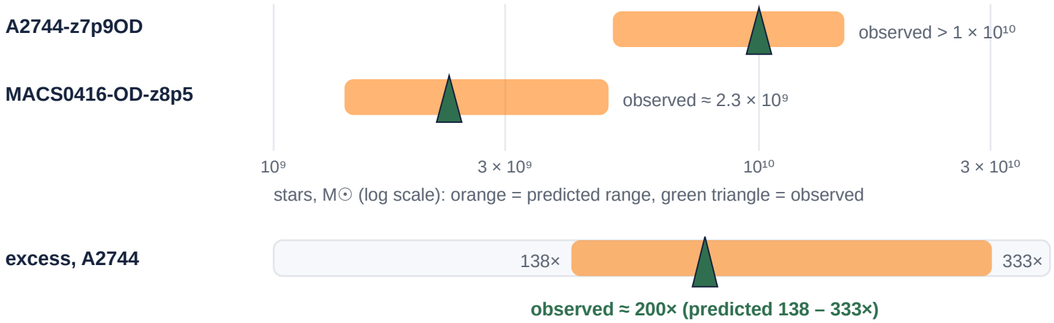
The last two columns are close at all four epochs: the feed stands to the stellar-mass census as JWST's own star-formation record does. The feed itself sits within 5% of that record (page 77).

IS THERE ENOUGH TO DELIVER? — COMPUTED

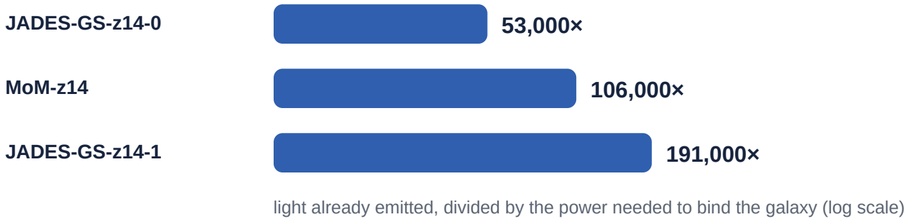


Share of the throat-feed ceiling (6.6 × 10⁸ M☉ per Mpc³) that JWST's highest inferred efficiencies would call on. At no epoch is more than 22% used: headroom of 4.4 to 29× across z ≈ 6 to 10.

PROTOCLUSTERS, PREDICTED BEFORE COMPARISON — PREDICTED



BINDING IS EASY; BUILDING IS THE TASK — COMPUTED



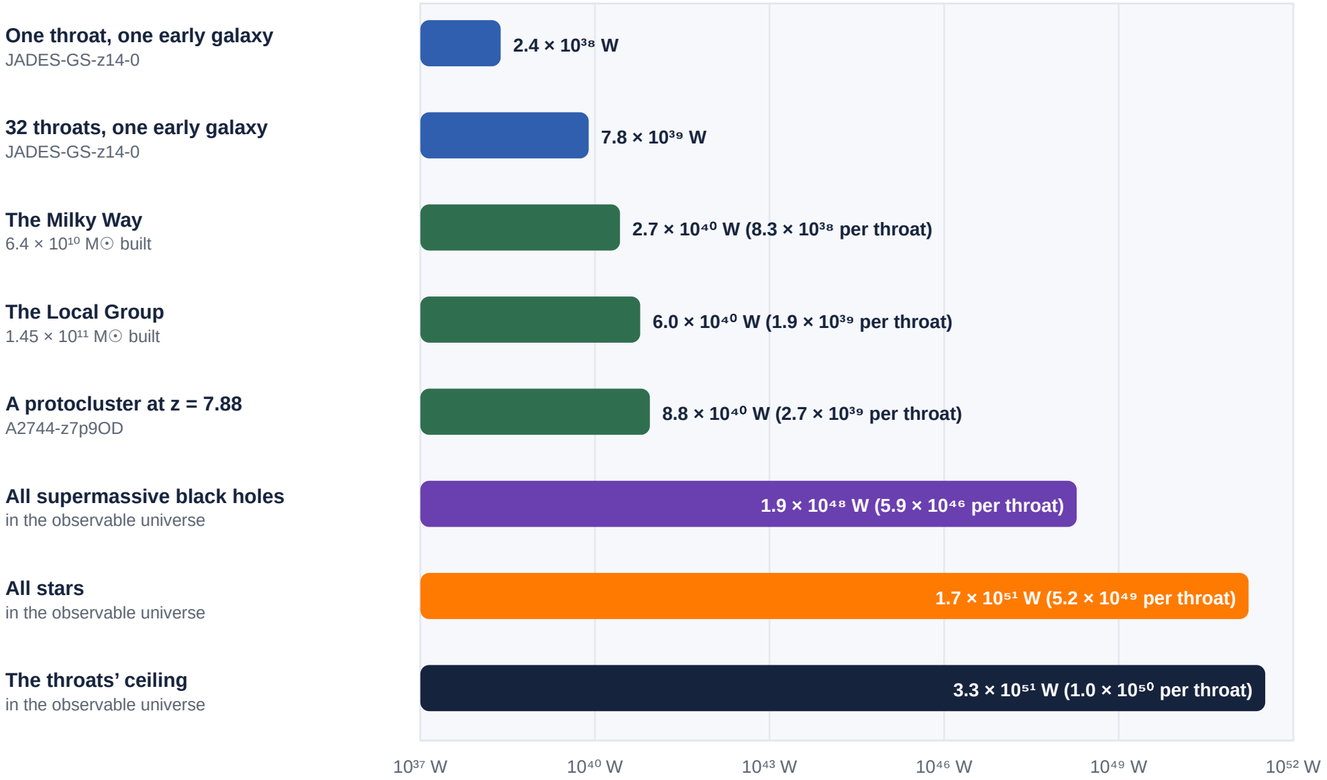
Binding takes about 10^{31} to 10^{32} W; the galaxies already radiate 10^{36} to 10^{37} W. The open problem is assembling the mass, the job the continuous feed performs.

How to read: all four are computed in the Continuous Throat Feed (pages 12, 16, 18 and 20); they were not yet on pages 75, 77 and 80.

From the Throats to the Cosmic Web: The Energy Audit

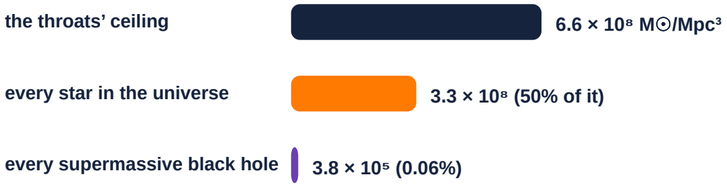
Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

THE POWER LADDER · ONE THROAT TO THE COSMIC WEB — COMPUTED



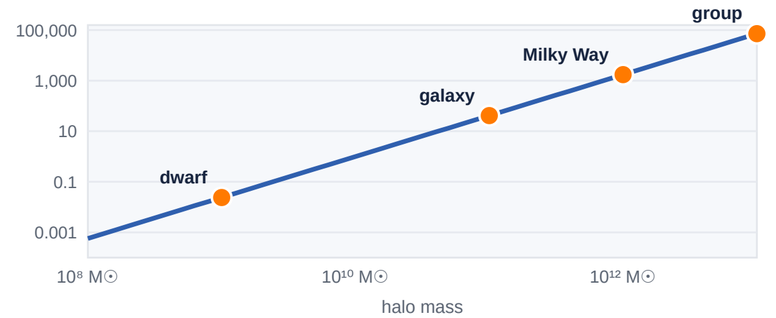
Power = stellar mass built × c² ÷ time since the feed switched on at z = 20 (13.6 Gyr). Per throat = ÷ 32. Masses are from the Throat Feed; the unit conversion is computed here.

IS IT ENOUGH? — COMPUTED



2.0× the ceiling is twice every star in the universe: 8.0 × 10²¹ M☉ allowed, 4.0 × 10²¹ M☉ in stars, out of 4.9 × 10²³ M☉ of matter. Stars and black holes together use half of it.

THE FEED FOLLOWS THE WEB — FITTED γ = 0.62



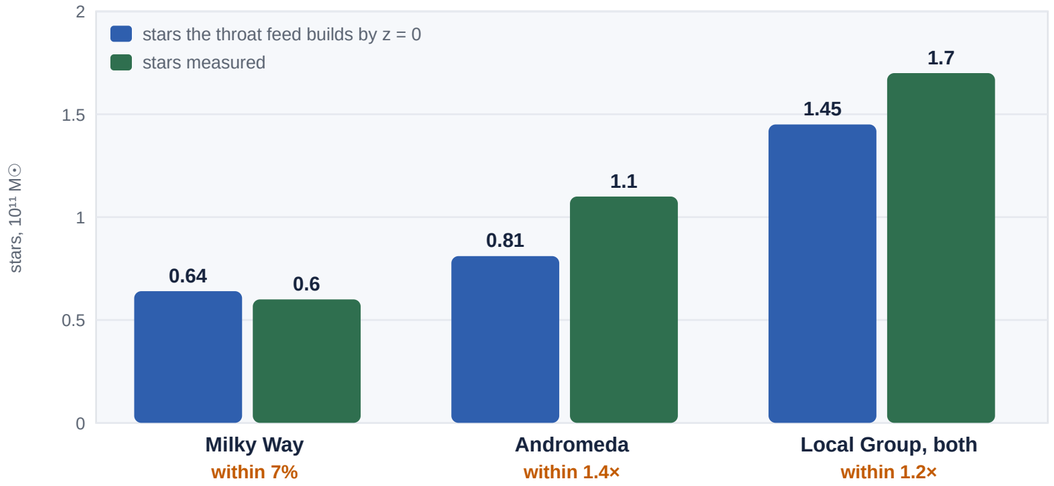
Delivery per halo ∝ M × (M / 10¹⁰ M☉)^0.62, set by the stellar-mass census at z ≈ 8. Compared with a 10¹⁰ M☉ halo, a Milky Way halo draws about 1,700 times more.

How to read: P = M★c² ÷ t, per throat = P ÷ 32; feed Q = 3 η H ρm f_halo with η = 0.053% (fitted). Sources: Continuous Throat Feed, pages 7–9, 11, 15, 16, 18, 19 and 21.

The Milky Way, and the Structure Behind the Numbers

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

THE MILKY WAY AND ITS NEIGHBOR — COMPARED



The delivery law was set using public JWST measurements at z ≈ 9 to 14.5 and never adjusted. Carried forward thirteen billion years, with the same η and γ = 0.62 and no number tuned to either galaxy, the model's feed gives 6.4 × 10¹⁰ M☉ for the Milky Way, against about 6 × 10¹⁰ measured, and gives Andromeda within 1.4×. In the Milky Way, Sagittarius A* holds 0.007% of the throat budget (4.3 × 10⁶ against 6.4 × 10¹⁰ M☉).

0.639	clustering of the architecture
0.613	clustering of a public galaxy catalogue
0.495	random sky, for scale

THE STRUCTURE AND THE COMPARISONS

THE STRUCTURE

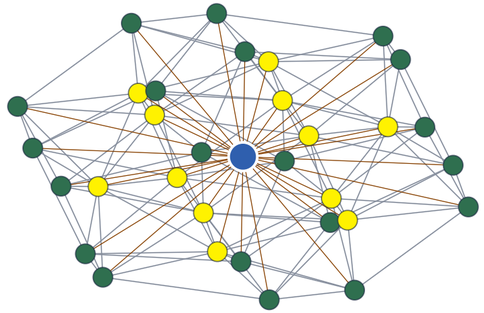
the 33-multiverse geometry, proved or computed

- 33 domains in exactly 3 groups: the hub, the twelve, the twenty
- 152 links and 120 symmetries
- 32 identical throats, 12 apex and 20 curvilinear
- two link counts only: 11 for an apex, 7 for a curvilinear
- apex over curvilinear conductance to the hub: 1.4298

THE COMPARISONS

what is set beside the public record, and what kind of number each is

first galaxies, 10^{39} W each	computed
paused gravity, p = 0.781	fitted
feed amplitude, η = 0.053%	fitted
protoclusters, 2 of 2	predicted
black holes, 14 of 14	compared
star-formation record, 1.00–1.05	compared
Milky Way, within 7%	compared



The comparisons are what the public record shows beside each number. The structure is what every number is computed on: the same 33 domains, the same two classes and the same 32 throats, from the inception spark to the Milky Way and Earth. It does not change from one comparison to the next, so it is the part that can be checked.

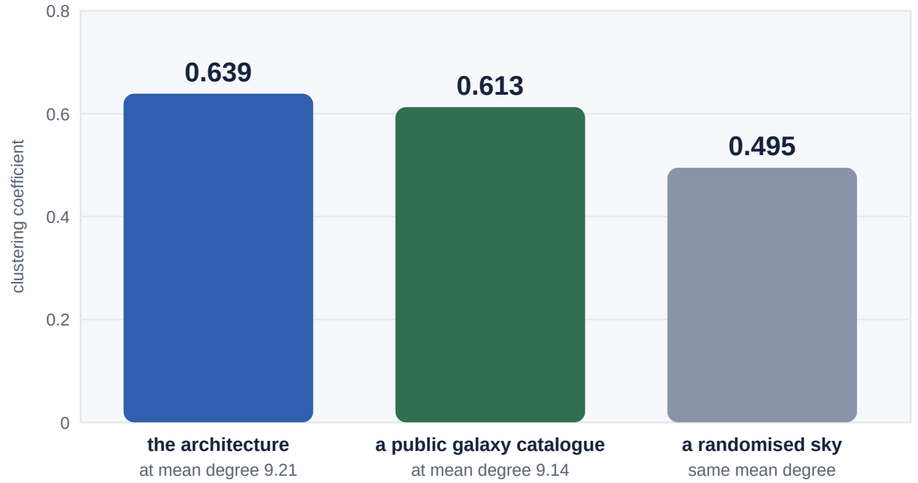
How to read: Milky Way and Andromeda from the Continuous Throat Feed (page 19); clustering from the Cosmic Web Atlas (page 8). Kinds: computed, fitted, predicted before the data, compared.

The Cosmic Web, Beside the Architecture

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

CLUSTERING, AT MATCHED DENSITY — COMPARED

0.639 against 0.613: four percent apart



120,606 galaxies from a public volume-limited sample, compared with the architecture at matched mean degree against a randomised-sky control. Both sit far above chance and on the same side of it. The clustering is stable at 0.612–0.621 across linking lengths.

WHERE IT MATCHES AND WHERE IT DIFFERS — REPORTED AS IT CAME OUT

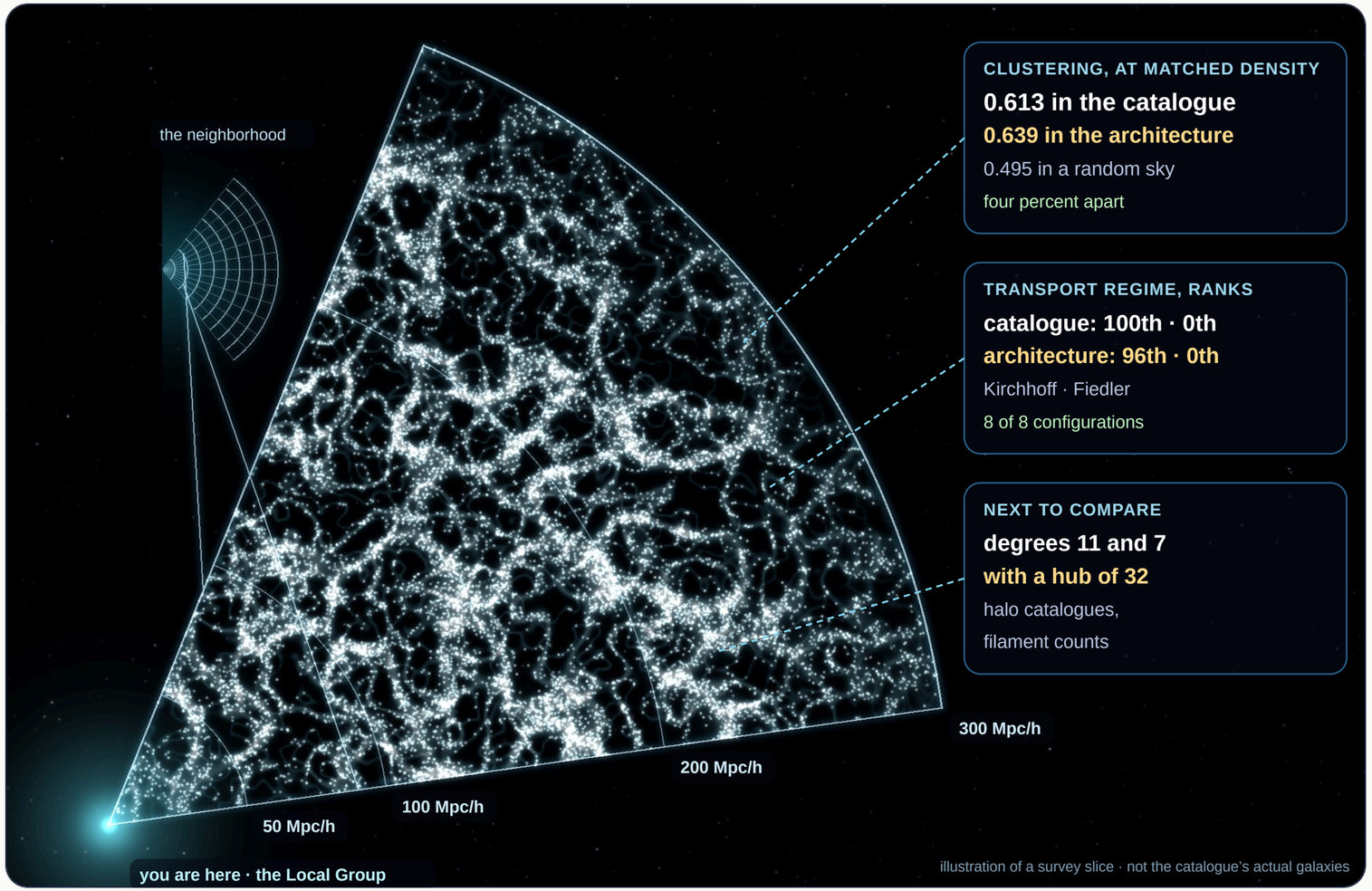
0.613 observed · 0.639 here	Clustering is close four percent apart; random sky 0.495	agrees
100th · 0th Kirchhoff · Fiedler rank	The observed galaxy network sits in the same transport regime: the architecture reads 96th and 0th 8 of 8 configurations; ranks against each network's own null	same regime
+0.72 assortativity · −0.15 here	Opposite sign: in the sky high-degree nodes join high-degree nodes; here apex domains join curvilinear ones follows from the two-sector wiring	reported
62% below mean density · 72% here	Volume fraction and energy fraction are different quantities; the comparison on Geometric Transport page 19 uses the second spherical occupation ratio 2.549 gives 71.8%	reported

The close quantities are ranks and a clustering value; the two that differ are shown beside them. The observed network and the architecture occupy the same transport regime (Geometric Transport, pages 24–25).

How to read: galaxy data are a public volume-limited sample at 50–300 Mpc/h; graphs built by comoving linking length so the mean degree matches. Compared as it came out, passes and differences together.

A Slice of the Cosmic Web, With the Architecture's Numbers

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly



CLUSTERING, AT MATCHED DENSITY
0.613 in the catalogue
0.639 in the architecture
0.495 in a random sky
four percent apart

TRANSPORT REGIME, RANKS
catalogue: 100th · 0th
architecture: 96th · 0th
Kirchhoff · Fiedler
8 of 8 configurations

NEXT TO COMPARE
degrees 11 and 7
with a hub of 32
halo catalogues,
filament counts

WHAT THE SLICE IS

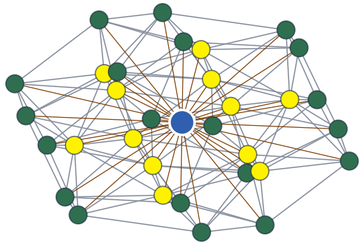
A volume-limited slice of a public galaxy catalogue: **120,606 galaxies** between 50 and 300 Mpc/h. Each galaxy is a point; two are joined when they lie within a linking length, chosen so the mean degree matches the architecture's: **9.14** against **9.21**.

COMPARED

0.613 catalogue clustering	**0.639** architecture
100th · 0th observed Kirchhoff and Fiedler ranks	**96th · 0th** the architecture's ranks

Reported: assortativity is +0.72 in the catalogue and −0.15 in the architecture; 62% of the surveyed volume lies below mean density, against 72% for the occupation ratio. Both are reported beside the close quantities (page 81).

THE ARCHITECTURE SIDE · 33 NODES, MEAN DEGREE 9.21

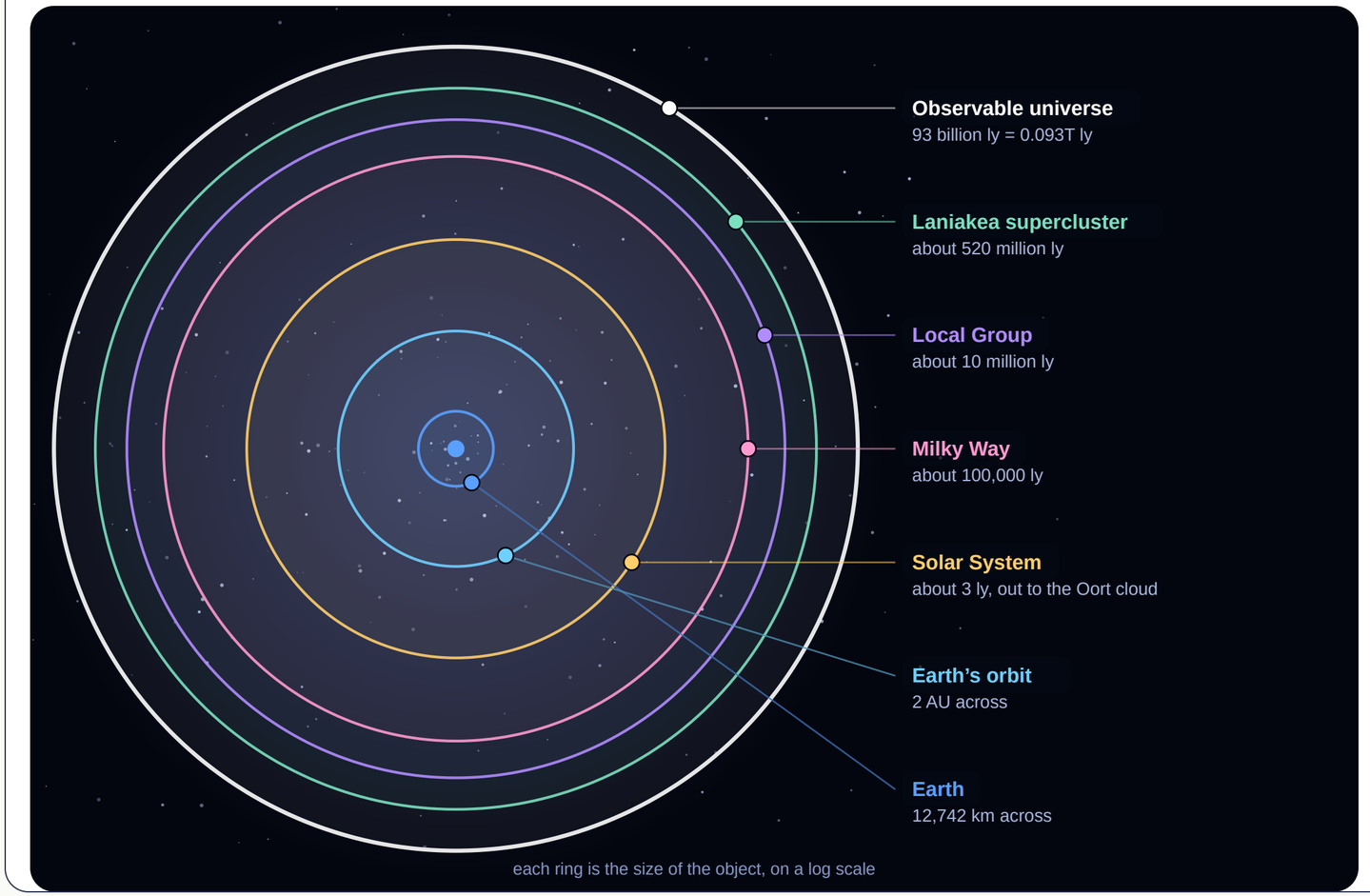


How to read: the sample, ranks and clustering are from the Geometric Transport file (pages 24–25); the picture is an illustration of such a slice.

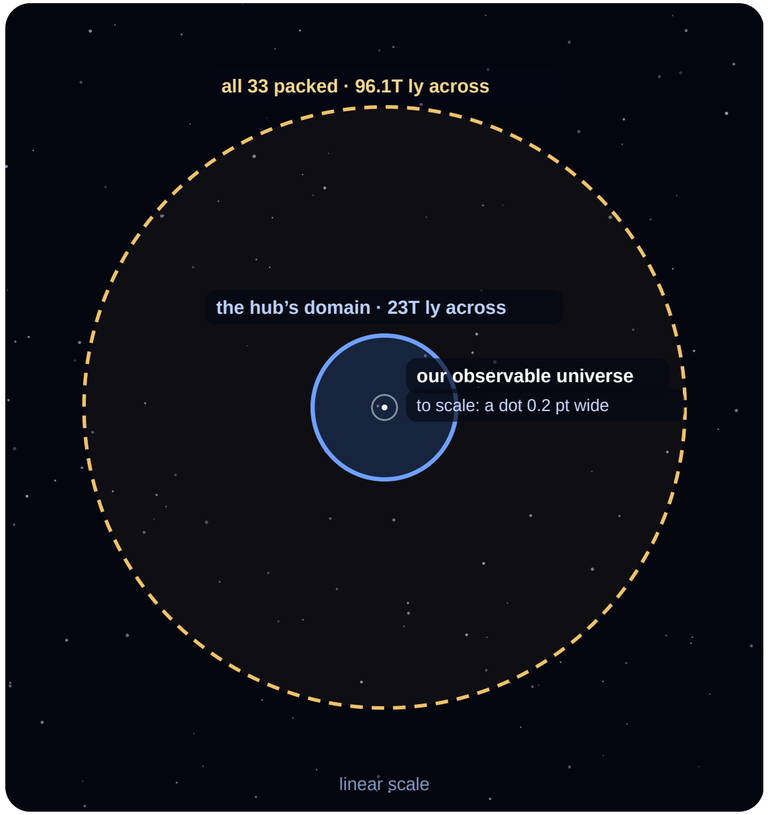
Zooming In: From Earth to the Whole Map

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

POWERS OF TEN · EARTH TO THE OBSERVABLE UNIVERSE — COMPUTED



TO SCALE · OUR UNIVERSE IN THE MAP — COMPUTED



Everything we can see is about **1 in 15 million** of the hub’s volume (page 2): 93 Gly across, against the hub’s 23,000 Gly.

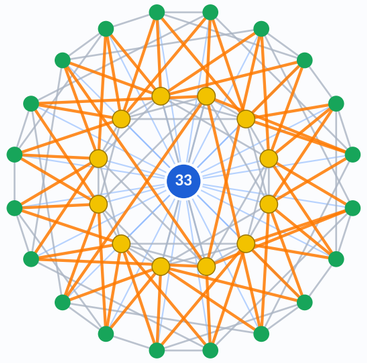
How to read: sizes are standard values; the hub and the full structure are from page 2. T = trillion; 1T ly = 1,000 Gly.

Loop Quantum, Computed on the Structure

COMPUTED

720° = 4π	33 of 33	4.4×10^{-16}	7.5×10^{-7}	20 of 20	0.4%
12 apex × 60° deficit, exact	nodes closed, no open leg	loop sign sequence −1 / 0 / +1, 17 regions	exact density = $\sin^2(\omega t)$, max deviation	throat-power reversals at hub zero crossings	curvilinear closure to spin 4

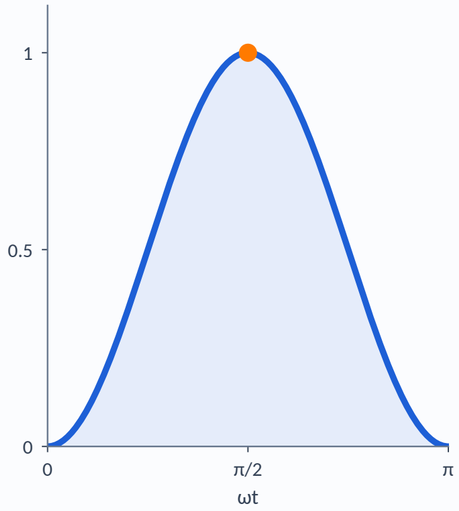
CURVATURE CARRIED BY THE LOOPS



12 × 60° = 720° = 4π

one 60° deficit at each of 12 apex · exact

THE BOUNCE: DENSITY = $\sin^2(\omega t)$



max deviation 7.5×10^{-7}

$(dx/dt)^2 = k\cdot x(1-x),\ k = 4\omega^2$

EVERY NODE CLOSED

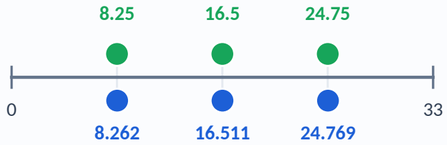


33 of 33 closed

spin ½ on 120 surface links · spin 1 on 32 throats

no open leg · totals hub 32 · apex 6 · curvilinear 4

THE CYCLE MEETS THE ZETA TURN



Rivero-Zeta cycle (step of 33)

hub wave of the structure

turn within 0.07% · zeros within 0.15%

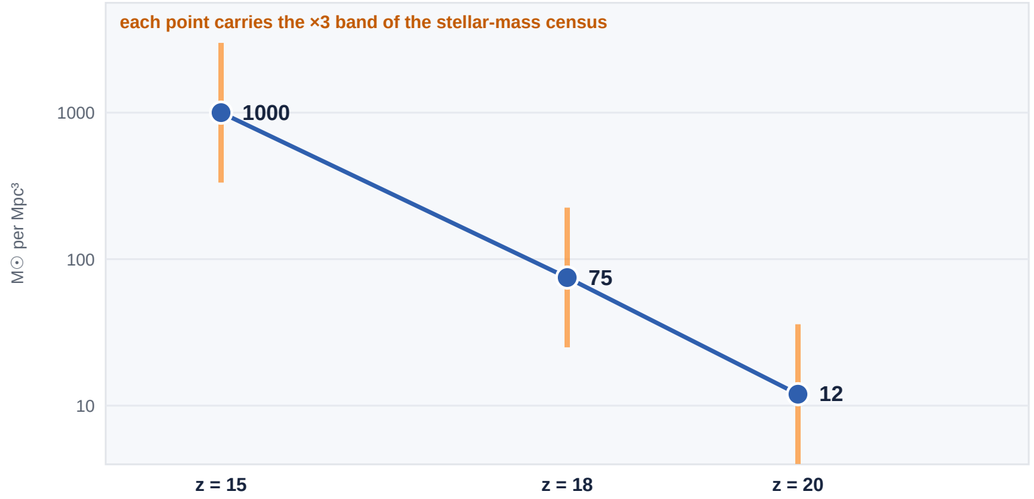
How to read: Fifteen published loop quantum ingredients, each beside the result computed on the whole 33-node object. Full table: 13 Quantum-Gravity Bridge pp. 30–32 and 14 Loop Quantum Gravity Bridge pp. 10–12.

AI33-MPOPT Compared with Our Universe

Gly = billion · 1 Gly = 1,000,000,000 light-years
T = trillion · 1T ly = 1,000 Gly

THE FIRST 200 MILLION YEARS — PREDICTED · RECORDED BEFORE THE DATA

Before any galaxy at z > 14.5 has a measured mass, the throat feed fixes the cosmic stellar mass density it can have built: **1.0 × 10³ M☉ per Mpc³ at z = 15, 7.5 × 10¹ at z = 18 and 1.2 × 10¹ at z = 20**.



Recorded in 2026 in the Continuous Throat Feed (page 26), ahead of public measurements at these redshifts.

ALSO COMPARED · LOOP QUANTUM GRAVITY

9 Spin networks and holonomies — computed · compared with published forms
33 of 33 nodes closed (spin ½ on the 120 surface links, spin 1 on the 32 throats); 720° = 4π; loop sign sequence matched to 4.4×10⁻¹⁶ over 17 regions.

10 The loop quantum bounce — computed · compared with the published bounce form
Density sin²(ωt) to 7.5×10⁻⁷ with k = 4ω²; 20 of 20 throat-power reversals; Rivero-Zeta turn at 16.5 of 33, wave turn at 16.511.

WHAT HAS BEEN COMPARED — COMPARED

1 The uncertainty minimum
Computed · the standard-physics floor, ħ/2
Product of the two radii 0.500000. Exact, computed and checked independently.

2 Dark energy ÷ dark matter
Compared with Planck
2.5490 in the structure against Planck's 2.5893 ± 0.0562 (0.72σ apart).

3 Early black holes
Compared with JWST · z ≈ 4 to 10.6
14 of 14 inside the black-hole-first ceiling from the feed (η fitted).

4 Star formation, z ≈ 9 to 6
Compared with the JWST star-formation record
Feed ÷ record 1.00–1.05, nothing re-fitted. Feed ÷ stellar-mass census 0.31, 0.32, 0.31, 0.40 against 0.31, 0.30, 0.30, 0.40 for the record.

5 The first galaxies
Compared with JWST star formation · z ≈ 14 to 7
The required feed rate rises about 17-fold, 1.4 to 24 M☉ a year: within an order of magnitude of the star formation JWST measures.

6 Protoclusters
Predicted before the data
Predicted 0.5–1.5 × 10¹⁰ and 1.4–4.9 × 10⁹ M☉; observed > 1 × 10¹⁰ and ≈ 2.3 × 10⁹ M☉: 2 of 2 inside. A2744 excess observed ≈ 200×, predicted 138–333×.

7 The Milky Way and its neighbors
Compared with measured stellar mass · same η and γ as the early galaxies
6.4 × 10¹⁰ M☉ against about 6 × 10¹⁰ (within 7%). Andromeda within 1.4×, Local Group within 1.2×.

8 The cosmic web
Compared with a public galaxy catalogue · 120,606 galaxies
Clustering 0.639 against 0.613 (4% apart; random sky 0.495). Transport ranks 96th · 0th against 100th · 0th in the catalogue (8 of 8 configurations).

How to read: Each row sets a number from the structure beside a published measurement. Fitted: η, p and γ. Sources: pages 66–84.

How the nine decks fit together

One line per deck: what each one computed, proved, compared or fitted

13	**Quantum-Gravity Bridge**	One operator, $T_p = L + pK + (1-p)Q$ on 33 nodes. An isolated 11-mode band; $L \cdot R = \kappa(F - \langle F \rangle)$ at $r = 0.84$ (perm. $p < 0.0005$).	COMPUTED
14	**Loop Quantum Gravity Bridge**	Spectral dimension $d_s = 2.56$ at 642 nodes. The $F \leftrightarrow R$ coupling with a plain graph Laplacian: +0.561 against +0.582.	COMPUTED
15	**Lawfulness**	97.4% in one component against a strongest-null maximum of 79.1% ($p < 0.0005$). 114 of 120 edges recovered (AUC 0.9965).	COMPUTED
16	**Super Formula Build-Up**	T_p built piece by piece (L, K, Q, p). $p = 0.781 \pm 0.013$ fitted to JWST chronometer ages; fixed-point test $p^* = 0.7577$ (1.8σ).	FITTED
17	**Geometric Transport Arch.**	$\lvert Aut \rvert = 120$, orbits 1, 12, 20. Throats deliver 9.487 vs 6.635 (1.43). SDSS DR7: 100th / 0th percentile; architecture 96th/0th.	PROVED · COMPARED
18	**Continuous Throat Feed**	Feed ceiling ≤ 1.65% of cosmic matter. Fitted $\eta = 0.053\%$, $\gamma = 0.62$ against star formation $z = 9$–14.5; 14/14 black holes inside.	FITTED · COMPARED
19	**Cosmic Web Atlas**	Occupation ratio 2.5168 exactly $(2\sqrt{15}/\sqrt{(5+2\sqrt{5})})$. Flow ratios 1.3963 and 1.5714. Delivery per unit mass rises as $M^{0.62}$.	PROVED · COMPUTED
20	**The Rivero-Zeta Curve**	Twelve equal 60° quanta total 4π. $R^2 = I$, eigenspaces +1 × 17 and −1 × 6, index 11.	PROVED
21	**The 33-Multiverse Map**	Clustering at matched mean degree: 0.639 here, 0.613 public catalogue, 0.495 random sky. All 528 pair entropies computed.	COMPARED